

Chapter 6: Automated and Emerging Technologies

6.1 Automated systems

6.2 Robotics

6.3 Artificial intelligence

Learning Objectives:

6.1 Automated systems

Candidates should be able to:

- 1 Describe how sensors, microprocessors and actuators can be used in collaboration to create automated systems
- 2 Describe the advantages and disadvantages of an automated system used for a given scenario

Notes and guidance

- Including scenarios from:
 - industry
 - transport
 - agriculture
 - weather
 - gaming
 - lighting
 - science

Learning Objectives:

6.2 Robotics

Candidates should be able to:

- 1 Understand what is meant by robotics
- 2 Describe the characteristics of a robot
- 3 Understand the roles that robots can perform and describe the advantages and disadvantages of their use

Notes and guidance

- Robotics is a branch of computer science that incorporates the design, construction and operation of robots
- Examples include factory equipment, domestic robots and drones
- Including:
 - a mechanical structure or framework
 - electrical components, such as sensors, microprocessors and actuators
 - programmable
- Robots can be used in areas including:
 - industry
 - transport
 - agriculture
 - medicine
 - domestic
 - entertainment

Learning Objectives:

6.3 Artificial intelligence

Candidates should be able to:

- 1 Understand what is meant by artificial intelligence (AI)
- 2 Describe the main characteristics of AI as the collection of data and the rules for using that data, the ability to reason, and can include the ability to learn and adapt
- 3 Explain the basic operation and components of AI systems to simulate intelligent behaviour

Notes and guidance

- AI is a branch of computer science dealing with the simulation of intelligent behaviours by computers
- Limited to:
 - expert systems
 - machine learning
- Expert systems have a knowledge base, a rule base, an inference engine and an interface
- Machine learning is when a program has the ability to automatically adapt its own processes and/or data

Resources:



<https://www.hoddereducation.co.uk/subjects/general/products/14-16/cambridge-igcse-and-o-level-computer-science-secon>

Resources:



Craig 'n' Dave For Students

CRAIG 'N' DAVE

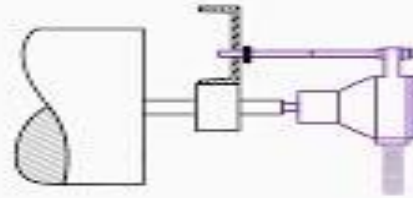
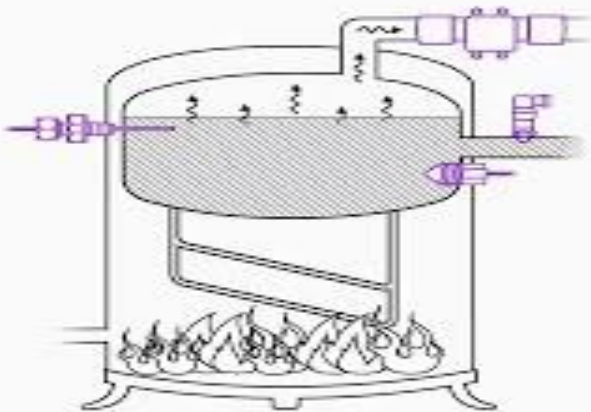
To explore the entire series,
visit <https://craigndave.org/free-videos>

6.1 Automated systems

Recap of Chapter 3 Hardware: Sensors

<https://www.youtube.com/watch?v=XI49uFm5HRE>

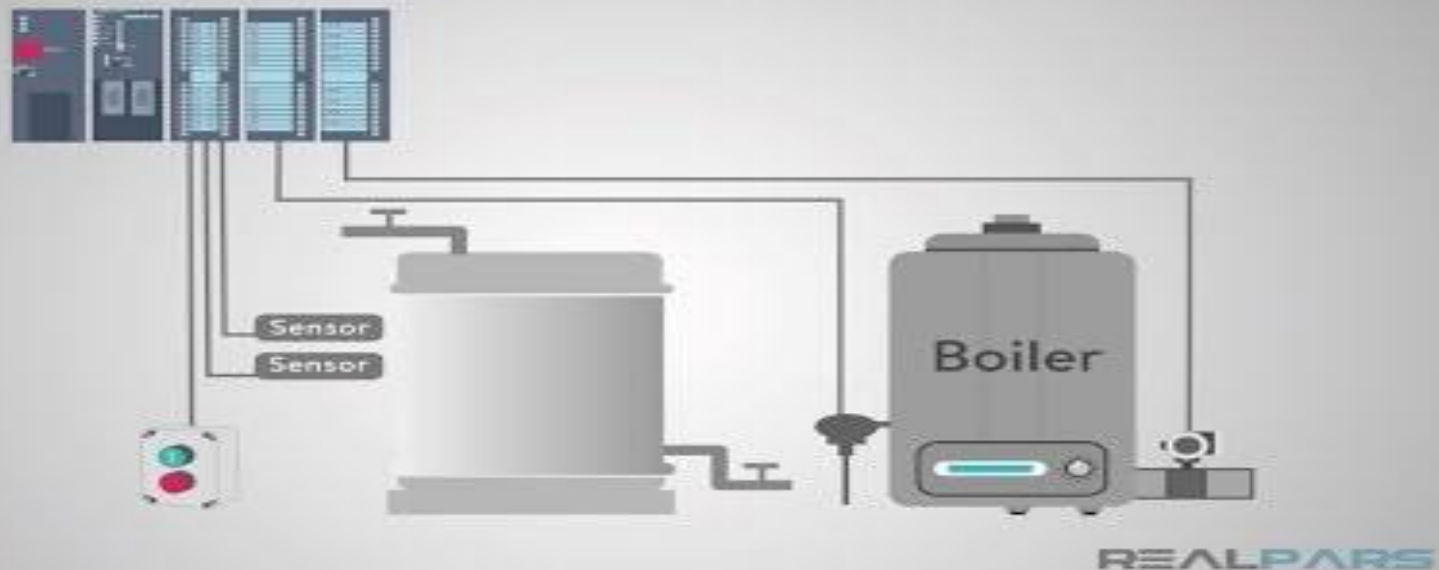
Types of Sensors



6.1 Automated systems

Recap of Chapter 3 Hardware: Sensors

https://www.youtube.com/watch?v=J_KoRp8SnoE



6.1 Automated systems

Recap of Chapter 3 Hardware

Sensors

Controlling stuff using Computers

The role of the sensor

we looked at how sensors could be used alongside computers to measure quantities (like temperature, light intensity etc) and then log changes in these quantities.

Sensors can also be used alongside computers to **control different devices** . The process goes like this:

Input - Sensor detects data in the environment around it

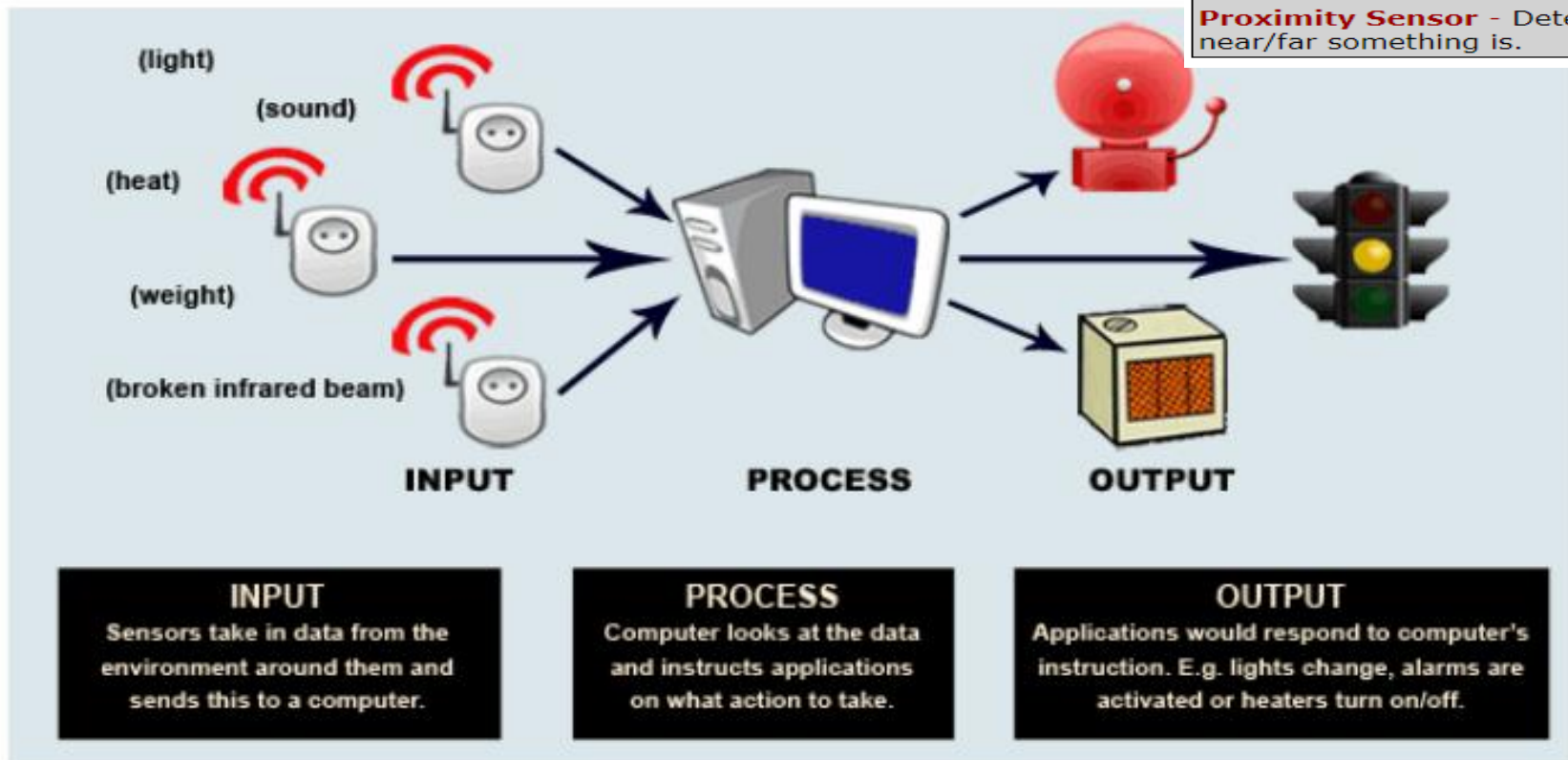
Process - Data passed to a computer (microprocessor) inside the device which analyses it and decides what action to take. The computer sends instructions to the device telling it what to do.

Output - The device would carry out the instructions.

6.1 Automated systems

Recap of Chapter 3 Hardware: Sensors

Picture Example:



Examples of Sensors

Temperature/Heat - Detect changes in temperature.

Light Sensor - Used to detect the brightness of light.

Sound Sensor - Detect the loudness of sound.

Humidity Sensor - Detect water in air.

Infra-red Sensor - Detect movement (Used in burglar alarms).

Load/Weight Sensor - Detect weight (Used in digital scales).

Proximity Sensor - Detect how near/far something is.

6.1 Automated systems

Recap of Chapter 3 Hardware: Sensors

Advantages of using a computer to control devices rather than people

Computers have certain **advantages over people** when it comes to controlling devices. For example:

Cheaper -	➔ If a computer is monitoring and controlling applications, you do not need to employ people .
Higher Work Rate -	➔ Computers can control applications all day, every day without getting tired or bored.
Safer -	➔ Computers can work in conditions that would be too dangerous for people . Examples include chemical plants, radioactive sites and extremely cold areas (antarctic).
Accuracy -	➔ Computers will respond to inputs from sensors accurately every time . E.g. a heater will be switched on as soon as the temperature falls below 10°C.
Speed -	➔ Computers will respond to data received from sensors very quickly . E.g. as soon as an infrared sensor detects an intruder the alarm will sound.



Radioactive sites can be too dangerous for people to work in.

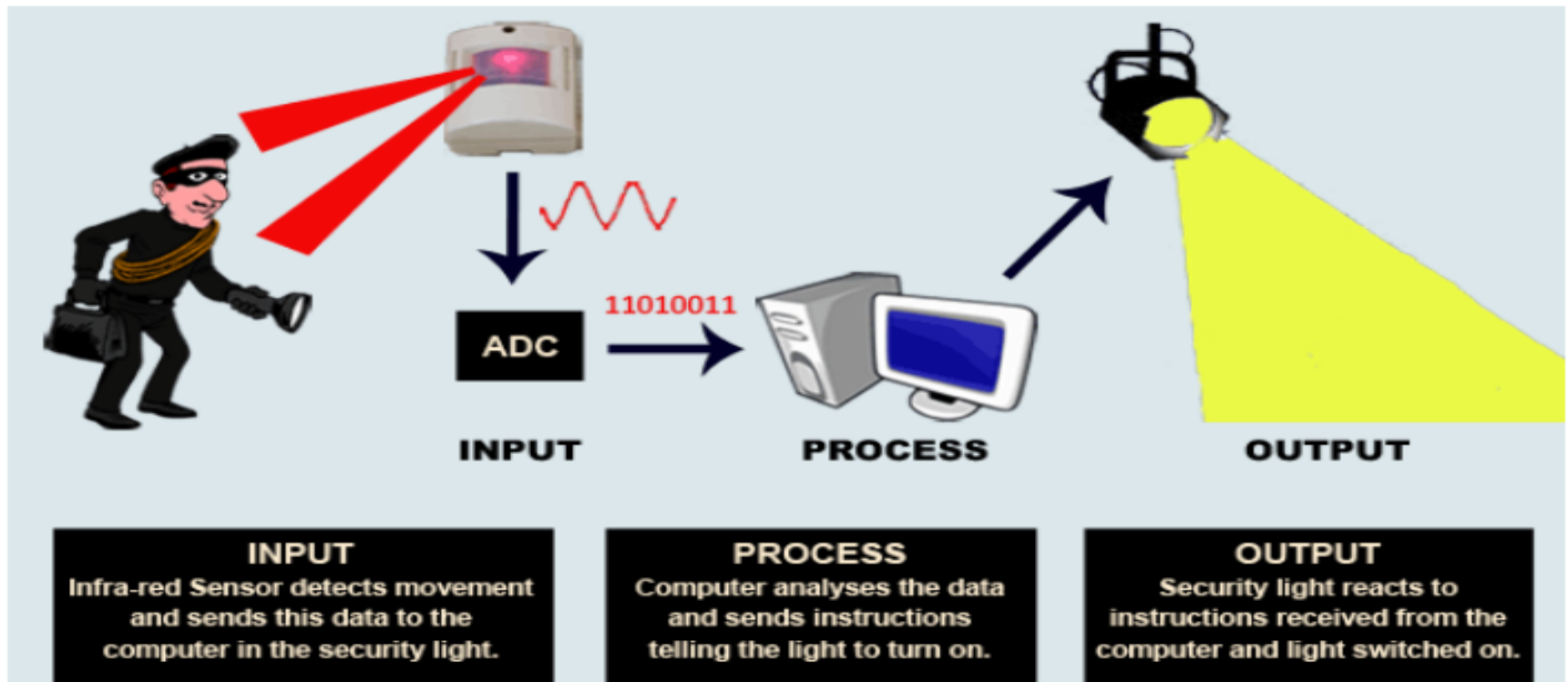
6.1 Automated systems

Sensors for Security Light

- # Security lights use **infra-red sensors** to **detect movement**.
- # Movement is detected when something (man or beast) **breaks the infra-red beam**. When this happens, the system will **turn on the light**.
- # If nothing else breaks the beam after a period of time, the system will turn the light off.



Security lights with built-in infra-red sensor



6.1 Automated systems

Recap of Chapter 3 Hardware

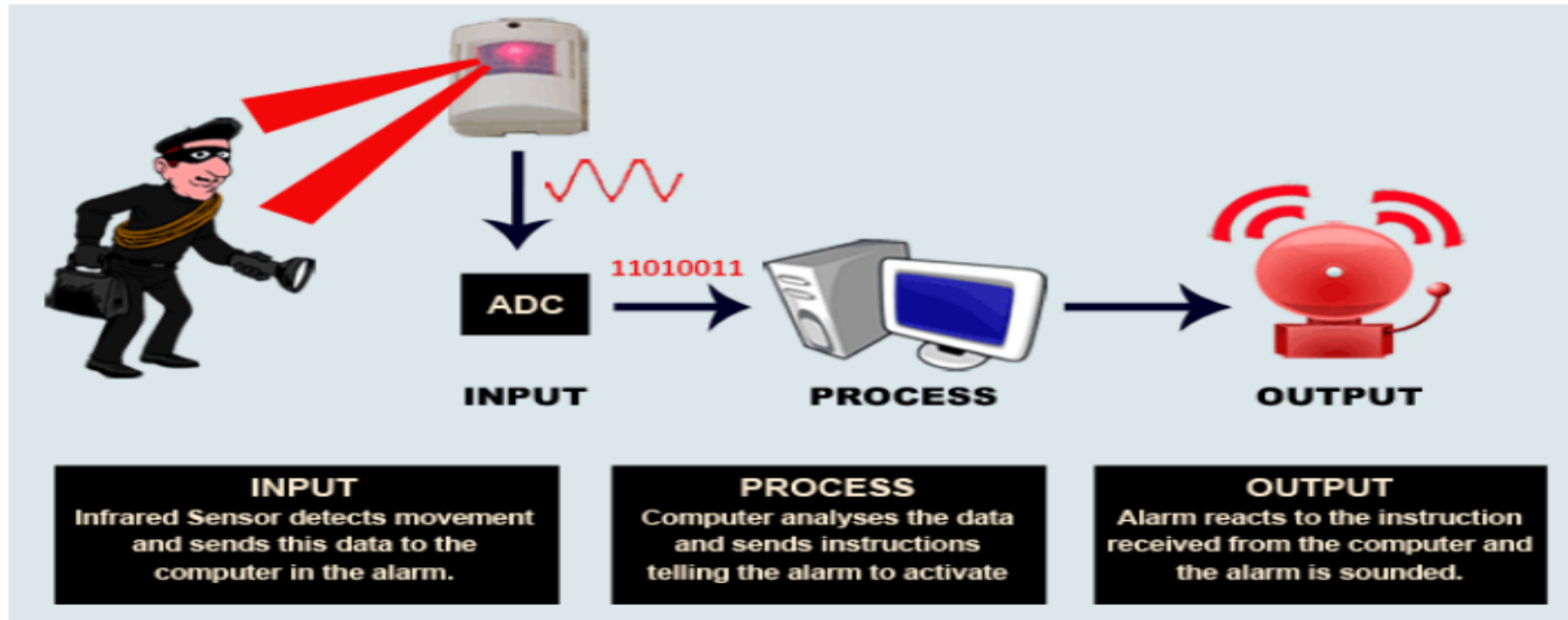
Sensors for Burglar Alarms

Burglar alarms work in almost the same way as a security light system.

- Input -** Infra-red sensor detects movement when the sensor is broken and this information is sent to the computer
- Process -** Computer makes the decision to sound the alarm
- Output -** Alarm is activated.



A modern alarm system with infra-red sensors



6.1 Automated systems

Recap of Chapter 3 Hardware

Sensors Central Heating



- # Modern central heating systems can be programmed to **maintain a constant desirable temperature.**

Let's imagine our perfect room temperature was set at **25°C** - we could use sensors and a computer to:

- **Maintain the optimum temperature of 25°C**
- **Turn heating off when it rises above 25°C**
- **Turn heating on when it falls below 25°C**

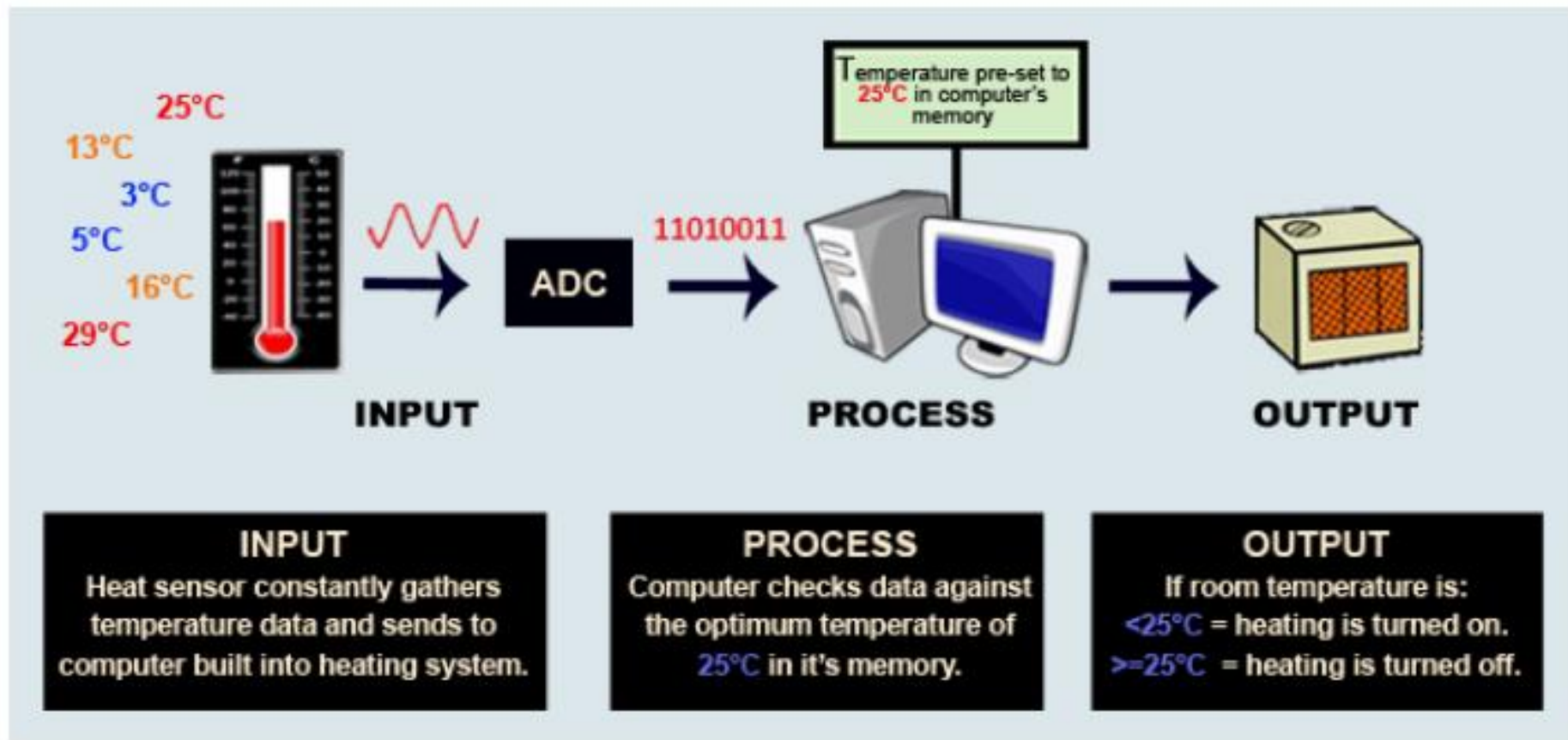
- # Here is how it would work:

- Input -** Heat sensor detects the current temperature and sends this data to the computer built into the heating system
- Process -** Computer would check the sensor heat data against the temperature setting stored in it's memory. Computer decides if the heating needs to be turned on (or off)
- Output -** Heating is switched on or off.

6.1 Automated systems

Recap of Chapter 3 Hardware

Sensors Central Heating



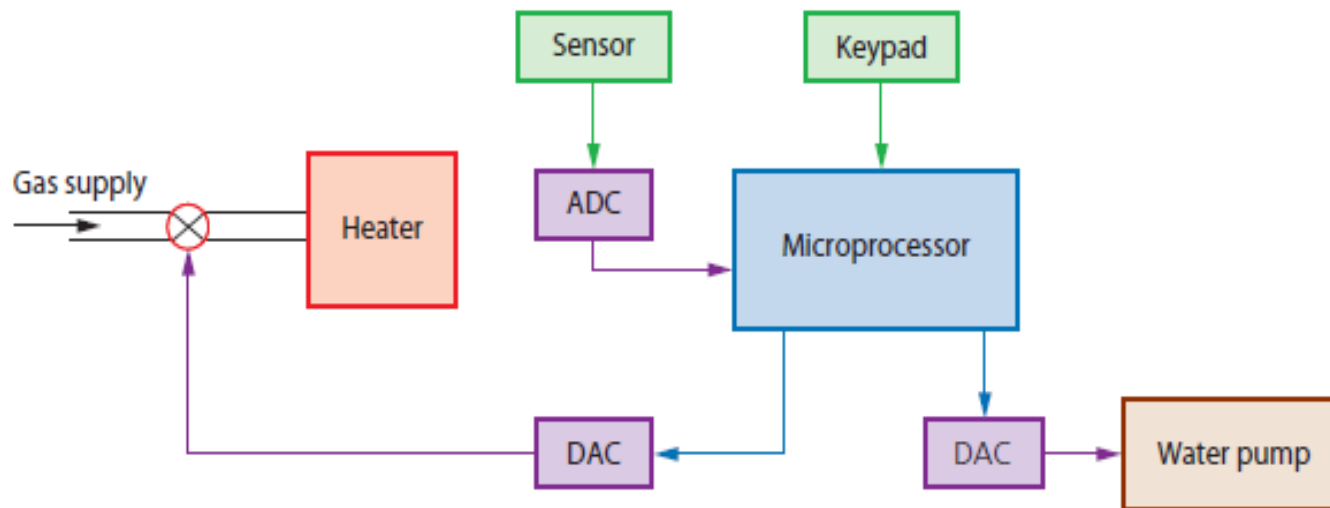
6.1 Automated systems

Recap of Chapter 3 Hardware

Sensors Central Heating



In this example, a gas supply is used to heat water using a heater. A valve on the gas supply is controlled by a microprocessor and is opened if the heating levels need to be increased. A water pump is used to pump hot water around the central heating system whenever the temperature drops below a pre-set value:



6.1 Automated systems

Recap of Chapter 3 Hardware

Sensors Greenhouse

- # Automatic greenhouses can provide the **optimum conditions** for growing plants by using computer control.
- # Computers can be used to **monitor conditions** and **control a range of applications** (devices) to keep the **perfect conditions constant**.

Sensors needed to collect data:

- Light sensor
- Moisture sensor
- Heat sensor

Control applications (devices) needed:

- Grow lights to make plants flourish
- Motor to turn sprinkler on if plants need water
- Heater to warm the greenhouse
- Motor to open window vent to cool greenhouse down if it gets too hot.

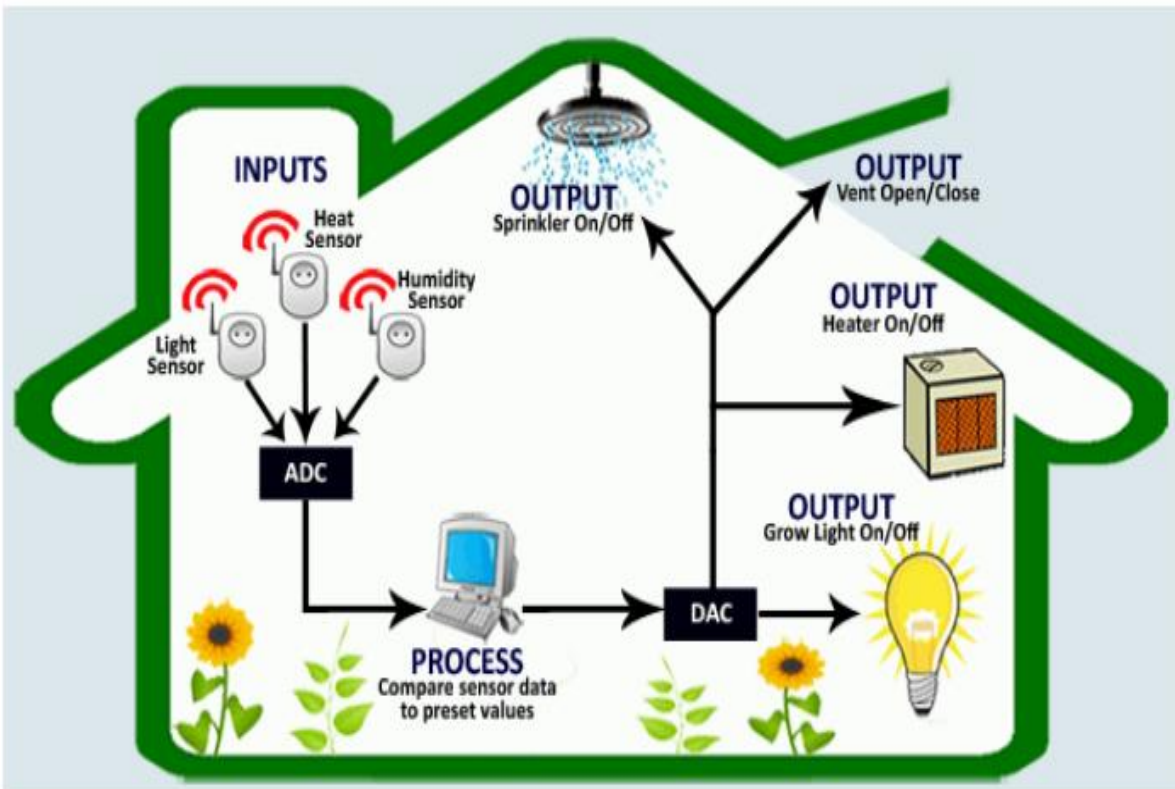


Computer controlled greenhouses **automatically detect humidity levels** and switch sprinklers on when the soil becomes too dry

6.1 Automated systems

Recap of Chapter 3 Hardware

Sensors Greenhouse



How the system works:

- 1 Computer **reads data** from sensors and **compares** it to **preset values**.
- 2 If it is **too hot** = heater turned off and **vent opened**.
- 3 If it is **too cold** = heater switched **on** and **vent is closed**.
- 4 If there is **enough light** = **grow light switched off**.
- 5 If it is **too dark** = **grow light switched on**.
- 6 **Wet soil** = **sprinkler switched off**.
- 7 **Dry soil** = **sprinkler switched on**.

NOTE!

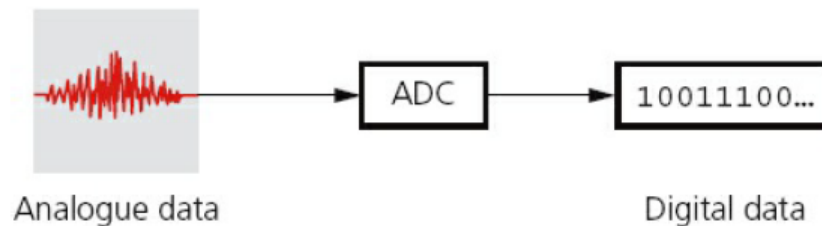
The process will **repeat constantly**. Sensors continually feed new data into the system and the computer will make decisions based on that data.

6.1 Automated systems

Recap of Chapter 3 Hardware: Sensors

A sensor is a hardware device that will take measurement of physical properties like temperature, pressure or acidity.

The data that is taken in / measured by a sensor is usually in **analogue form**. computers cannot make any sense of these physical quantities and the data needs to be converted into a digital format. This is usually achieved by an **ANALOGUE TO DIGITAL CONVERTER (ADC)**. This device converts physical values into discrete digital values.



When the computer is used to control devices, such as a motor or a valve, it is necessary to use a **DIGITAL TO ANALOGUE CONVERTER (DAC)** since these devices need analogue data to operate

There are many different sensors available each designed to measure different physical properties. Some of these sensors are included in the table below:

6.1 Automated systems

Recap of Chapter 3 Hardware Sensors

Sensor	Description of sensor	Example applications
Temperature	measures temperature of the surroundings by sending signals; these signals will change as the temperature changes	• control of a central heating system • control/monitor a chemical process • control/monitor temperature in a greenhouse
Moisture	measures water levels in, for example, soil (it is based on the electrical resistance of the sample being monitored)	• control/monitor moisture levels in soil in a greenhouse • monitor the moisture levels in a food processing factory
Humidity	this is slightly different to moisture; this measures the amount of water vapour in, for example, a sample of air (based on the fact that the conductivity of air will change depending on the amount of water present)	• monitor humidity levels in a building • monitor humidity levels in a factory manufacturing microchips • monitor/control humidity levels in the air in a greenhouse
Light	these use photoelectric cells that produce an output (in the form of an electric current) depending on the brightness of the light	• switching street lights on or off depending on light levels • switch on car headlights automatically when it gets dark
Infrared (active)	these use an invisible beam of infrared radiation picked up by a detector; if the beam is broken, then there will be a change in the amount of infrared radiation reaching the detector (sensor)	• turn on car windscreen wipers automatically when it detects rain on the windscreen • security alarm system (intruder breaks the infra-red beam)
Infrared (passive)	these sensors measure the heat radiation given off by an object, for example, the temperature of an intruder or the temperature in a fridge	• security alarm system (detects body heat) • monitor the temperature inside an industrial freezer or chiller unit
Pressure	a pressure sensor is a transducer and generates different electric currents depending on the pressure applied	• weighing of lorries at a weighing station • measure the gas pressure in a nuclear reactor
Acoustic/sound	these are basically microphones that convert detected sound into electric signals/pulses	• pick up the noise of footsteps in a security system • detect the sound of liquids dripping at a faulty pipe joint

6.1 Automated systems

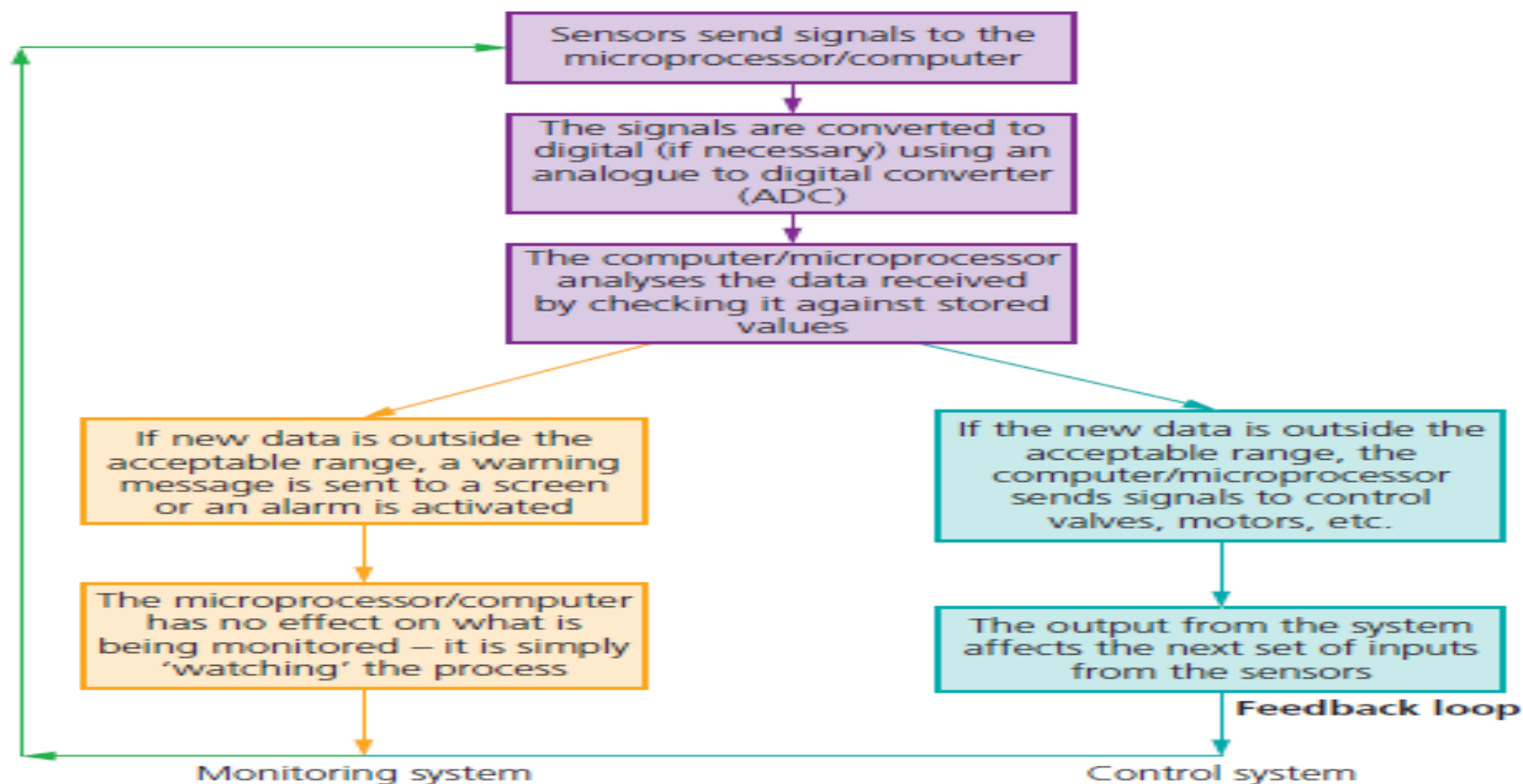
Recap of Chapter 3 Hardware: Sensors

Gas	most common ones are oxygen or carbon dioxide sensors; they use various methods to detect the gas being monitored and produce outputs that vary with the oxygen or carbon dioxide levels present	• monitor pollution levels in the air at an airport • monitor oxygen and carbon dioxide levels in a greenhouse • monitor oxygen levels in a car exhaust
pH	these measure acidity through changes in voltages in, for example, soil	• monitor/control acidity levels in the soil in a greenhouse • control acidity levels in a chemical process
Magnetic field	these sensors measure changes in magnetic fields – the signal output will depend on how the magnetic field changes	• detect magnetic field changes (for example, in mobile phones and CD players) • used in anti-lock braking systems in cars
Accelerometer	these are sensors that measure acceleration and motion of an application, i.e. the change in velocity (a piezoelectric cell is used whose output varies according to the change in velocity)	• used in cars to measure rapid deceleration and apply air bags in a crash • used by mobile phones to change between portrait and landscape mode
Proximity	these sensors detect the presence of a nearby object	• detect when a face is close to a mobile phone screen and switches off screen when held to the ear
Flow (rate)	these sensors measure the flow rate of a moving liquid or gas and produce an output based on the amount of liquid or gas passing over the sensor	• used in respiratory devices and inhalers in hospitals • measure gas flows in pipes (for example, natural gas)
Level	these sensors use ultrasonics (to detect changing liquid levels in, for example, a tank) or capacitance/conductivity (to measure static levels (for example, height of water in a river) – note, level sensors can also be optical or mechanical in nature	• monitor levels in a petrol tank in a car • in a pharmaceutical process where powder levels in tablet production need to be monitored • leak detection in refrigerant (air conditioning)

6.1 Automated systems

Recap of Chapter 3 Hardware: Sensors

Sensors are used in both monitoring and control applications. There is a subtle difference between how these two methods work (the flowchart is a simplification of the process):
Monitoring and control systems using sensors:



6.1 Automated systems

Recap of Chapter 3 Hardware

Monitoring Applications:

Monitoring systems do not make any changes to the process, instead it simply reports the information / values to the user e.g. a life support system in a hospital will sound an alarm if the patient's heart stops...it will not take action to re start the heart, it only informs doctors that they need to take action.

Examples

Burglar alarm system

Monitoring a patients vitals in intensive care

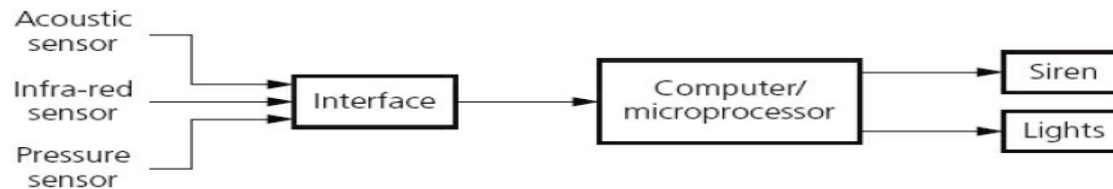
Checking for intruders in an alarm system

Monitoring pollution levels

6.1 Automated systems

Recap of Chapter 3 Hardware: Monitoring Applications:

Let's look at burglar alarm monitoring system will carry out the following actions:



- The system is activated by keying in a password on a keypad.
- The infra-red sensor picks up the movement of an intruder in the building.
- The acoustic sensor picks up sounds such as footsteps or breaking glass.
- The pressure sensor picks up the weight of an intruder coming through a door or through a window.
- The sensor data is passed through an ADC if it is in an analogue form to produce digital data.
- The computer/microprocessor will sample the digital data coming from these sensors at a given frequency (e.g. every five seconds); the data is compared with the stored values by the computer/microprocessor.
- If any of the incoming data values are outside the acceptable range, then the computer sends a signal to: a siren to sound the alarm, or a light to start flashing.
- A DAC is used if the devices need analogue values to operate them.
- The alarm continues to sound/lights continue to flash until the system is reset with a password.

6.1 Automated systems

Recap of Chapter 3 Hardware : Control Applications:

Control systems can take control and alter how a process is altering e.g. turning up fans on an air-conditioning machine because the temperature has become too high.

Examples:

Air conditioning / heating systems

Traffic light control

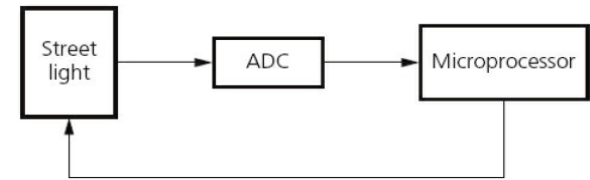
Automatically turning street lights on and off at night

6.1 Automated systems

Recap of Chapter 3 Hardware; Control Applications:

Let's look at Control of street lighting:

This next sequence shows how a microprocessor is used to control the operation of a street lamp. The lamp is fitted with a light sensor which constantly sends data to the microprocessor. The data value from the sensor changes according to whether it is sunny, cloudy, raining or it is night time etc.



- The light sensor sends data to the ADC interface.
- This digitises the data and sends it to the microprocessor.
- The microprocessor samples the data every minute (or at some other frequency rate).
- If the data from the sensor $<$ value stored in memory:
 - a signal is sent from the microprocessor to the street lamp
 - and the lamp is switched on.
- The lamp stays switched on for 30 minutes before the sensor readings are sampled again (this prevents the lamp flickering off and on during brief heavy cloud cover, for example).
- If the data from the sensor $\geq$ value stored in memory:
 - a signal is sent from the microprocessor to the street lamp
 - and the lamp is switched off.
- The lamp stays switched off for 30 minutes before sensor readings are sampled again (this prevents the lamp flickering off and on during heavy cloud



6.1 Automated systems

Recap of Chapter 3 Hardware:

CONTROL / MONITORING SYSTEM

Examples of monitoring	Examples of control
• monitoring a patient in a hospital for vital signs such as heart rate, temperature, and so on • checking for intruders in a burglar alarm system • checking the temperature levels in a car engine • monitoring pollution levels in a river	• turning street lights on at night and turning them off again during daylight • controlling the temperature in a central heating/air conditioning system • controlling the traffic lights at a road junction • operating anti-lock brakes on a car when necessary • controlling the environment in a greenhouse

Learning Objectives:

6.1 Automated systems

Candidates should be able to:

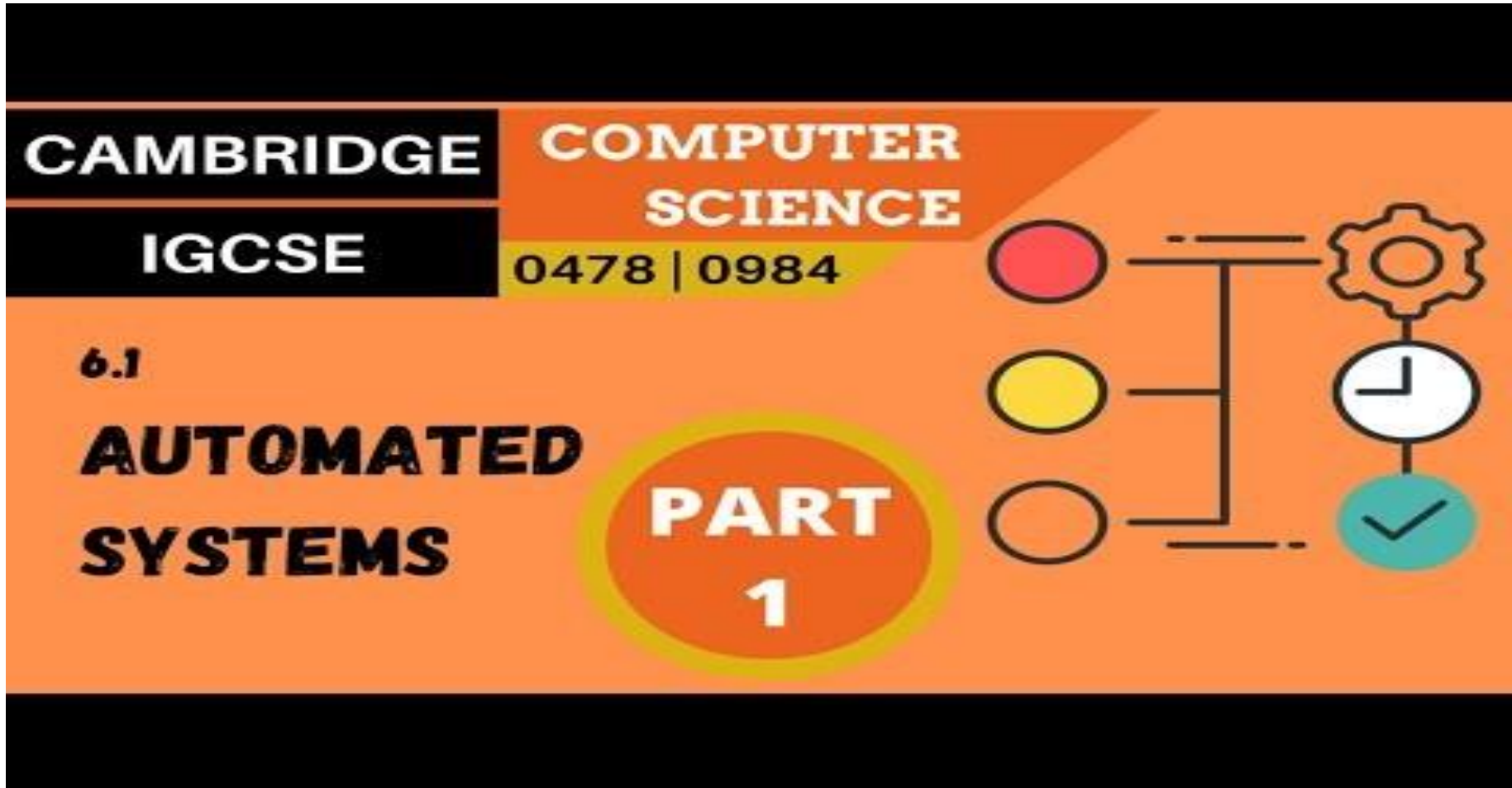
- 1 Describe how sensors, microprocessors and actuators can be used in collaboration to create automated systems
- 2 Describe the advantages and disadvantages of an automated system used for a given scenario

Notes and guidance

- Including scenarios from:
 - industry
 - transport
 - agriculture
 - weather
 - gaming
 - lighting
 - science

6.1 Automated systems

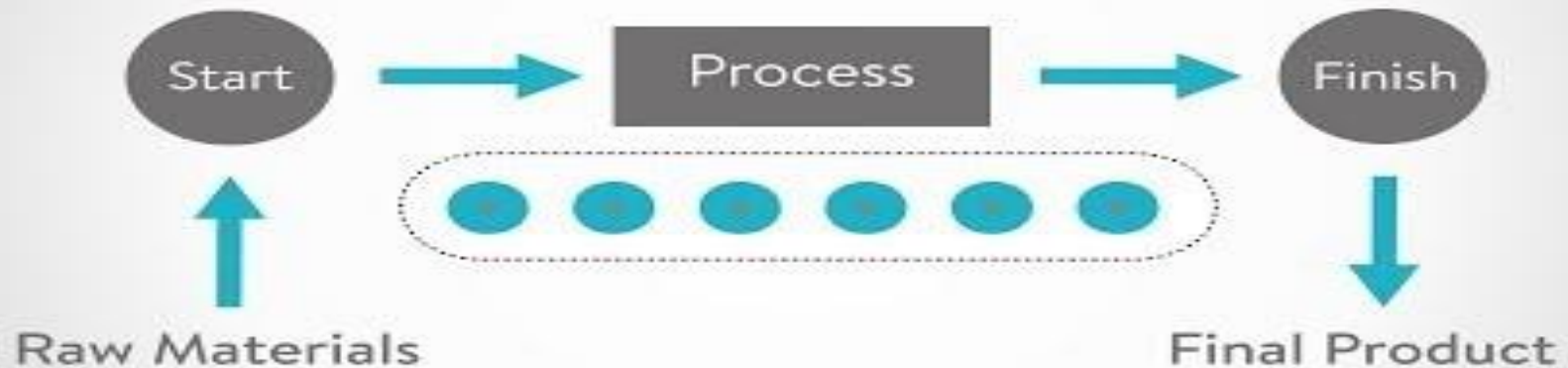
<https://www.youtube.com/watch?v=krleObyk8ac>



6.1 Automated systems

What Automation?

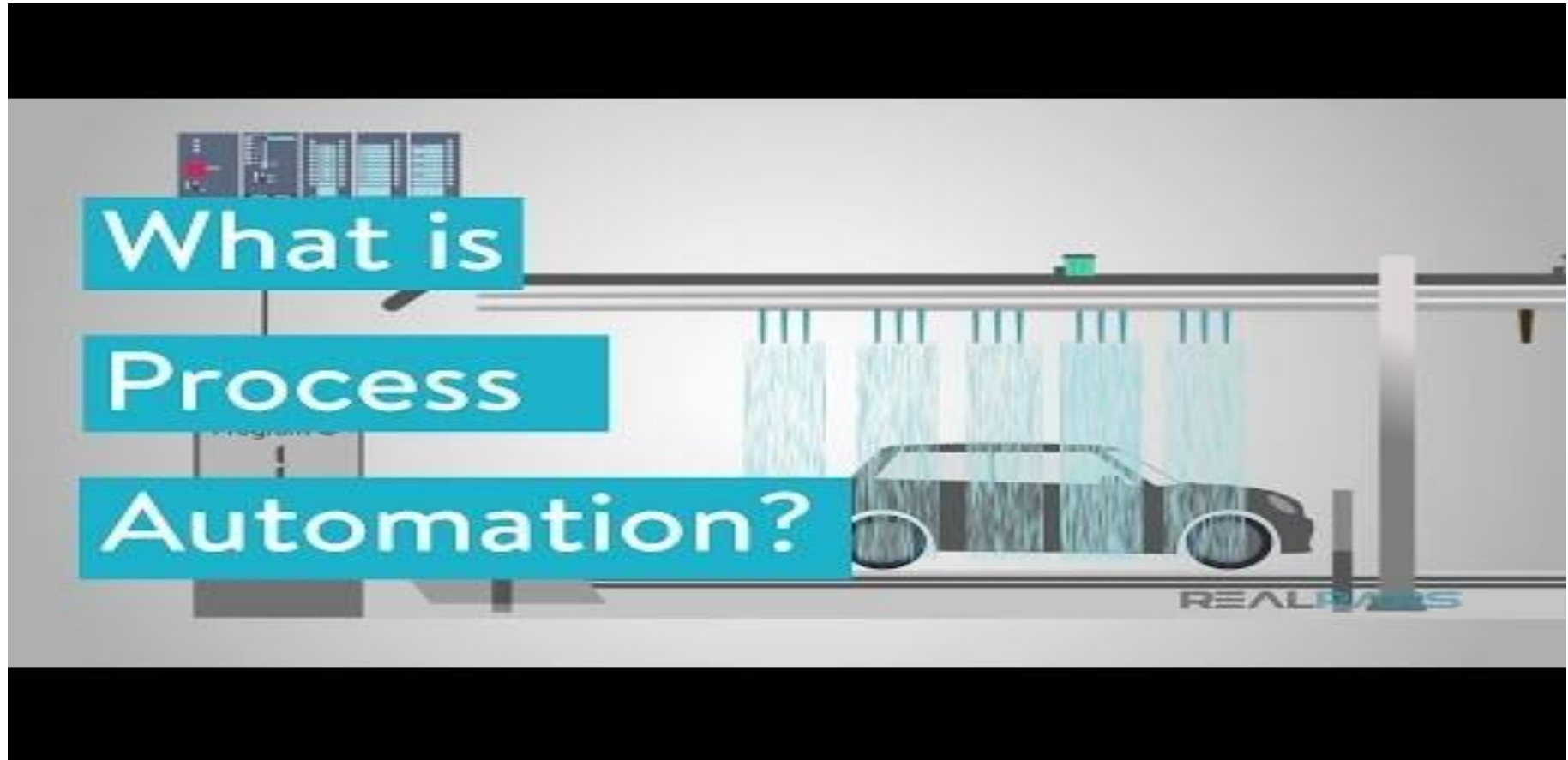
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6.1 Automated systems

What is Process Automation?

<https://www.youtube.com/watch?v=uEhuxYXPTOE>



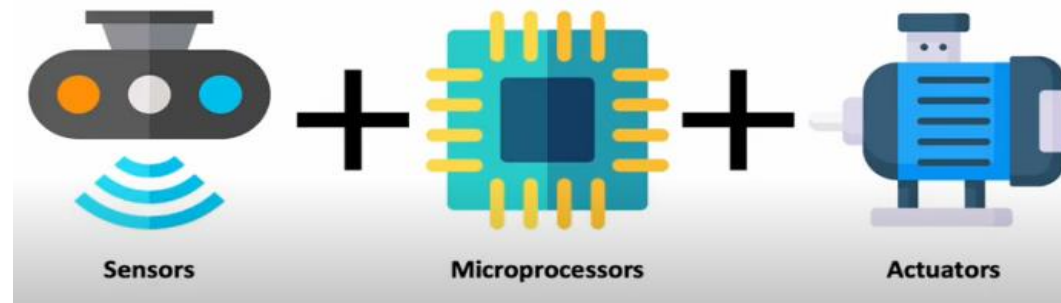
6.1 Automated systems

An **automated system** is a combination of software and hardware (for example, sensors, microprocessors and actuators) that is designed and programmed to work automatically without the need of any human intervention. However, such systems often involve human monitoring.

» **Sensors** are input devices that take readings from their surroundings and send this data to a microprocessor or computer. If the data is analogue, it is first converted into a digital format by an analogue-digital converter (ADC).

» The **microprocessor** will process the data and take the necessary action based on programming.

» This will involve some form of output, usually involving signals sent to **actuators** to control motors, wheels, solenoids, and so on.



6.1 Automated systems

Automated systems

A vast array of different sensors exist – they are used to take measurements from their environmental surroundings.



Acoustic



Accelerometer



Flow



Gas



Humidity



Infrared



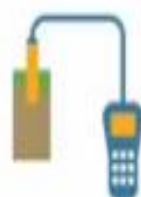
Level



Light



Magnetic field



Moisture



pH



Pressure



Proximity

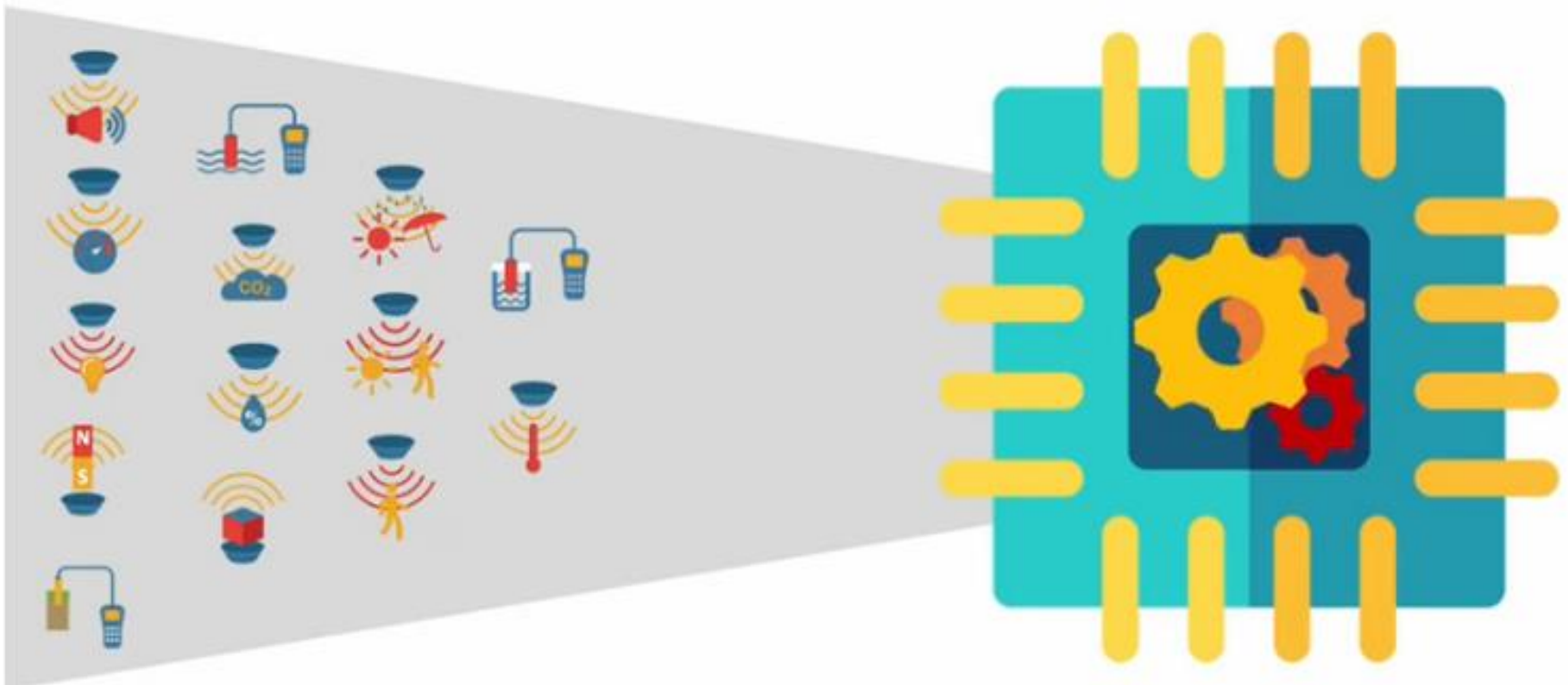


Temperature

6.1 Automated systems

Automated systems

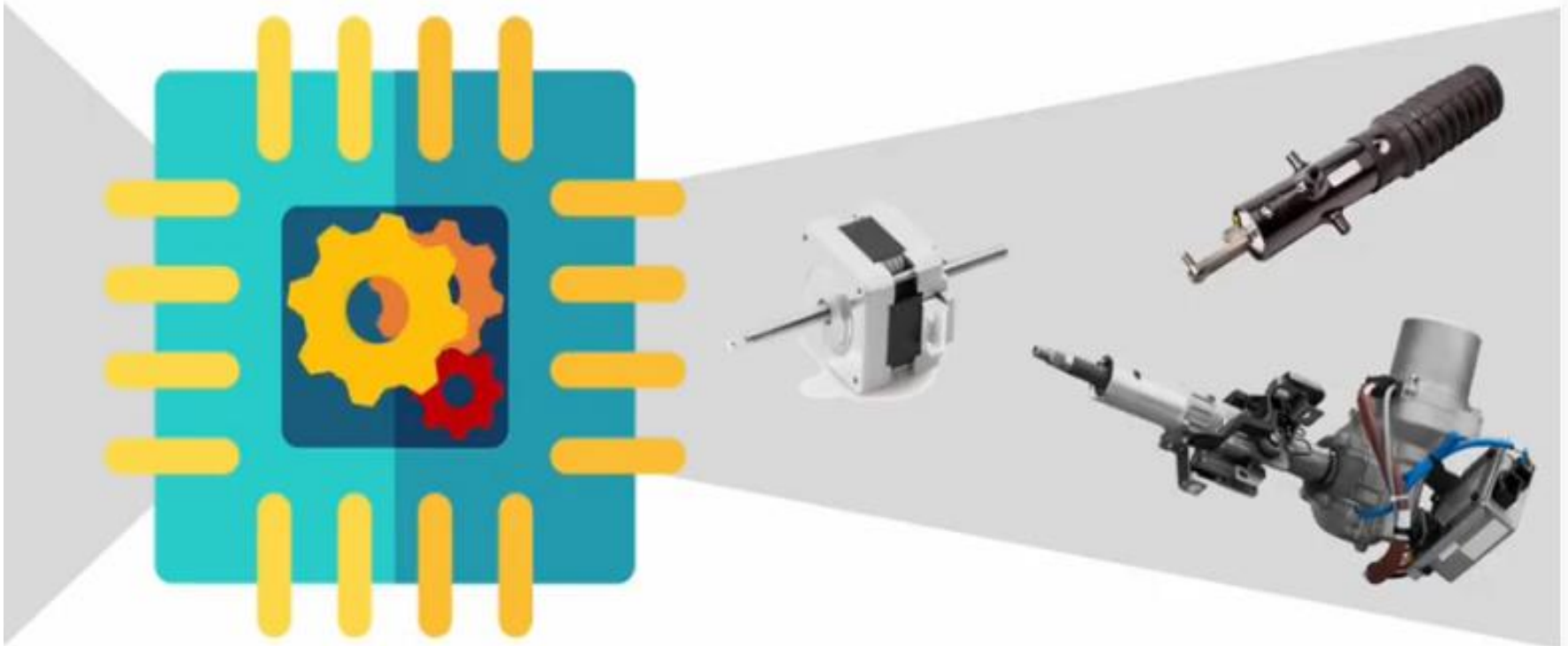
Sensors provide these measurements as inputs to microprocessors, which then process the information and make decisions based on predefined rules.



6.1 Automated systems

Automated systems

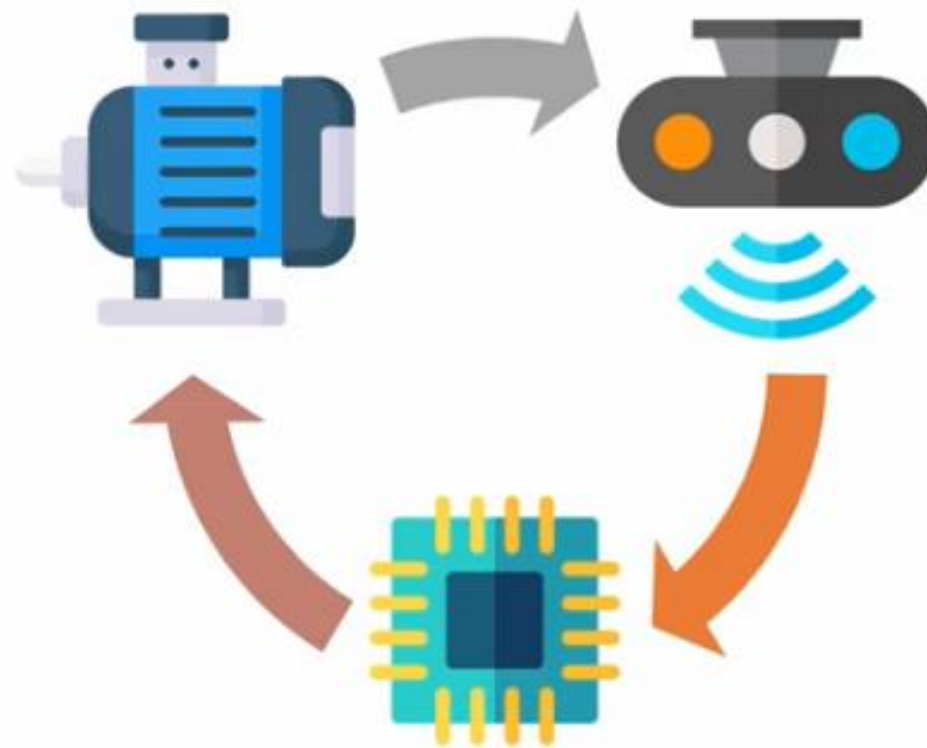
Decisions made by microprocessors typically result in instructions to control an actuator and make physical changes
– e.g., activating an alarm, opening or closing a door or valve, etc.



6.1 Automated systems

Feedback

These components are often used to create a feedback loop where changes (outputs) from the automated system affect the following set of input values received from the sensors.



6.1 Automated systems



Automated systems

Automated systems usually comprise three distinct components:

- Sensors
- Microprocessors
- Actuators

Sensors take measurements from their environmental surroundings.

Sensor measurements are fed in as inputs to microprocessors, which then process the information and make decisions.

Decisions made by microprocessors typically result in instructions to control an actuator and make physical changes – e.g., activating an alarm, opening or closing a door or valve, etc.

6.1 Automated systems

Sensors

https://www.youtube.com/watch?v=ZmdXZMO_lx0



6.1 Automated systems

Sensors

Sensors

Sensors are **input devices**. They come in different shapes and sizes, but they all measure physical properties of the environment and translate these measurements into digital information that a computer can use.



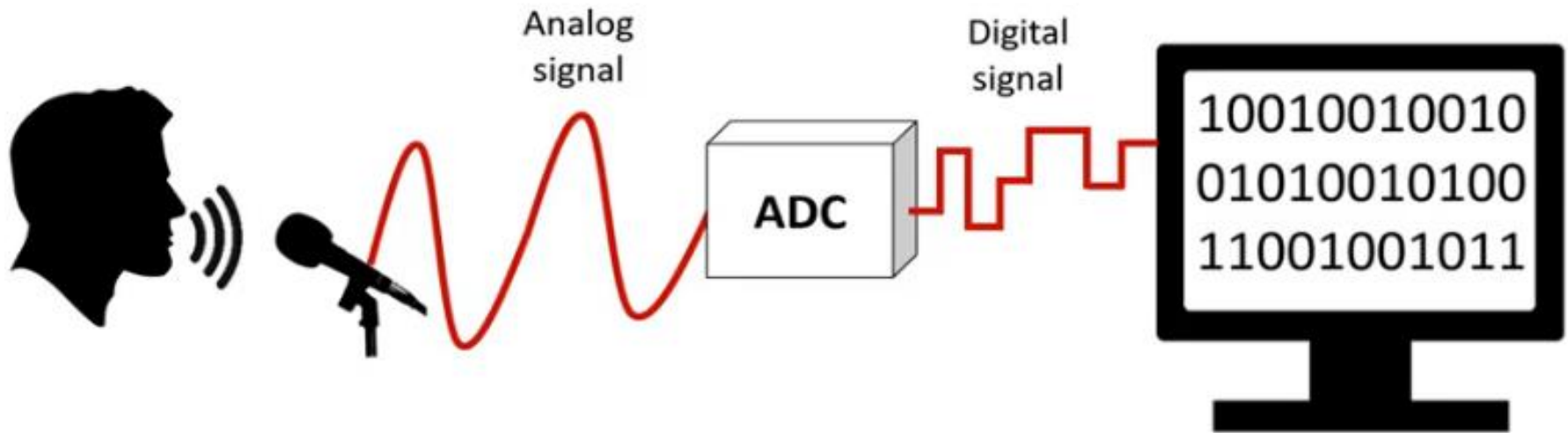
6.1 Automated systems

Sensors

From analog to digital

To make sense of physical, real-world quantities, all sensors convert the information they collect into a digital format – this is typically achieved using some form of **analog-to-digital converter (ADC)**.

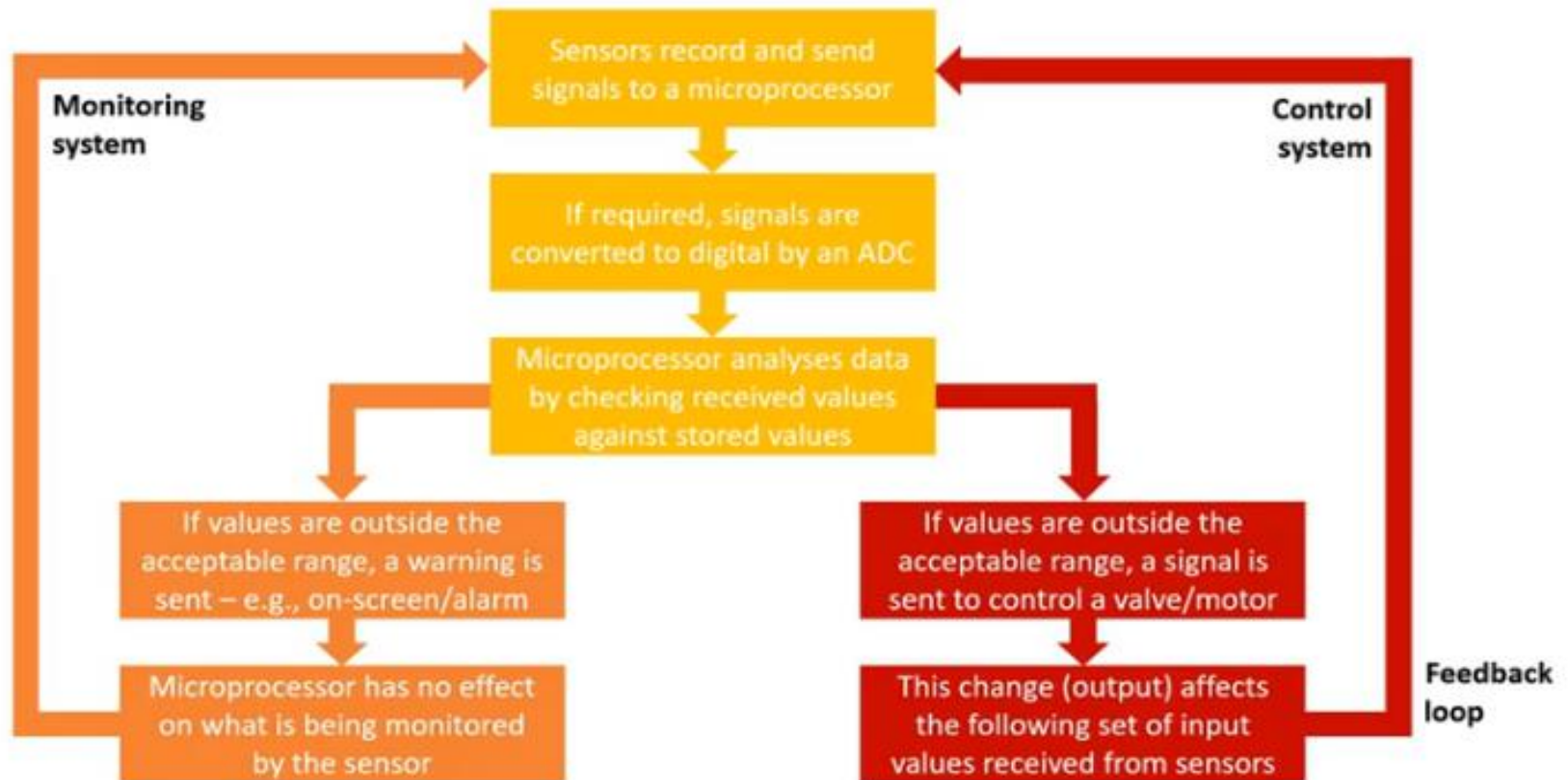
These devices convert physical, real-world information into discrete digital values.



6.1 Automated systems

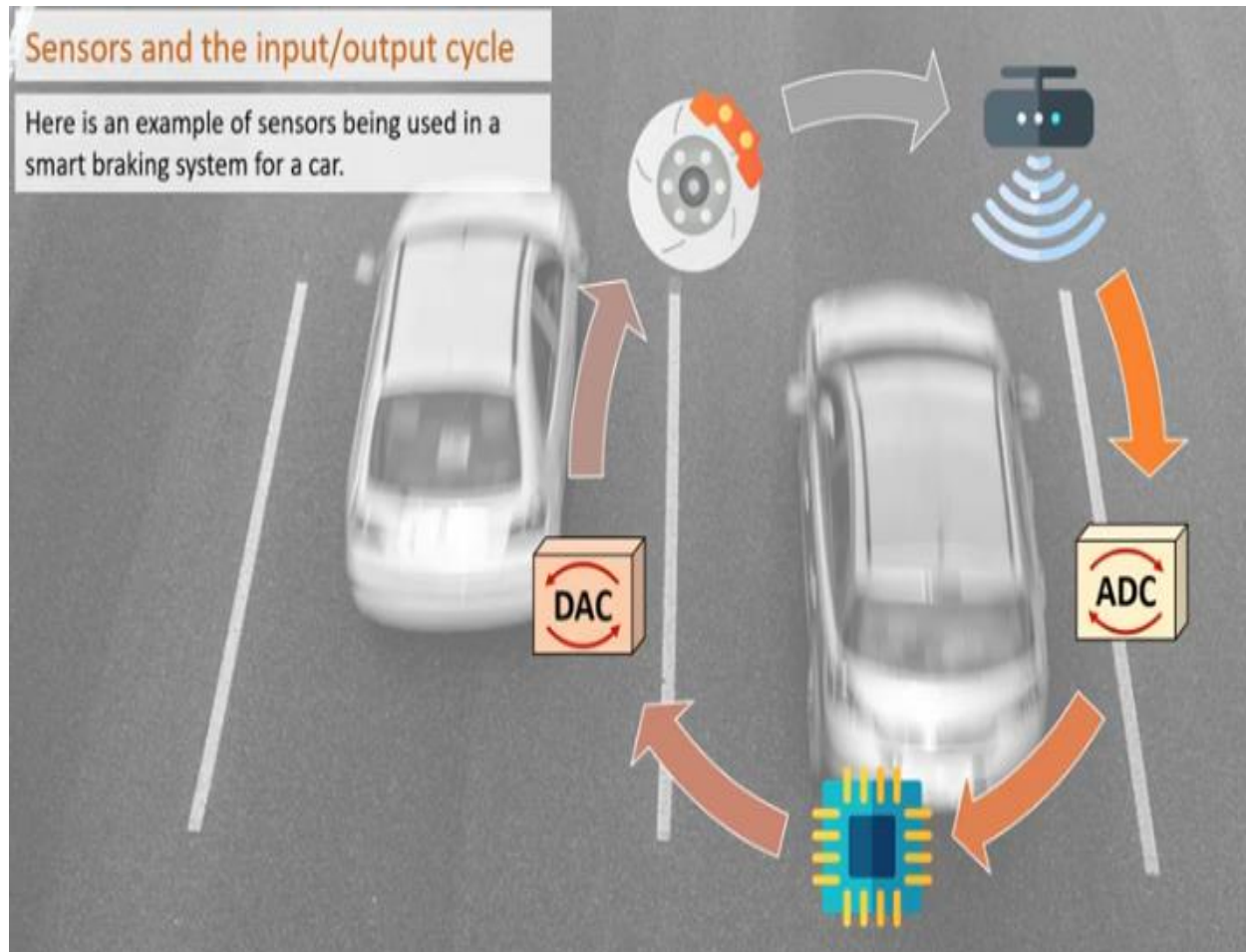
Sensors

Using sensors for monitoring and control



6.1 Automated systems

Sensors



1. A sensor takes constant readings of the car's distance to objects ahead of it.
2. These analog, real-world values are converted to a format a microprocessor can understand using an analog-to-digital convertor (ADC).
3. A microprocessor takes this input and calculates the speed the car is travelling and safe braking distances. If it detects the car is travelling too fast and a collision is likely, it issues commands to brake.
4. Digital brake signals are converted into an analog form – these can then be sent to a motor (actuator), which will apply the brake via a digital-to-analog convertor (DAC).
5. The brake slows the car down.
6. The sensor continues to send out values. These will have changed due to the slowing of the car – this is known as feedback.

6.1 Automated systems

Sensors

Sensors

Sensors can be designed to detect the levels of a wide variety of conditions.

You are required to know about the following sensors for the exam.



Acoustic



Accelerometer



Flow



Gas



Humidity



Infrared



Level



Light



Magnetic field



Moisture



pH



Pressure



Proximity



Temperature

6.1 Automated systems

Sensors

Sensors

All sensors are input devices. They come in a variety of different shapes and sizes.

Sensors measure physical properties of the environment. They then need to translate the information they collect into a digital form that a computer can use, achieved via an analog-to-digital convertor (ADC).

Sensors that you may be examined on are limited to:

- Acoustic
- Accelerometer
- Flow
- Gas
- Humidity
- Infrared
- Level
- Light
- Magnetic field
- Moisture
- pH
- Pressure
- Proximity
- Temperature



6.1 Automated systems

Sensors

<https://www.youtube.com/watch?v=gtDqaoCaUVk>



6.1 Automated systems

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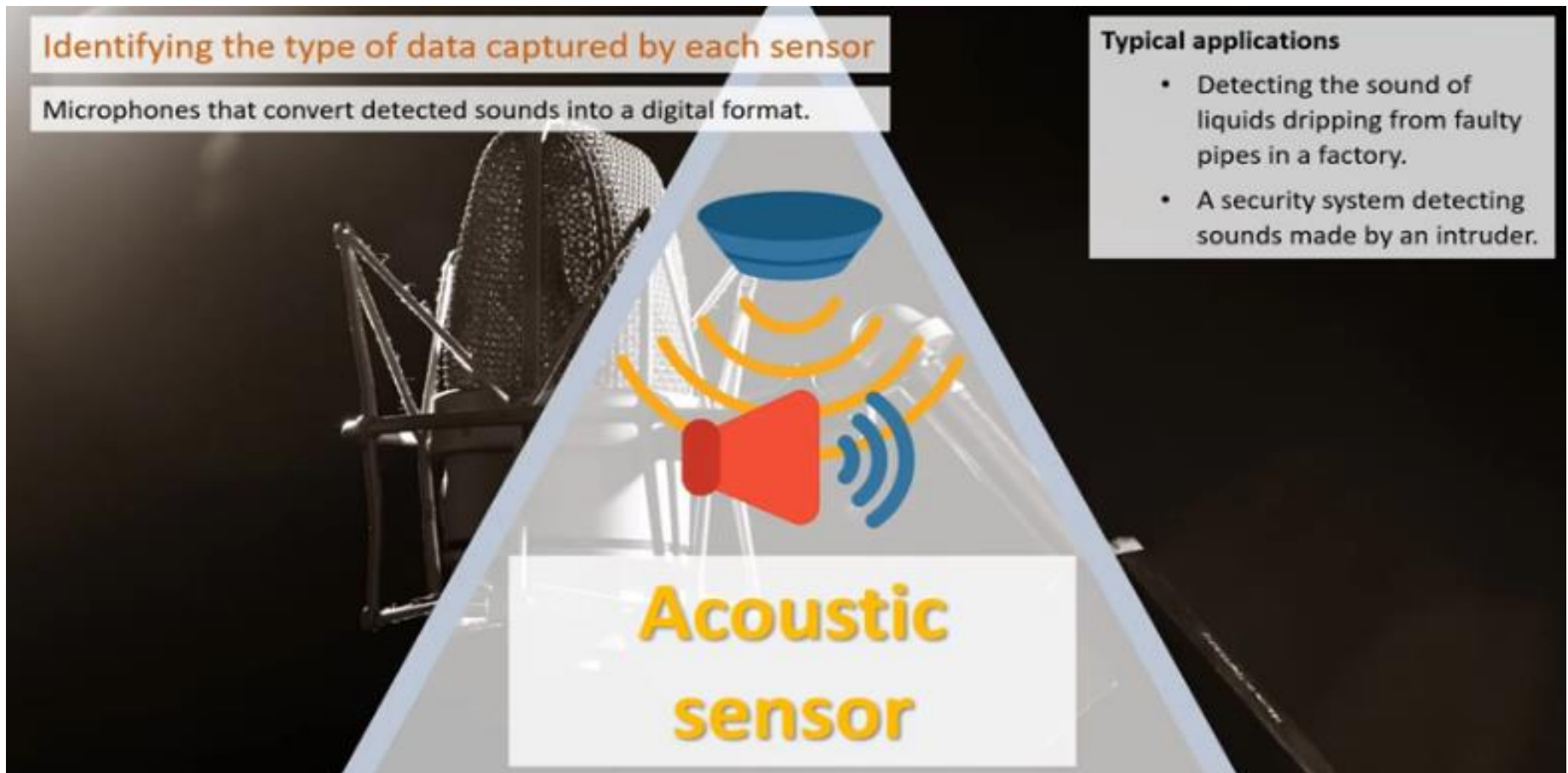
Proximity



Temperature

6.1 Automated systems

Sensors

A diagram illustrating an acoustic sensor. It features a large, professional studio microphone on the left. In the center, a blue bowl is shown with yellow curved lines representing sound waves emanating from it. Below the bowl, a red speaker icon with blue concentric arcs is shown. The entire diagram is framed by a large, light blue triangle. Text boxes provide information about the sensor's function and applications.

Identifying the type of data captured by each sensor

Microphones that convert detected sounds into a digital format.

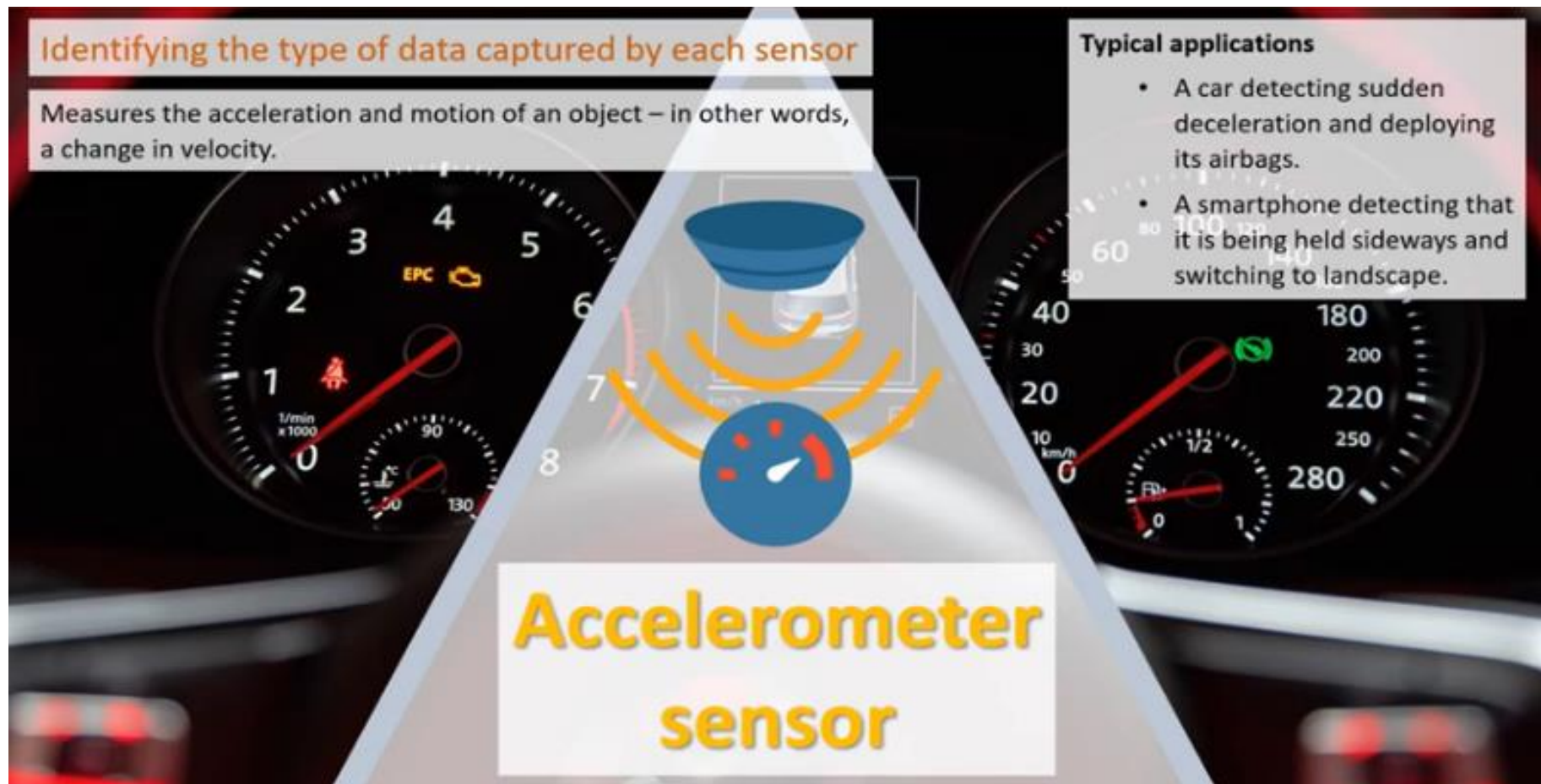
Acoustic sensor

Typical applications

- Detecting the sound of liquids dripping from faulty pipes in a factory.
- A security system detecting sounds made by an intruder.

6.1 Automated systems

Sensors



The diagram illustrates an accelerometer sensor within a car's dashboard. A central blue circular sensor icon with yellow curved lines representing motion is positioned over a speedometer and tachometer. A large white triangle points downwards towards the sensor icon. The background shows a car's instrument cluster with a tachometer on the left and a speedometer on the right.

Identifying the type of data captured by each sensor

Measures the acceleration and motion of an object – in other words, a change in velocity.

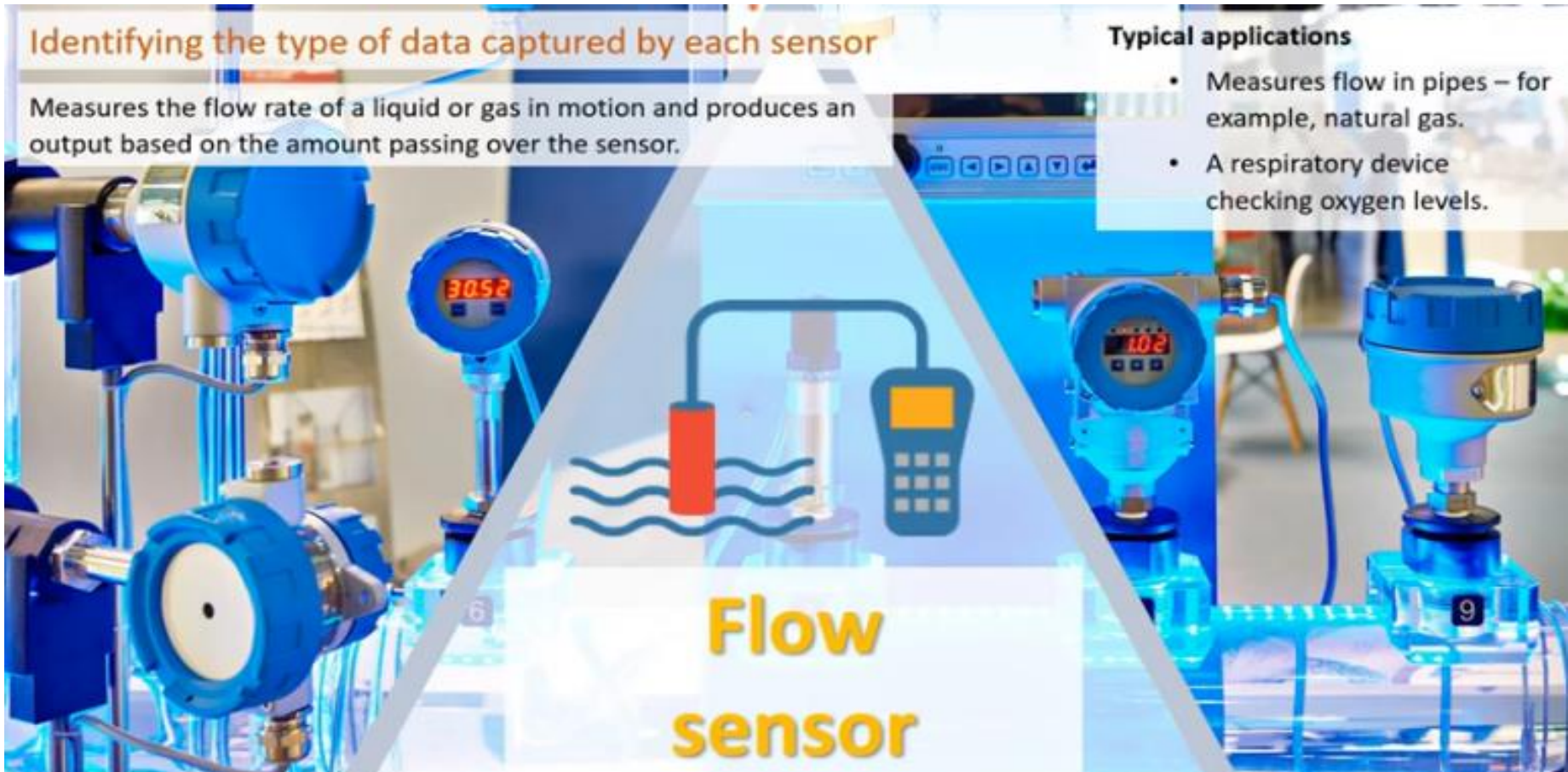
Typical applications

- A car detecting sudden deceleration and deploying its airbags.
- A smartphone detecting that it is being held sideways and switching to landscape.

Accelerometer sensor

6.1 Automated systems

Sensors



Identifying the type of data captured by each sensor

Measures the flow rate of a liquid or gas in motion and produces an output based on the amount passing over the sensor.

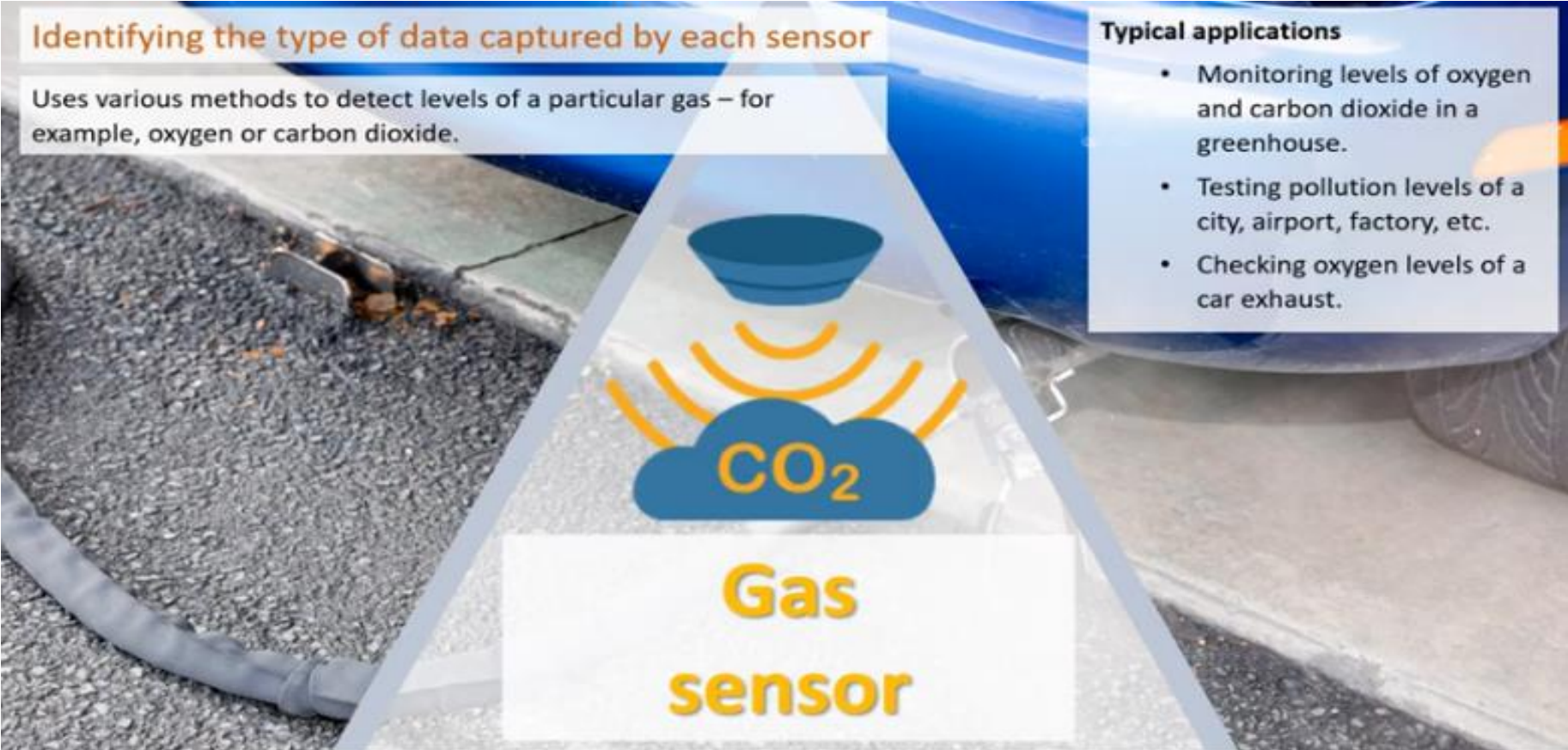
Typical applications

- Measures flow in pipes – for example, natural gas.
- A respiratory device checking oxygen levels.

Flow sensor

6.1 Automated systems

Sensors



Identifying the type of data captured by each sensor

Uses various methods to detect levels of a particular gas – for example, oxygen or carbon dioxide.

Typical applications

- Monitoring levels of oxygen and carbon dioxide in a greenhouse.
- Testing pollution levels of a city, airport, factory, etc.
- Checking oxygen levels of a car exhaust.

Gas sensor

6.1 Automated systems

Sensors



Identifying the type of data captured by each sensor

Not to be confused with a sensor that detects moisture, these sensors measure water vapour – e.g., in an air sample.

Typical applications

- Monitoring humidity levels in a factory producing microchips.
- Checking the level of humidity in a greenhouse.

Humidity sensor

6.1 Automated systems

Sensors

Identifying the type of data captured by each sensor

Uses an invisible beam of infrared radiation picked up by a detector. If the beam is broken by an object, the detector senses the change.



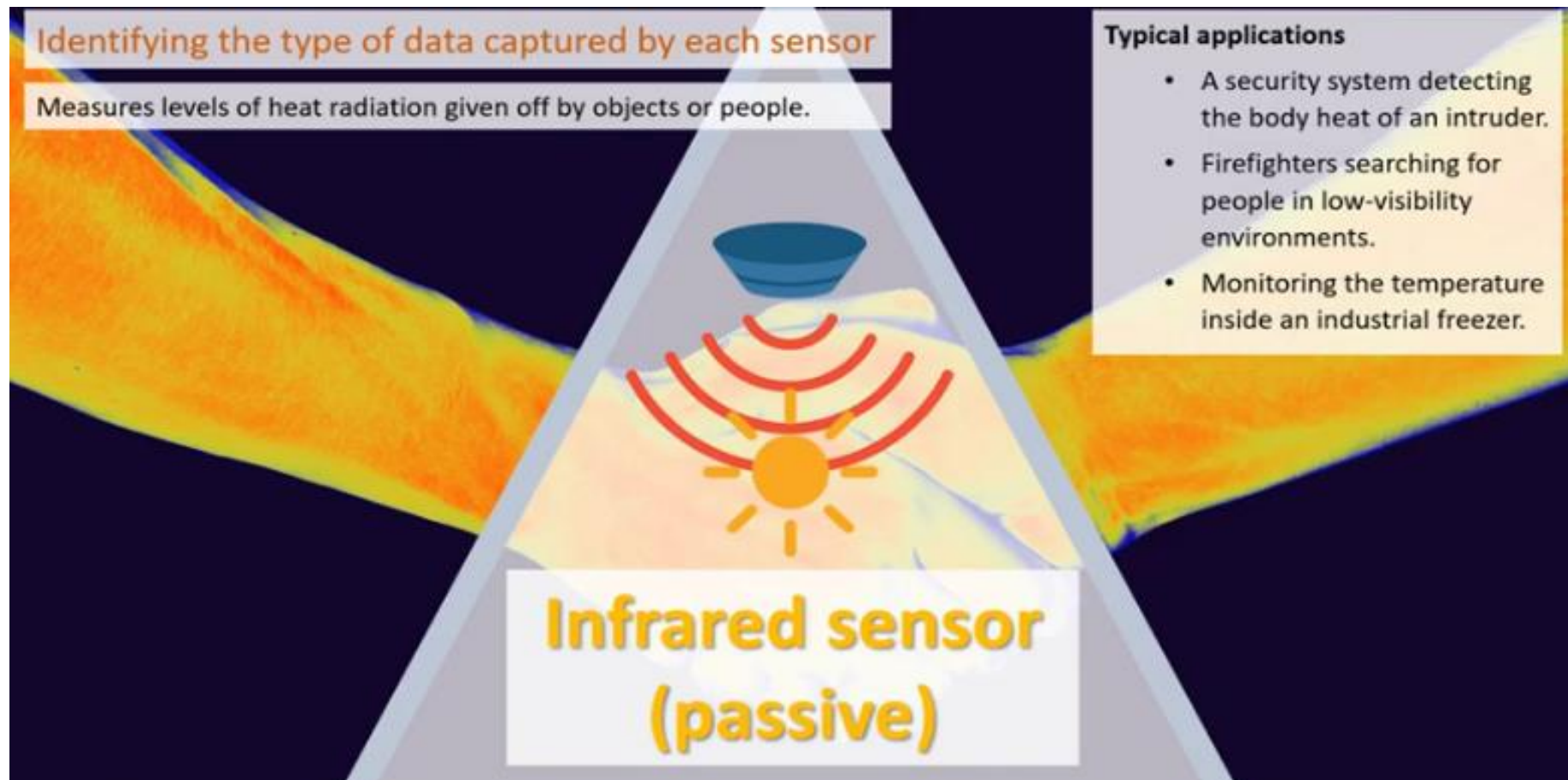
**Infrared sensor
(active)**

Typical applications

- A security system detecting when an intruder breaks the infrared beam.
- A car automatically turning on its windscreen wipers when rain is detected.

6.1 Automated systems

Sensors



6.1 Automated systems

Sensors

The diagram illustrates a level sensor system. A red sensor probe is submerged in a tank of liquid, connected by a blue cable to a handheld electronic device. The background features a large, semi-transparent fuel gauge with markings for 20, 80, and 200, and a needle pointing towards the 20 mark. A white triangular highlight points to the sensor probe.

Identifying the type of data captured by each sensor

These can be ultrasonic, conductive, optical or mechanical in nature, but they all serve the same purpose – to detect levels of a substance such as water in a tank.

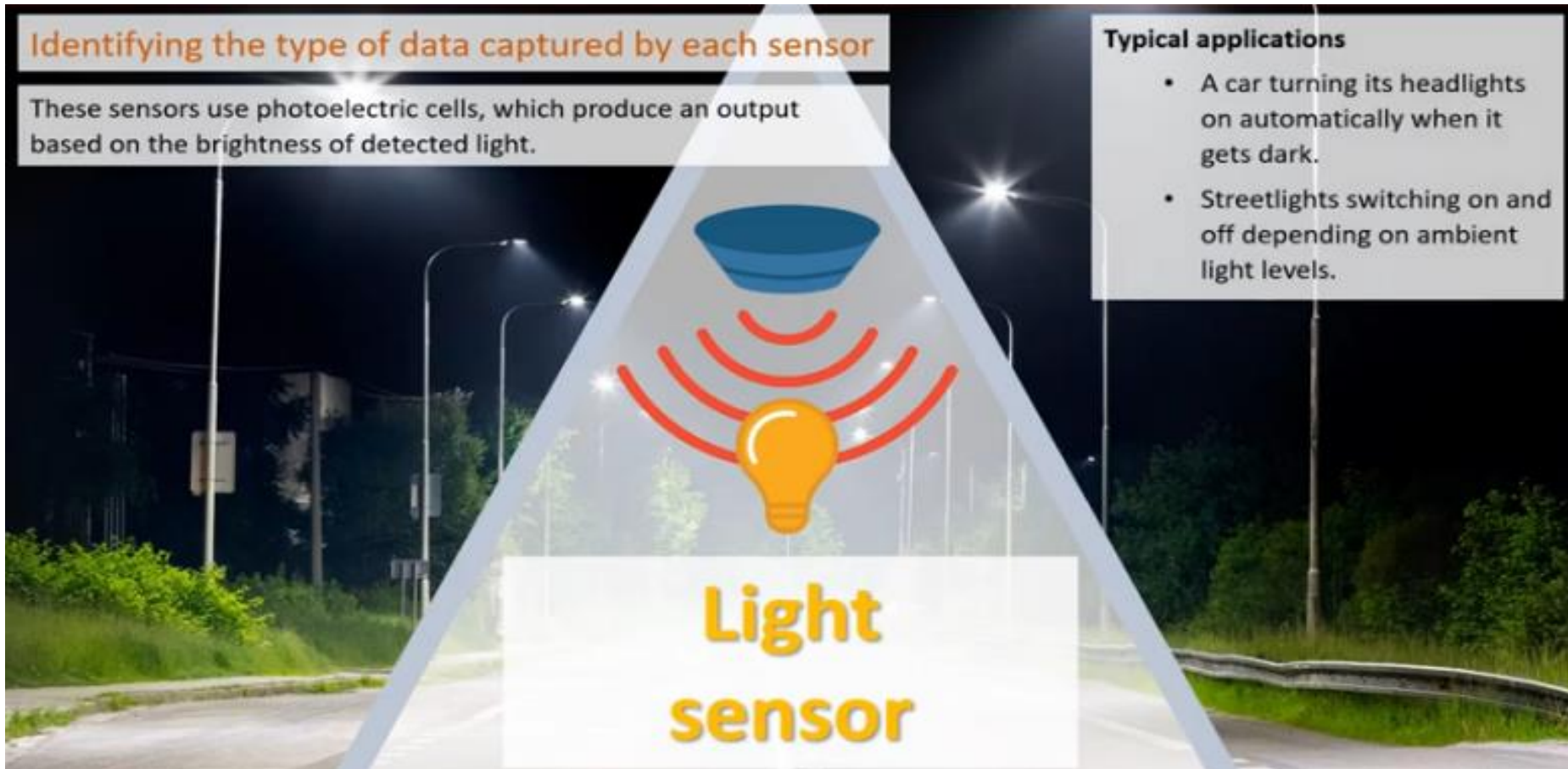
Typical applications

- A car monitoring the level of petrol left in its tank.
- Pharmaceutical companies monitoring powder levels when making tablets.
- Detecting a drop in liquid levels caused by leaks.

Level sensor

6.1 Automated systems

Sensors



Identifying the type of data captured by each sensor

These sensors use photoelectric cells, which produce an output based on the brightness of detected light.

Typical applications

- A car turning its headlights on automatically when it gets dark.
- Streetlights switching on and off depending on ambient light levels.

Light sensor

6.1 Automated systems

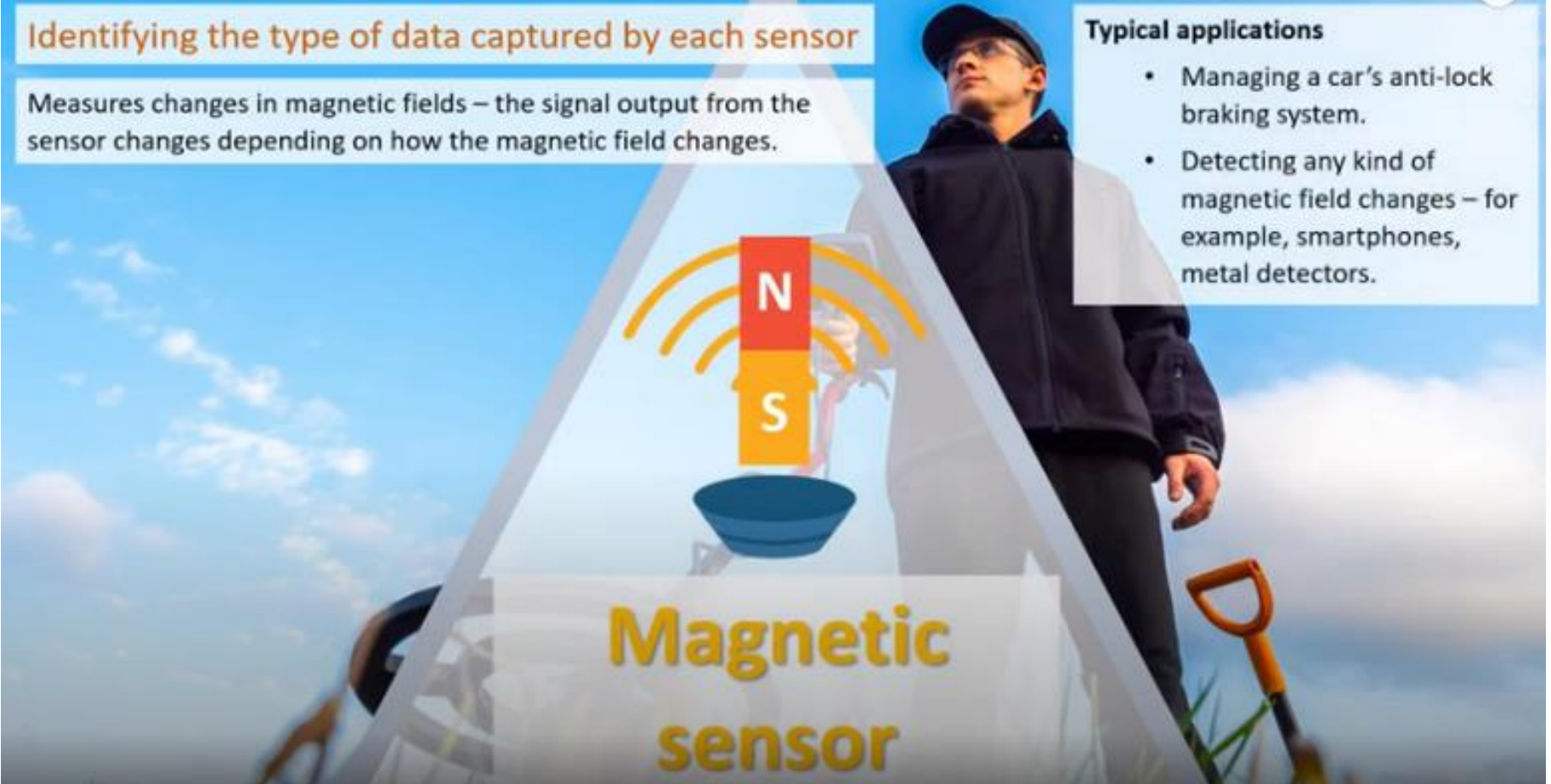
Sensors

Identifying the type of data captured by each sensor

Measures changes in magnetic fields – the signal output from the sensor changes depending on how the magnetic field changes.

Typical applications

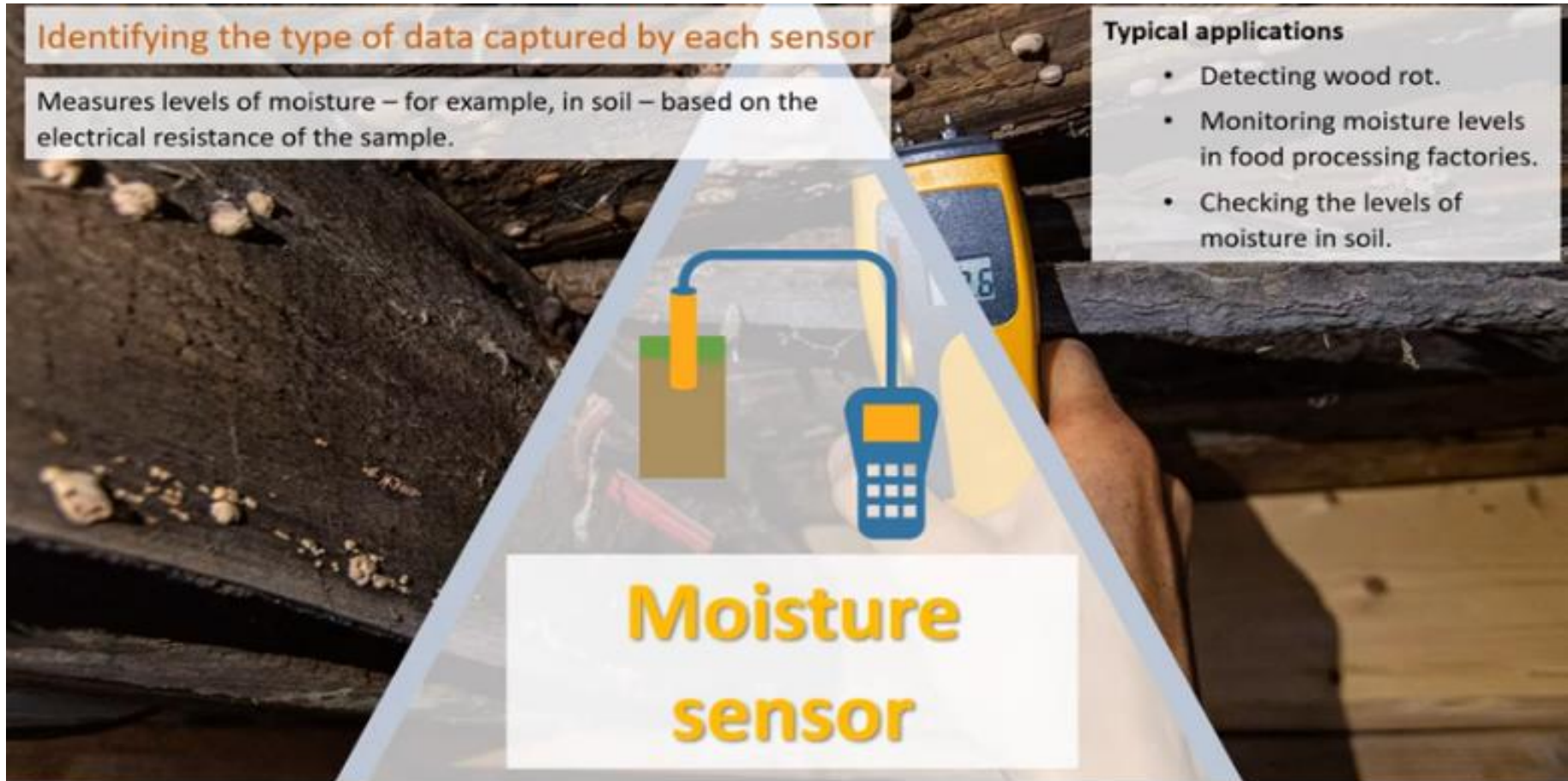
- Managing a car's anti-lock braking system.
- Detecting any kind of magnetic field changes – for example, smartphones, metal detectors.

A photograph of a person wearing a cap and safety glasses, using a metal detector in a field. A semi-transparent graphic is overlaid on the image. It features a vertical bar with a red top half labeled 'N' and an orange bottom half labeled 'S', representing a magnet. Yellow curved lines radiate from the magnet, indicating a magnetic field. Below the magnet is a blue bowl-shaped icon representing a sensor. The text 'Magnetic sensor' is written in large yellow letters at the bottom of the graphic.

**Magnetic
sensor**

6.1 Automated systems

Sensors



Identifying the type of data captured by each sensor

Measures levels of moisture – for example, in soil – based on the electrical resistance of the sample.

Typical applications

- Detecting wood rot.
- Monitoring moisture levels in food processing factories.
- Checking the levels of moisture in soil.

Moisture sensor

6.1 Automated systems

Sensors



Identifying the type of data captured by each sensor

Measures levels of acidity – for example, in soil – by detecting changes in voltages.

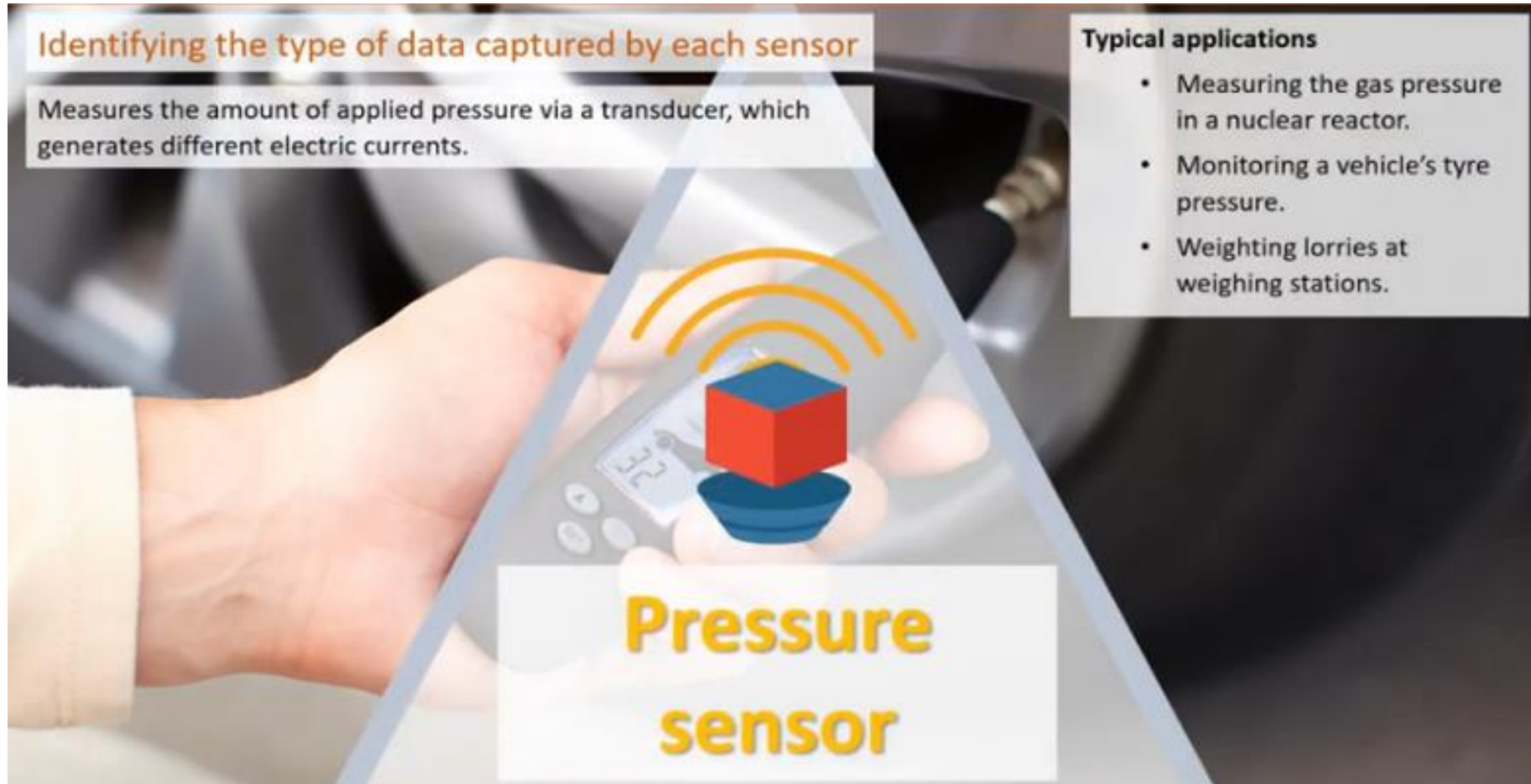
Typical applications

- Monitoring acidity levels of soil for agriculture.
- Checking the level of acidity in a chemical process.

pH sensor

6.1 Automated systems

Sensors



Identifying the type of data captured by each sensor

Measures the amount of applied pressure via a transducer, which generates different electric currents.

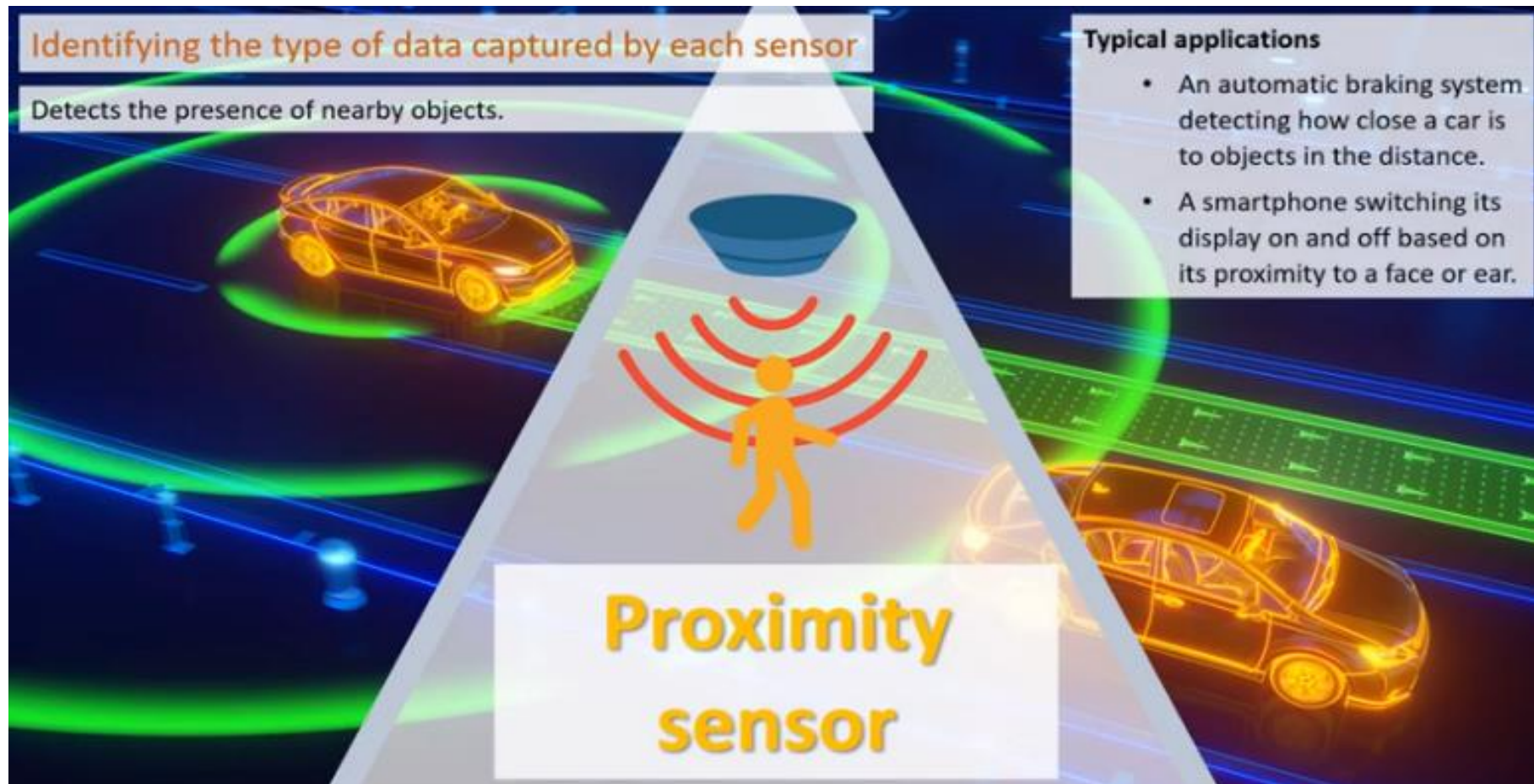
Typical applications

- Measuring the gas pressure in a nuclear reactor.
- Monitoring a vehicle's tyre pressure.
- Weighting lorries at weighing stations.

Pressure sensor

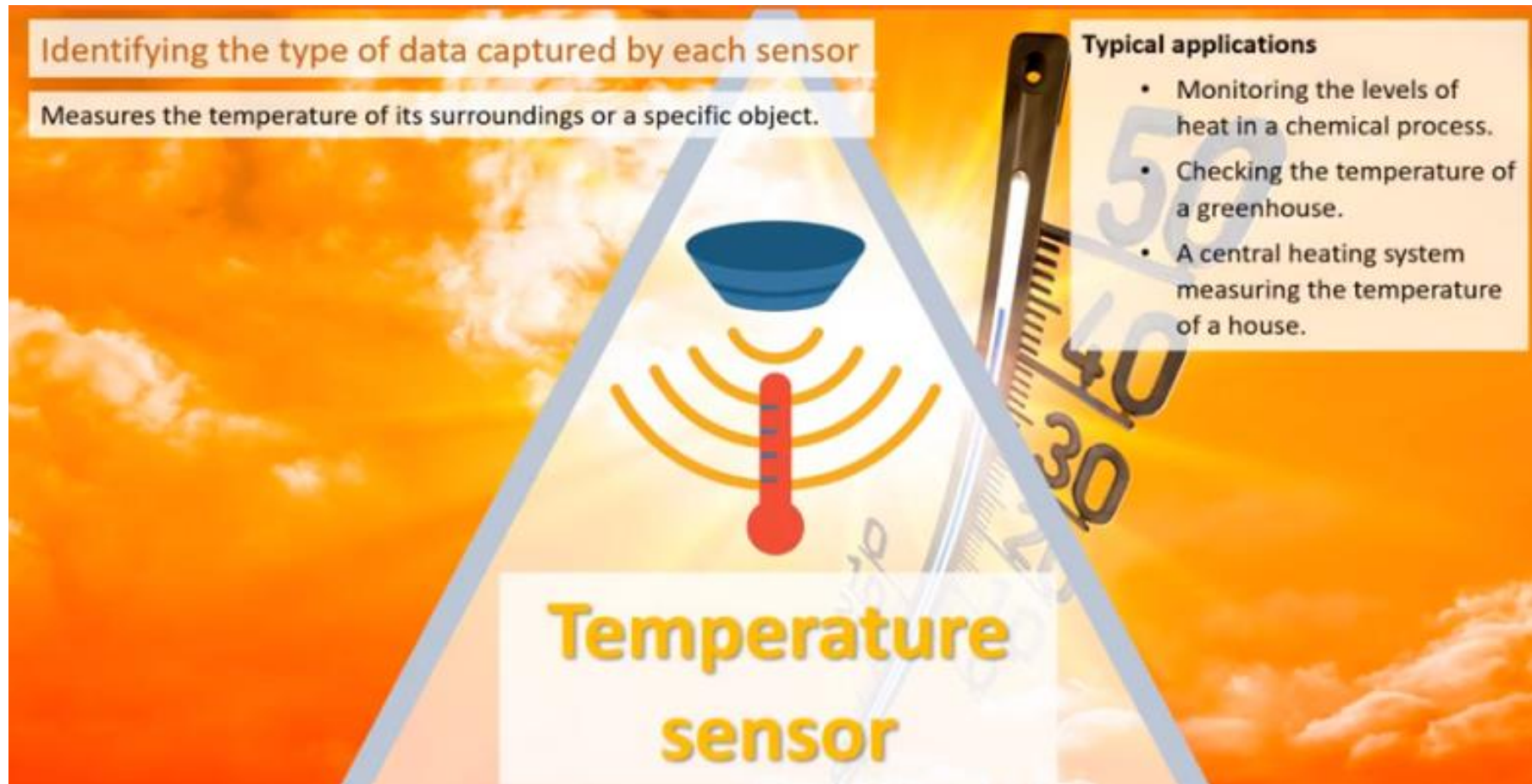
6.1 Automated systems

Sensors



6.1 Automated systems

Sensors



The diagram features a large thermometer on the right side, with a blue bowl positioned above a red thermometer icon. The background is a bright orange sky with white clouds. A large, semi-transparent number '50' is visible in the background.

Identifying the type of data captured by each sensor

Measures the temperature of its surroundings or a specific object.

Typical applications

- Monitoring the levels of heat in a chemical process.
- Checking the temperature of a greenhouse.
- A central heating system measuring the temperature of a house.

Temperature sensor

6.1 Automated systems

Sensors

Selecting the most suitable sensor for a given context

Security monitoring system

What sensors do you think might be involved in this situation?

Infrared

Acoustic

Proximity

Pressure

6.1 Automated systems

Sensors



6.1 Automated systems

Sensors

Sensors

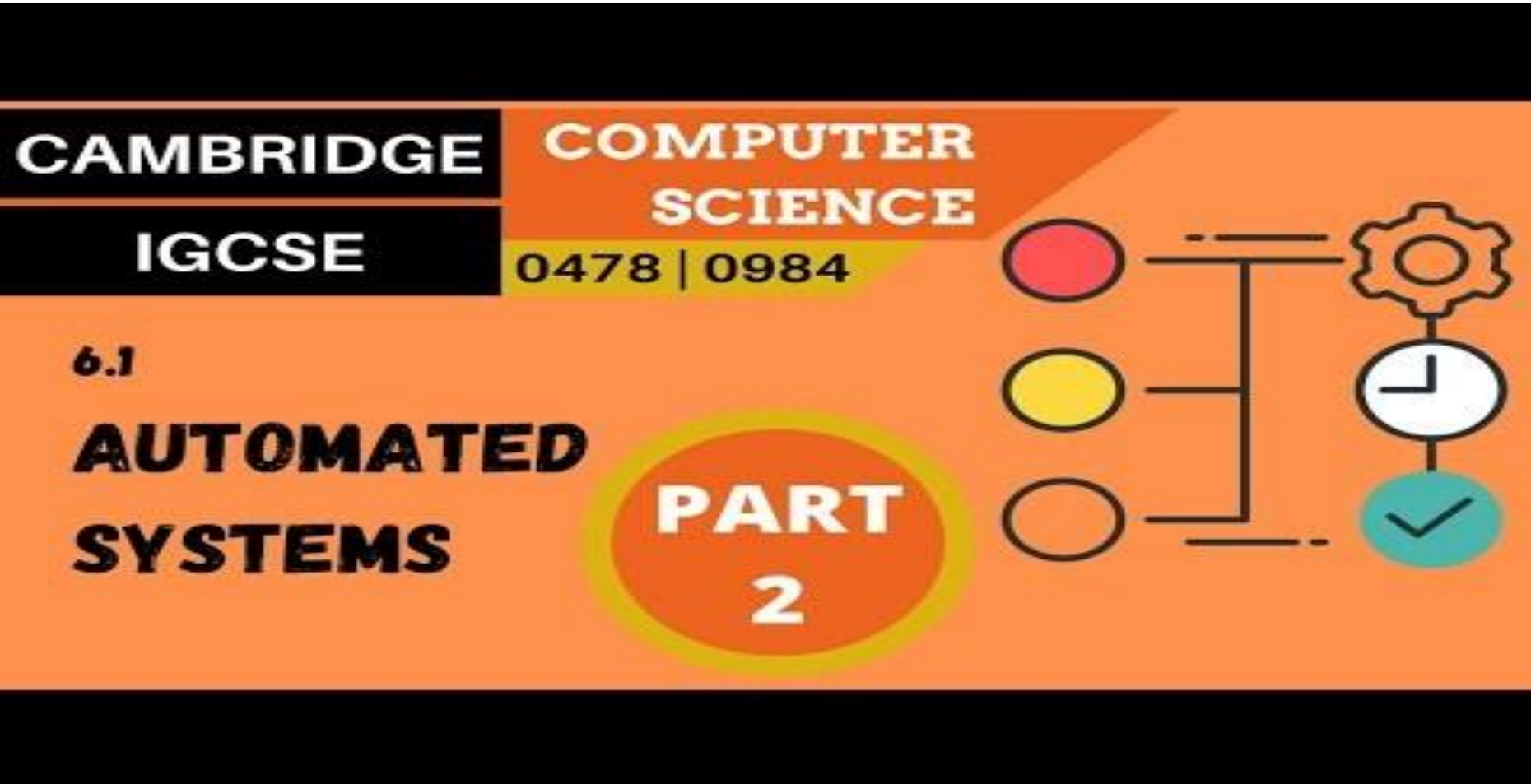
Sensors measure the real-world, physical properties of their environment and convert these analog readings into discrete digital values:

- **Acoustic:** Sound levels.
- **Accelerometer:** Acceleration, vibration, tilt.
- **Flow:** Rate of gas, liquid or powder flow.
- **Gas:** Presence of gas – e.g., carbon monoxide.
- **Humidity:** Levels of water vapour.
- **Infra-red:** Detecting motion or heat.
- **Level:** Liquid levels.
- **Light:** Light levels.
- **Magnetic field:** Presence and strength.
- **Moisture:** Presence and levels of moisture.
- **pH:** Acidity or alkalinity.
- **Pressure:** Gas, liquid or physical pressure.
- **Proximity:** Distance – e.g., parking sensor.
- **Temperature:** Heat levels.



6.1 Automated systems

<https://www.youtube.com/watch?v=8B-TUjl3mA0>



6.1 Automated systems

Advantages and Disadvantages of Automated Systems

Automation is a set of tools, such as software and computer programs, that are designed to perform repetitive tasks. Automation is a process of doing something through the use of machines rather than humans. There are many places that automation can help businesses, such as in manufacturing. However, automation can also save time and money when it comes to marketing by automating individual tasks. There are two types of automation: batch and concurrent.

Advantages commonly attributed to automation include higher production rates and increased productivity, more efficient use of materials, better product quality, improved safety, shorter workweeks for labour, and reduced factory lead times.

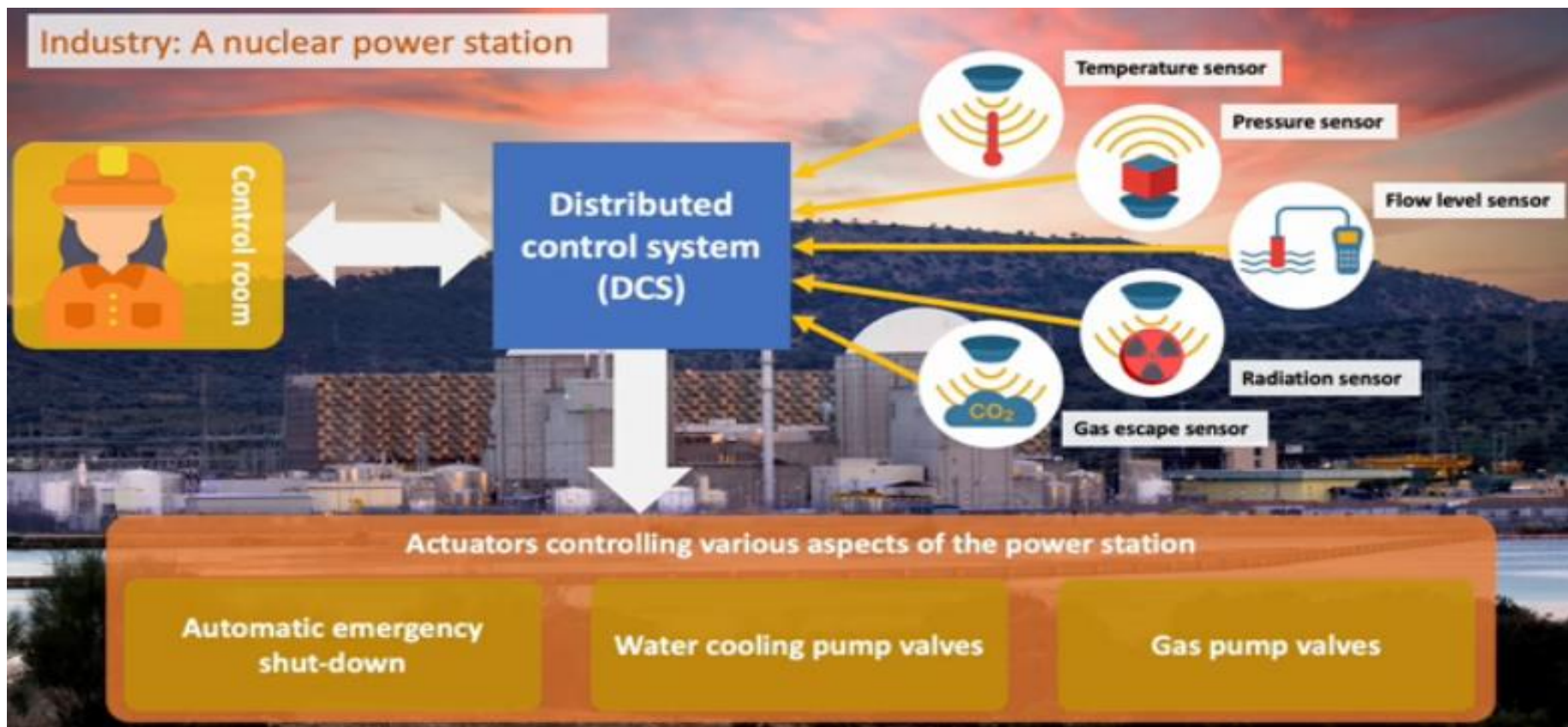
Disadvantages of automation are the cost to replace human labour with automated systems. Other Advantages and disadvantages of automation are as follows:

Advantages	Disadvantages
Higher production rates	Worker displacement
Increased productivity	Needs large capital expenditure
More efficient use of materials	Can become redundant
Better product quality	Could introduce new safety hazards

6.1 Automated systems

Advantages and Disadvantages of Automated Systems

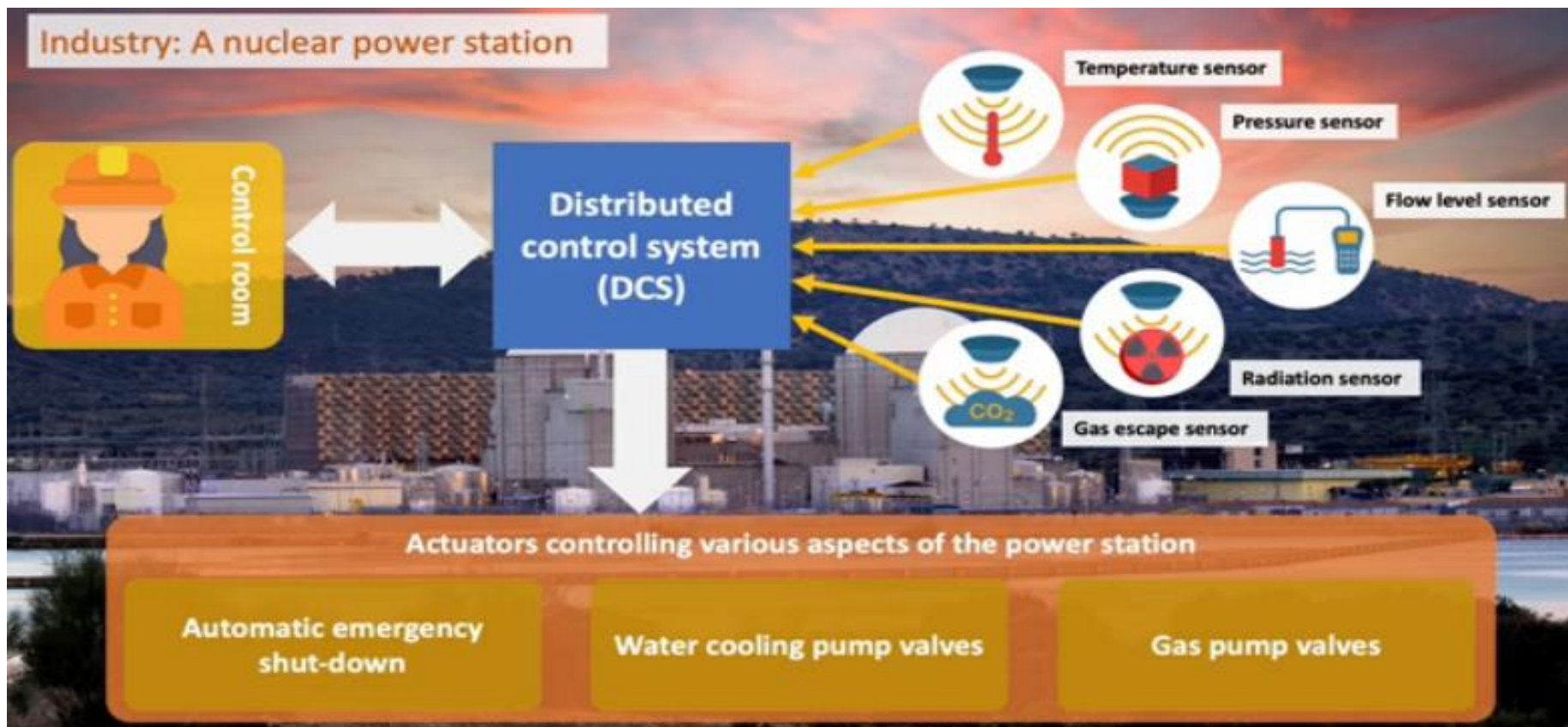
Describe the advantages and disadvantages of an automated system used for a given scenario



6.1 Automated systems

Advantages and Disadvantages of Automated Systems

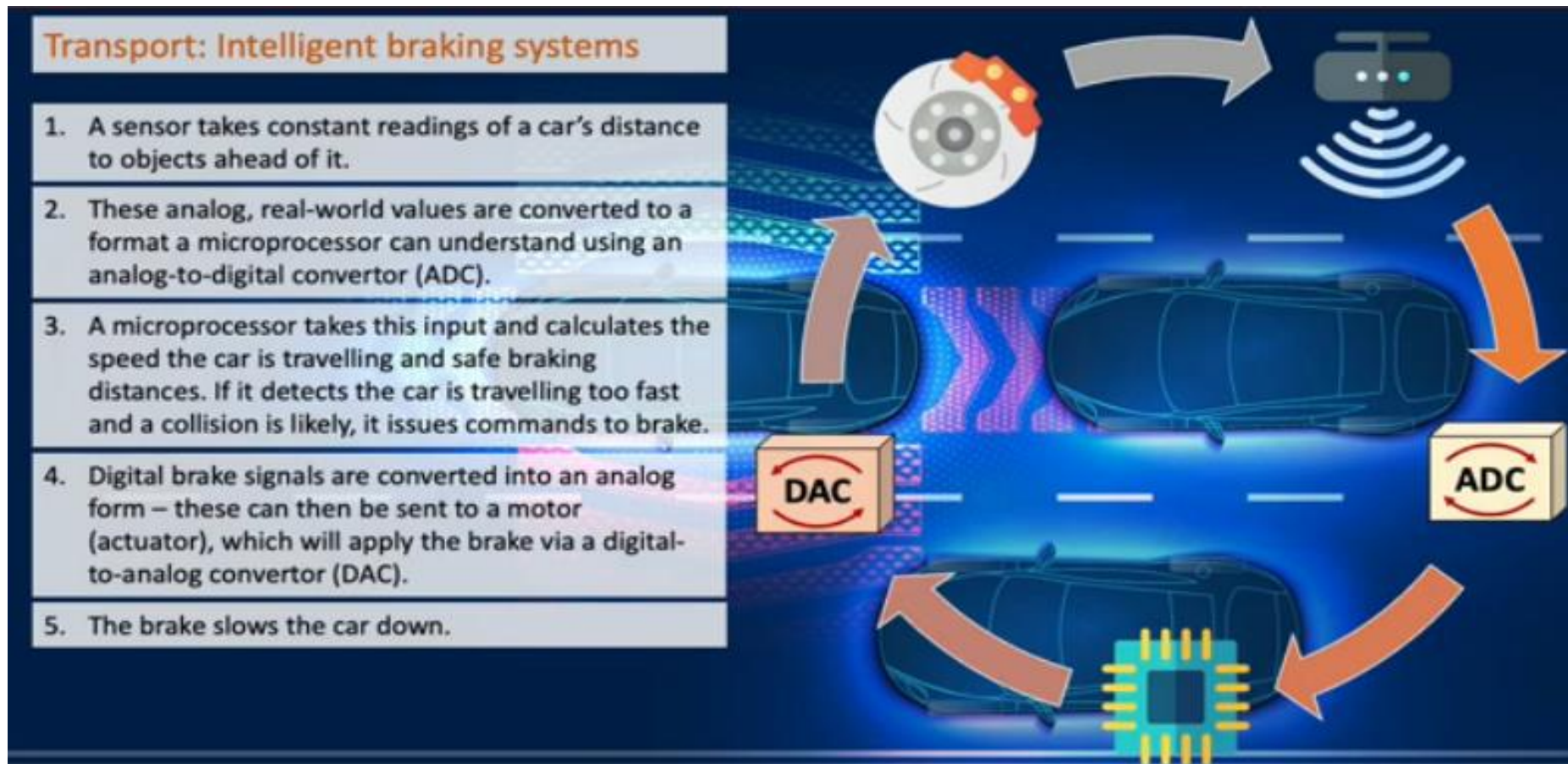
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6.1 Automated systems

Advantages and Disadvantages of Automated Systems

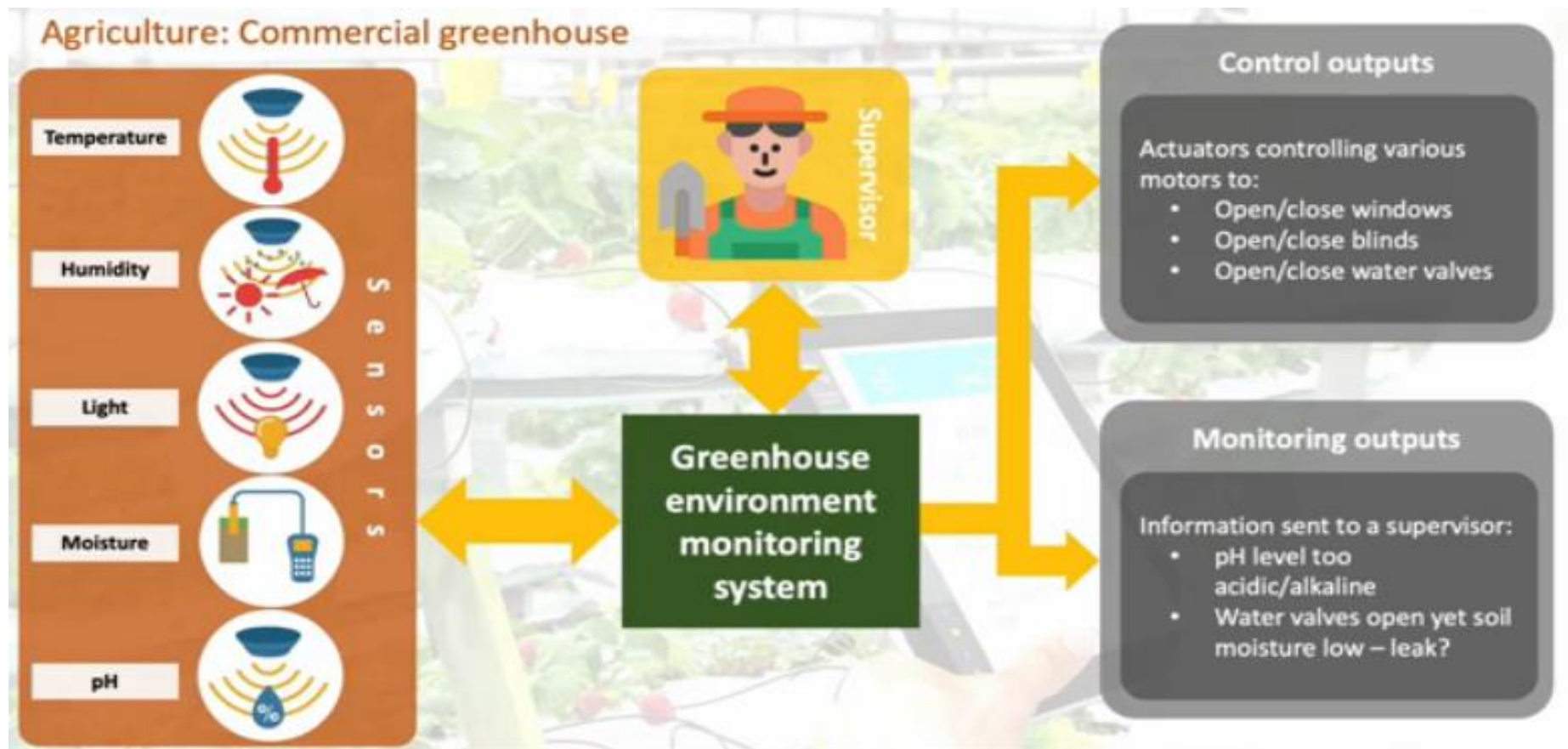
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6.1 Automated systems

Advantages and Disadvantages of Automated Systems

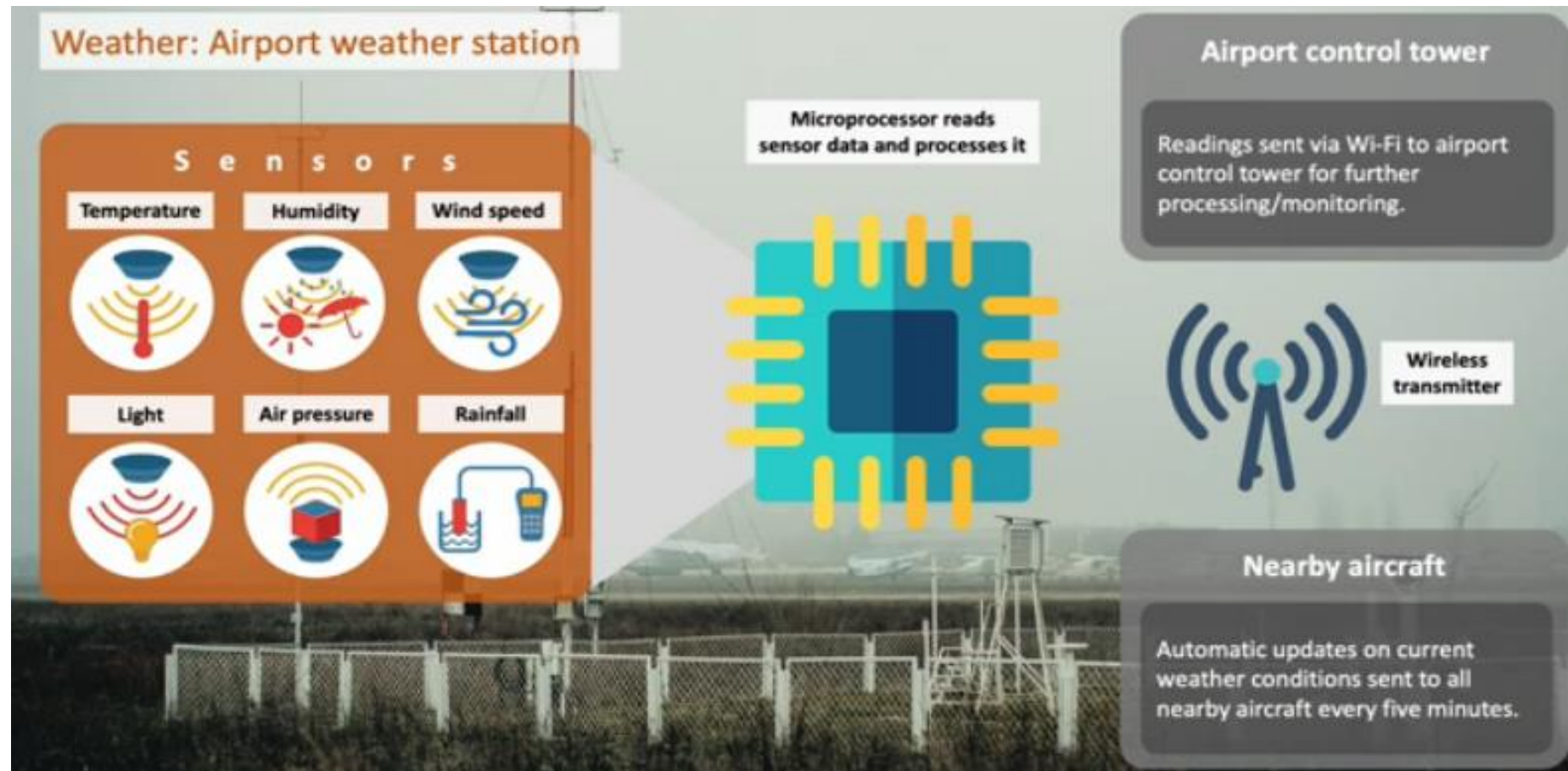
Describe the advantages and disadvantages of an automated system used for a given scenario



6.1 Automated systems

Advantages and Disadvantages of Automated Systems

Describe the advantages and disadvantages of an automated system used for a given scenario



6.1 Automated systems

Advantages and Disadvantages of Automated Systems

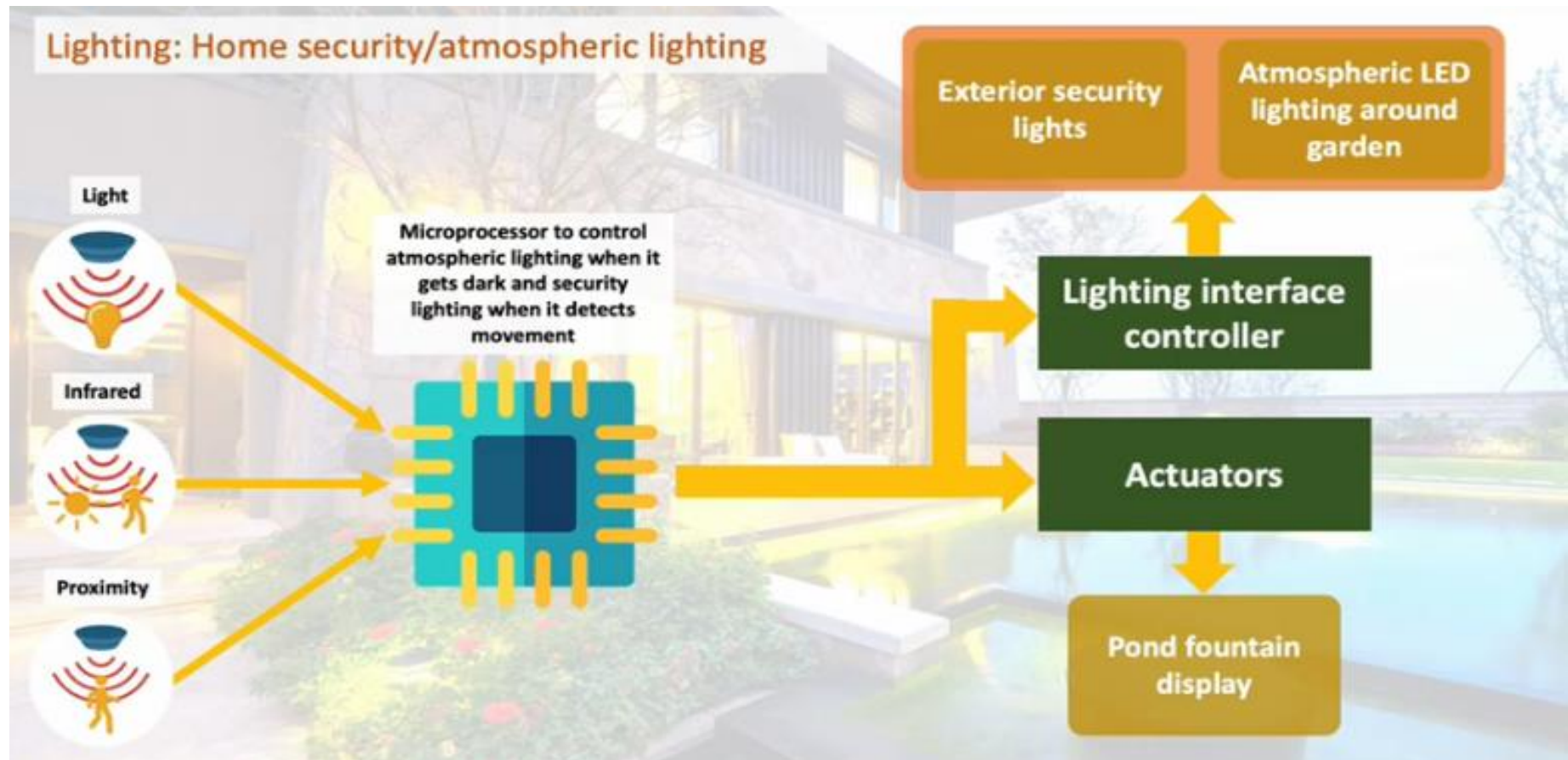
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6.1 Automated systems

Advantages and Disadvantages of Automated Systems

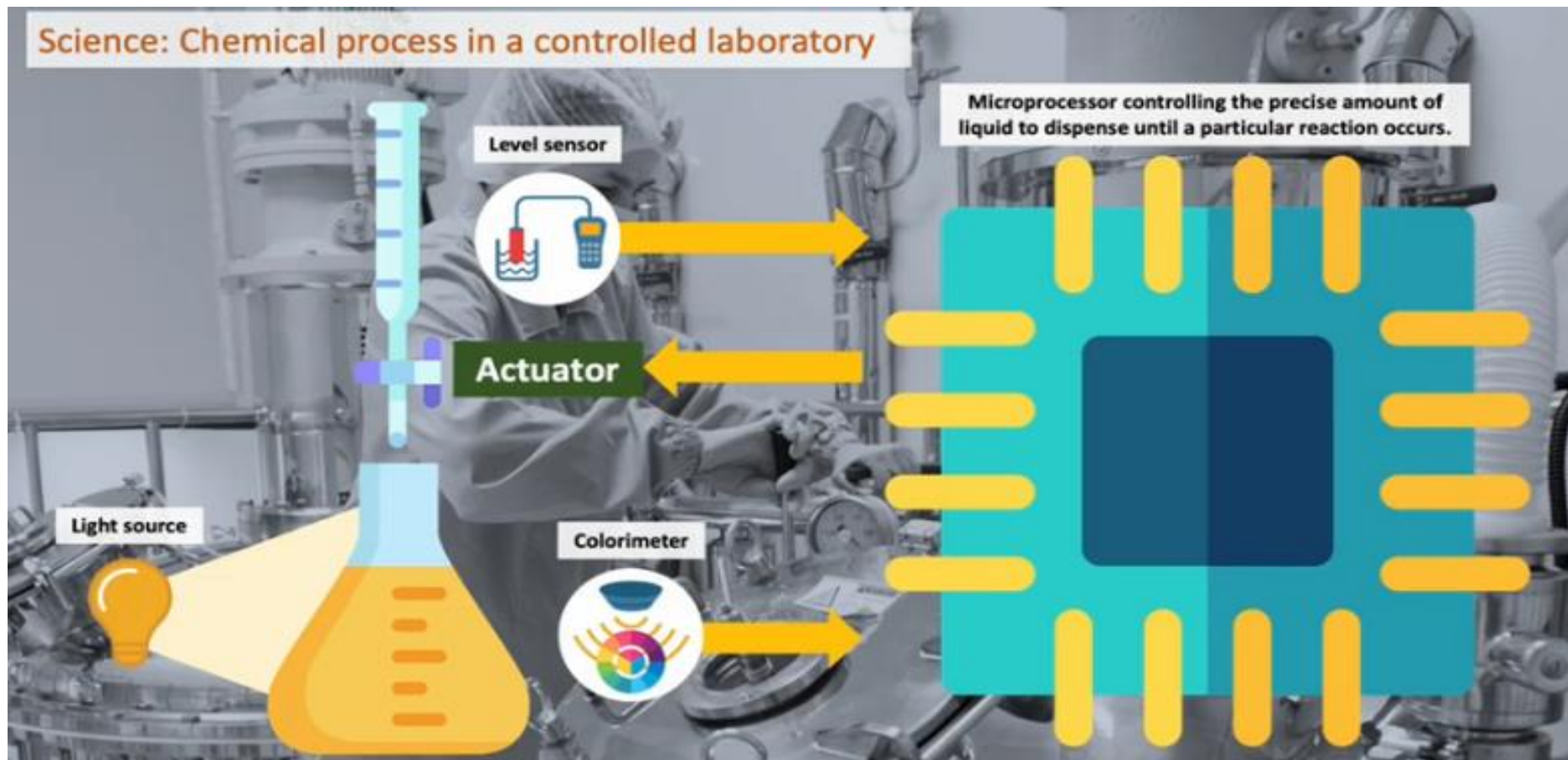
Describe the advantages and disadvantages of an automated system used for a given scenario



6.1 Automated systems

Advantages and Disadvantages of Automated Systems

Describe the advantages and disadvantages of an automated system used for a given scenario



6.1 Automated systems

Advantages and Disadvantages of Automated Systems

Describe the advantages and disadvantages of an automated system used for a given scenario

Advantages and disadvantages of automated systems



- Operations can be carried out faster than a human operator.
- Safer because an automated system is more likely to make timely interventions.
- Allows humans to keep away from hazardous environments.
- Processes are more likely to run under optimal conditions for longer.
- Long-term cost savings versus a large human workforce.
- Efficient use of materials and resources leads to increased productivity.
- More consistent and repeatable results.



- Initially expensive to set up.
- Requires significant testing and calibration to ensure it works as intended.
- Any computerised system is subject to cyber attacks.
- Requires highly specialised and ongoing maintenance.
- Only able to respond to specific situations within certain parameters – therefore, not as flexible as trained humans able to use intuition and experience.
- Can result in significant job losses.

6.1 Automated systems

Advantages and Disadvantages of Automated Systems

Describe the advantages and disadvantages of an automated system used for a given scenario



Automated systems

Automated systems can be used in many different situations and scenarios.

They can support partial or full automation using a variety of sensors, microprocessors and actuators/motors.

- Industry: Automation in factories using robotics and hazardous environments such as power plants.
- Transport: Self-drive cars, intelligent braking, automatic parking systems, adaptive cruise control.
- Agriculture: Commercial greenhouses, automated irrigation systems.
- Weather: Remote weather stations, airport weather stations.
- Gaming: Various gaming devices for providing an enhanced user experience.
- Lighting: House security lighting, automatic road lighting, smart ambient and atmospheric lighting.
- Science: Thousands of applications for automating aspects of scientific research, testing and production.

6.1 Automated systems

Actuators:

<https://www.youtube.com/watch?v=LHn7O6PUaoY>

What is an Actuator?



6.1 Automated systems

Actuators:

When a computer is used to control devices, such as a conveyer belt or a valve, it is usually necessary to use an **actuator** to, for example, start/stop the conveyer belt or open/close the valve. An actuator is a mechanical or electromechanical device such as a relay, solenoid or motor. We will consider a solenoid as the example; this converts an electrical signal into a magnetic field producing linear motion:

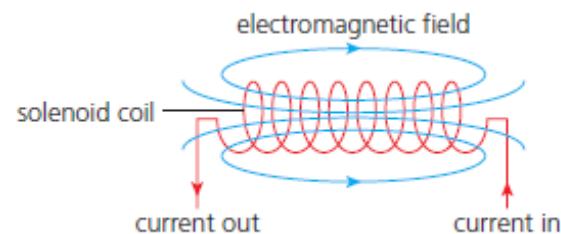
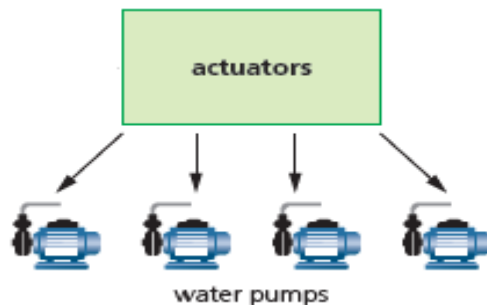


Figure 1

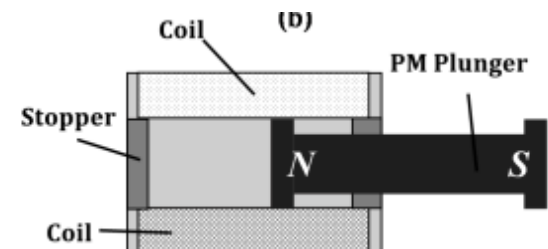


Figure 2

If a plunger (for example, a magnetized metal bar see figure 2) is placed inside the coil, it will move when a current is applied to the coil (see Figure 1). This would allow the solenoid to operate a valve or a switch, for example. There are also examples of rotary solenoids where a cylindrical coil is used. In this case, when a current is supplied to the coil, it would cause a rotational movement of the plunger.

6.2 Robotics

<https://youtu.be/fUyU3lKzoio>



6.2 Robotics

<https://youtu.be/dDsmbwOrHJs>



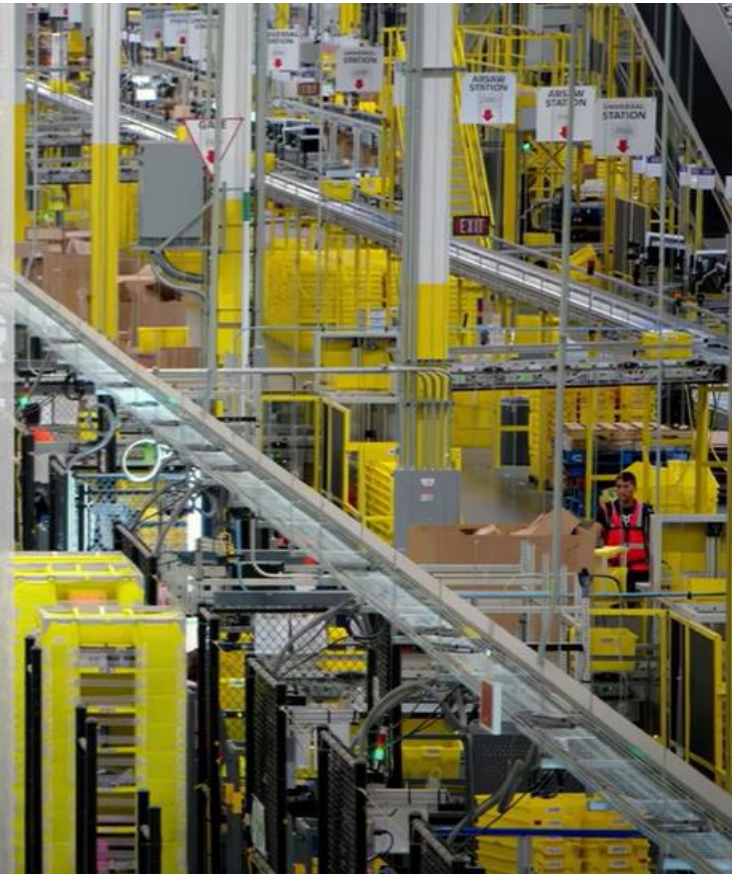
6.2 Robotics

Amazon Robotics Deploys First Fully Autonomous Robot With NVIDIA

https://www.youtube.com/watch?v=LUnZXBL_IqA

NVIDIA Isaac Sim

Amazon Robotics Deploys
First Fully Autonomous Robot



6.2 Robotics

Robots and robotics

<https://www.youtube.com/watch?v=DModfuQRA3w>



6.2 Robotics

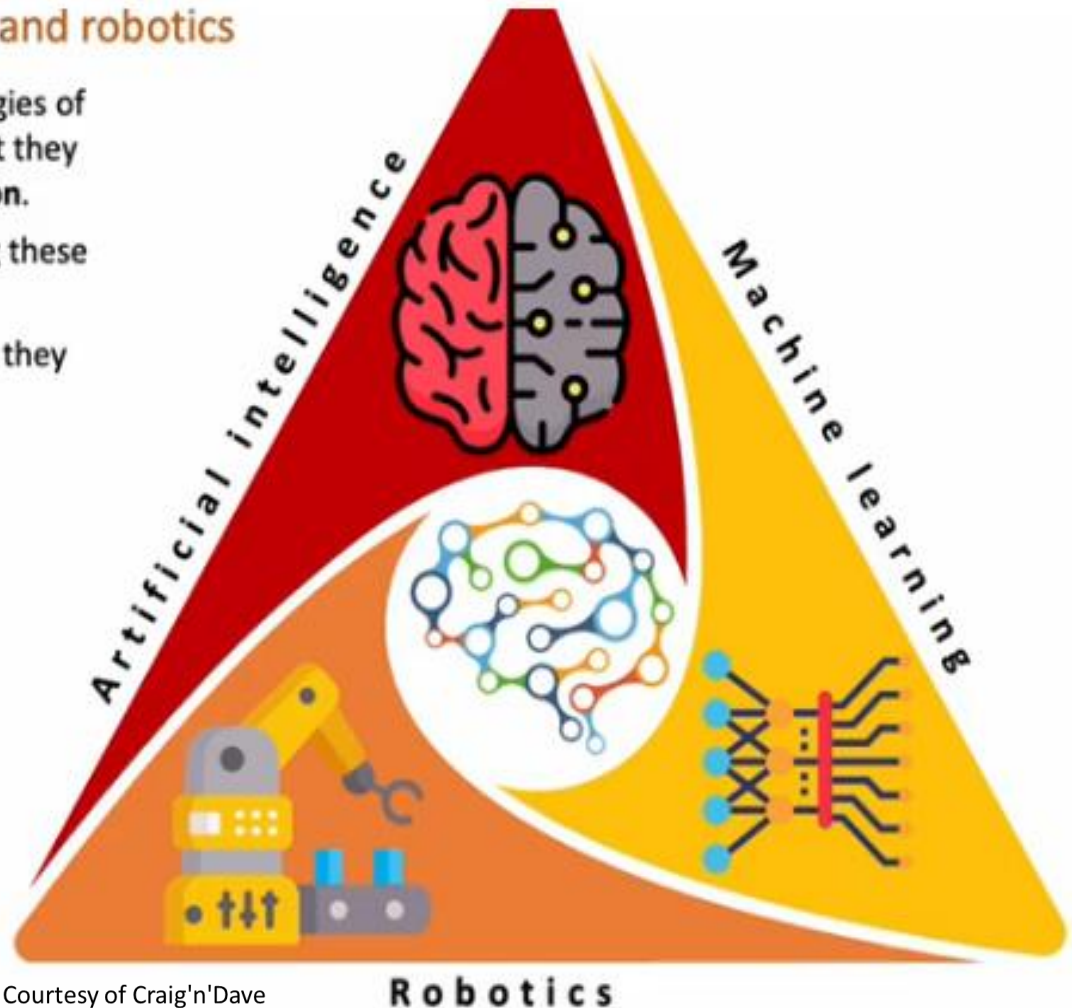
Robots and robotics

Artificial intelligence, machine learning and robotics

There is a lot of hype around the emerging technologies of **artificial intelligence** and **machine learning** and what they could mean for the future of **robotics** and **automation**.

However, there is also a lot of confusion surrounding these terms because they are often used interchangeably.

So, what does each term actually mean, and how do they relate to each other?



Courtesy of Craig'n'Dave

6.2 Robotics

Robots and robotics

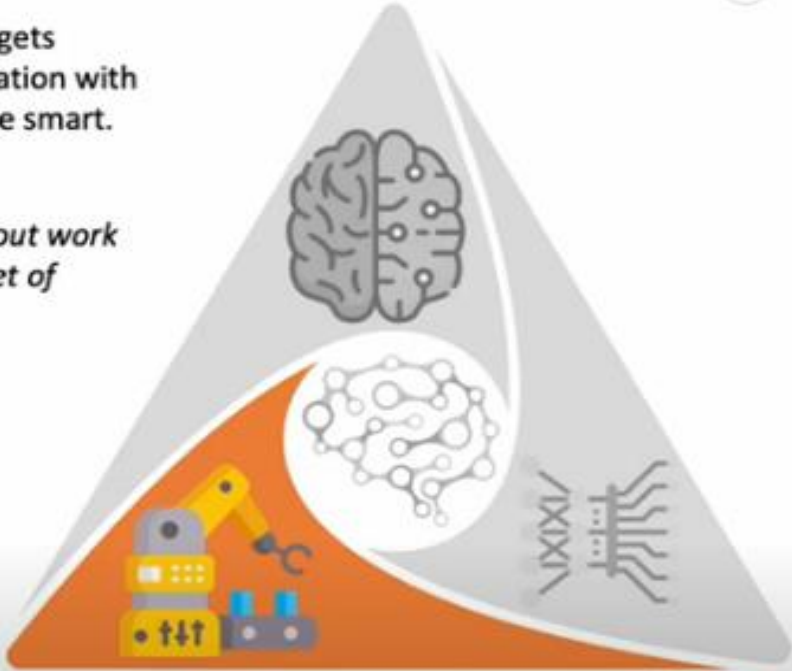


Robotics

The term *robotics* often gets thrown around in association with AI – but not all robots are smart.

What is a robot?

“A machine that carries out work by itself by following a set of programmed rules.”



Dumb robots	Smart robots
Simply repeat the same tasks over and over – no AI involved.	Trained to learn, adapt and carry out progressively more complex tasks.
Car assembly line	A robot designed to beat a human at chess

6.2 Robotics

What is robotics?

Robotics is a branch of engineering that involves the conception, design, manufacture and operation of robots. The objective of the robotics field is to create intelligent machines that can assist humans in a variety of ways.

what is a robot in the real world? Robotics is a branch of (computer) science that brings together the design, construction and operation of robots. Robots can be found in:

<u>factories</u>	<u>Home</u>	<u>Drones</u>
– welding parts together – spray-painting panels on a car – fitting windscreens to cars – cutting out metal parts to a high precision – bottling and labelling plants – warehouses (automatic location of items)	– autonomous floor sweepers – autonomous lawn mower – ironing robots (for example, 'dress man') – pool cleaning – automatic window cleaners – entertainment ('friend' robots)	– unmanned aerial vehicles (UAVs) are drones that are either remotely controlled or totally autonomous using embedded systems – can be used in reconnaissance (for example, taking aerial photographs) – can be used to make parcel deliveries (for example, Amazon).

6.2 Robotics

Characteristics of a robot

Characteristics of a robot

In very simple terms, you can think of the broad characteristics of all robots as falling into three categories.

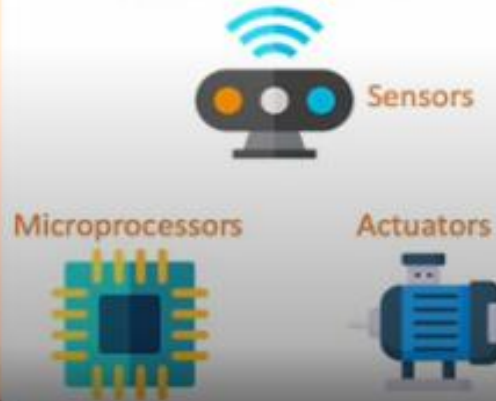
1

Mechanical structure or framework



2

Electrical components



3

Programmability



6.2 Robotics

Characteristics of a robot

To be correctly called a robot, they need to have the following characteristics:

1 Ability to sense their surroundings:

- this is done via sensors (such as light, pressure, temperature, acoustic, and so on)
- sensors allow a robot to recognize its immediate environment and gives it the ability to determine things like size, shape or weight of an object, detect if something is hot or cold, and so on; all sensor data is sent to a microprocessor or computer.

2 Have a degree of movement:

- they can make use of wheels, cogs, pistons, gears (etc.) to carry out functions such as turning, twisting, moving backwards/forwards, gripping or lifting
- they are **mechanical structures** made up of many parts (for example, motors, hydraulic pipes, actuators and circuit boards)
- they contain many **electrical components** to allow them to function
- can make use of **end effectors** (different attachments to allow them to carry out specific tasks such as welding, spraying, cutting or lifting).

3 Programmable:

- they have a 'brain' known as a **controller** that determines the action to be taken to perform a certain task (the controller relies on data sent from sensors or cameras, for example)
- controllers are **programmable** to allow the robots to do certain tasks.

6.2 Robotics

Characteristics of a robot



Robots and robotics

Robotics is a branch of computer science that incorporates the design, construction and operation of robots.

- Includes factory equipment, domestic robots and drones.

A robot is a machine that works by itself by following a set of programmed rules.

- Some robots can be *dumb* – they simply repeat the same tasks over and over.
- Some robots exhibit AI/machine learning – they are trained to carry out more complex tasks and adapt.

Robots share similar characteristics:

- A mechanical structure or framework
- Electrical components
- Programmability

6.2 Robotics

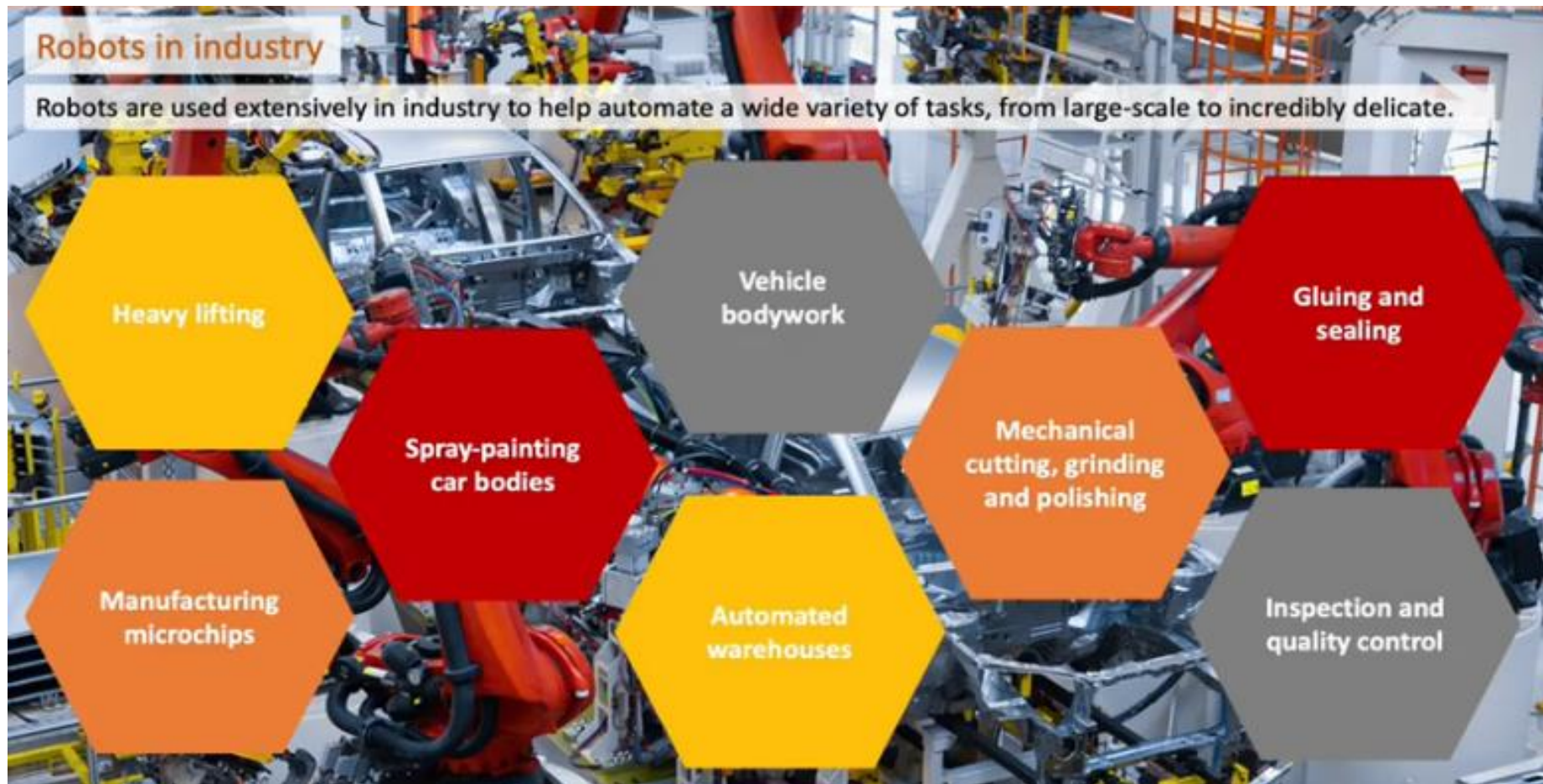
Characteristics of a robot

<https://www.youtube.com/watch?v=U70nr2bRxAE>



6.2 Robotics

Characteristics of a robot



6.2 Robotics

Characteristics of a robot

Robots in transport

Robotics is being used more and more throughout the transport industry, from partially automated parking to fully autonomous cars and trains.

These systems utilise a wide range of sensors to constantly gather and process data and perform physical actions such as:

- Changing gear
- Applying brakes
- Steering

We looked at this in more detail back in our videos on automation.

Self-parking cars

Adaptive cruise control

Airplane autopilot and self-landing

Autonomous trains

Driverless taxis



6.2 Robotics

Characteristics of a robot



6.2 Robotics

Characteristics of a robot



6.2 Robotics

Characteristics of a robot



6.2 Robotics

Characteristics of a robot

Robots in entertainment



Autonomous robots in theme parks and venues to interact with visitors.

Robotic cameras to replicate complex film shots many times in a row.

Bringing attractions to life and making them more immersive and exciting.

Making dangerous, action-heavy film sequences more realistic.

Enhancing live events, concerts and music festivals.

6.2 Robotics

The role of robots and their advantages and disadvantages

Advantages and disadvantages of robots and robotic systems



- Robots can work in conditions that may be dangerous for humans.
- They are well-suited to boring, repetitive tasks, producing consistent results over time.
- Reduced running costs in terms of heating, lighting, pension, sick/holiday pay.
- Robots can operate 24/7 without breaks, lunch or sleep, vastly increasing productivity.
- Improved safety because human error is largely or completely removed.



- Robotic systems are initially very expensive to set up.
- They have replaced a large number of manual jobs, resulting in unemployment.
- As with any computer-based technology, there is potential for cyber attacks.
- Robots tend to not cope as well with unexpected situations that require a degree of human intuition.

6.2 Robotics

The role of robots and their advantages and disadvantages in Industry

Advantages	Disadvantages
robots are capable of working in conditions that may be hazardous to humans	robots can find it difficult to do 'non-standard' tasks (for example, windscreen being fitted to a car is cracked)
robots work 24/7 without the need to stop	
robots are less expensive in the long run (since there will be fewer salaries to pay)	robots can lead to higher unemployment amongst manual labour tasks
robots are more productive than humans (higher productivity)	there is a risk of deskilling when robots take over certain tasks (for example, welding and paint spraying)
although not necessarily more accurate, robots are more consistent	
robots are better suited to boring, repetitive tasks than humans (therefore less likely to make mistakes)	factories can now be moved to anywhere in the world where operation costs are lower (leading again to unemployment in some countries)
there will be less cost in heating and lighting (robots don't need good light or warmth)	robots are expensive to buy and set up in the first place

6.2 Robotics

The role of robots and their advantages and disadvantages in Transport

A self-driving car, also known as an autonomous car, driver-less car, or robotic car (robo-car), is a car incorporating vehicular automation, that is, a ground vehicle that is capable of sensing its environment and moving safely with little or no human input. Self-driving cars combine a variety of sensors to perceive their surroundings, such as thermo graphic cameras, radar, sensor ..etc.

Advantages of autonomous vehicles	Disadvantages of autonomous vehicles
safer since human error is removed leading to fewer accidents	very expensive system to set up in the first place (high technology requirements)
better for the environment since vehicles will operate more efficiently	the ever-present fear of hacking into the vehicle's control system
reduced traffic congestion (humans cause 'stop-and-go' traffic known as ' <i>the phantom traffic jam</i> ', autonomous vehicles will be better at smoothing out traffic flow reducing congestion in cities)	security and safety issues (software glitches could be catastrophic; software updates would need to be carefully controlled to avoid potential disasters)
increased lane capacity (research shows autonomous vehicles will increase lane capacity by 100% and increase average speeds by 20%, due to better braking and acceleration responses together with optimized distance between vehicles)	the need to make sure the system is well-maintained at all times; cameras need to be kept clean so that they don't give false results; sensors could fail to function in heavy snowfall or blizzard conditions (radar or ultrasonic signals could be deflected by heavy snow particles)
reduced travel times (for the reasons above) therefore less commuting time	driver and passenger reluctance to use the new technology
stress-free parking for motorists (the car will find car parking on its own and then self-park)	reduction in the need for taxis could lead to unemployment (imagine New York without its famous yellow cabs!)

6.2 Robotics

The role of robots and their advantages and disadvantages in Autonomous trains

Advantages of autonomous trains	Disadvantages of autonomous trains
this improves the punctuality of the trains	the ever-present fear of hacking into the vehicle's control system
reduced running costs (fewer staff are required)	system doesn't work well with very busy services (at the moment)
improves safety since human error is removed	high capital costs and operational costs initially (that is, buying the trains, expensive signalling and control equipment and the need to train staff)
minimises energy consumption since there is better control of speed and an optimum service requires less energy (trains stuck in stations still use energy)	ensuring passenger behaviour is acceptable particularly during busy times (for example, jamming doors open on trains, standing too near the edge of platforms and so on)
it is possible to increase the frequency of trains (automated systems allow for shorter times between trains)	passenger reluctance to use the new technology
it is easier to change train scheduling (for example, more trains during busier times)	no drivers mean there will be a need for CCTV to monitor railway stations

6.2 Robotics

Characteristics of a robot



Roles that robots can perform

Robots come in a vast variety of shapes and sizes, from simple factory equipment through to domestic robots and drones. They are used extensively in:

- Industry: Factories, assembly plants, warehouses.
- Transport: Self-drive cars and trains, airplane autopilot.
- Agriculture: Cutting, pruning, weeding, seeding, harvesting, picking, fertilising.
- Medicine: Assisting with surgical procedures, cleaning and disinfecting, robotic prosthetics.
- Domestic: Cleaning, lawn-mowing, home assistants.
- Entertainment: Theme parks, concerts, festivals, film industry.

6.1 Automated systems and 6.2 Robotics

Questions 1:

What is the main difference between monitoring and controlling of a process?

Questions 2:

Describe how a computer and sensors are used to monitor the greenhouse environment; the temperature must be between 25° and 30° and the soil pH must be between 4 and 5.

Questions 3:

Name three other sensors and describe three different applications which use these sensors.

Questions 4:

What is an Actuator?

Questions 5:

Robots are being used to spray some metal components with paint.

Describe:

- (i) Two problems the robots might encounter when spraying these metal components.
- (ii) How these problems could be overcome.

Questions 6:

In general, what are the advantages and disadvantages of using robots rather than using human workers in a manufacturing company?

Questions 7:

A chemical process is being monitored using temperature sensors and pH sensors. These sensors send data to a microprocessor. If the temperature in the process drops to below 50° (then a heater is switched on and if the acidity rises to a pH of 5, then a valve is opened to admit more acid (**pH must be kept below 5**).

Describe how the sensors and microprocessor are used to control the chemical process.

6.1 Automated systems and 6.2 Robotics

Question 8:

6 Robots are used in a factory to build cars.

(a) One characteristic of a robot is its mechanical structure.

State **two other** characteristics of a robot.

1

2 [2]

(b) Suggest **two** advantages of using robots, instead of humans, to build cars in the factory.

1

2 [2]

6.1 Automated systems and 6.2 Robotics

Answer 1:

Monitoring a process:

- Sensors send data
- Data converted to digital data (using ADC)
- Computer analyses/compares data
- If outside the range (when compared to stored values) it sends a signal to sound alarm or produce warning message on a screen

Controlling a process:

- (First four lines are the same as above)
it sends signals to actuators
- Signals converted to analogue using DAC
- Pumps/valves (etc.) are opened/closed (etc.) to alter the process conditions
- The output affects the next input.

6.1 Automated systems and 6.2 Robotics

Answer 2:

- Data gathered from temperature and pH sensors
- Signals sent to ADC where they are converted to digital
- Information is sent to the computer where it is compared to pre-set/stored values
- If the temperature 30°C , a signal is sent by the computer to the motors to open the windows (again DAC and actuators used) and to switch off the heaters
- If the temperature is between 25°C and 30°C , no action is taken
- If the pH 5, a signal is sent by the computer to open valves to add acidic water (again use of DAC and actuators)
- If the pH is between 4 and 5, no action is taken
- In all cases, the output from the system will affect the next input to the system
- Alarm is sounded if there is an error in the system
- Monitoring and control continues until system turned off.

6.1 Automated systems and 6.2 Robotics

Answer 3:

Light sensor - automatic doors in a hotel

Acoustic sensor - picks up sounds. for example: a burglar alarm

Infrared sensor - counting items every time the beam is broken.

Answer 4:

Actuators are used in many control applications involving sensors and computers or microprocessors. They are usually electromechanical devices such as a motor, relay or solenoid. Actuators can be digital (solenoid) but others require a digital to analogue converter (DAC) since the actuator needs an electric current/voltage to operate a motor or a valve, for example.

Answer 5:

(i) Two problems - out of paint

- component is missing

- something is obstructing the paint spray gun.

(ii) Use of sensors to detect out of paint, item is in the correct position, something is obstructing the spray gun, etc.

Use of CCTV/cameras to show the status of the process (e.g. these can check paint surface of a car for imperfections).

6.1 Automated systems and 6.2 Robotics

Answer 6:

Advantages:

- Can work in conditions hazardous/unpleasant to humans;
- They can work non-stop (no breaks);
- Less expensive in the long run (no wages required);
- More productive (takes less time to do a task than a human);
- More consistent (everything is always made to the same standard);
- Removes the need for humans to do boring/ repetitive tasks;
- Less factory costs (can reduce heating, lighting levels, etc.).

Disadvantages:

- Difficulty dealing with non-standard situations;
- Can lead to unemployment;
- Possible risk of deskilling;
- Production process can be more easily moved to 'less expensive' countries.

Answer 7:

- sensors continually send data to the microprocessor

if the output from the sensors is analogue, then the data is converted to digital, using an ADC, before it is sent to the microprocessor

- the microprocessor contains optimum temperature and acid/pH values (or they are stored on a storage device connected to the microprocessor)
- if the temperature reading is = 50° (then no action is taken
- if the pH reading is > 5, then a signal is sent to a valve to open and allow acid to enter chemical process

6.1 Automated systems and 6.2 Robotics

Answer 8:

Question	Answer	Marks
6(a)	• It has electrical components. • It is programmable.	2
6(b)	Any two from: • more efficient than a human • more accurate than a human • can lift larger and heavier equipment than a human • can replace humans working in a dangerous environment • reduces running costs, such as wages and overheads • can produce consistent results/output • can work 24 hours a day // do not need to take breaks.	2

6.3 Artificial intelligence

6.3 Artificial intelligence

Candidates should be able to:

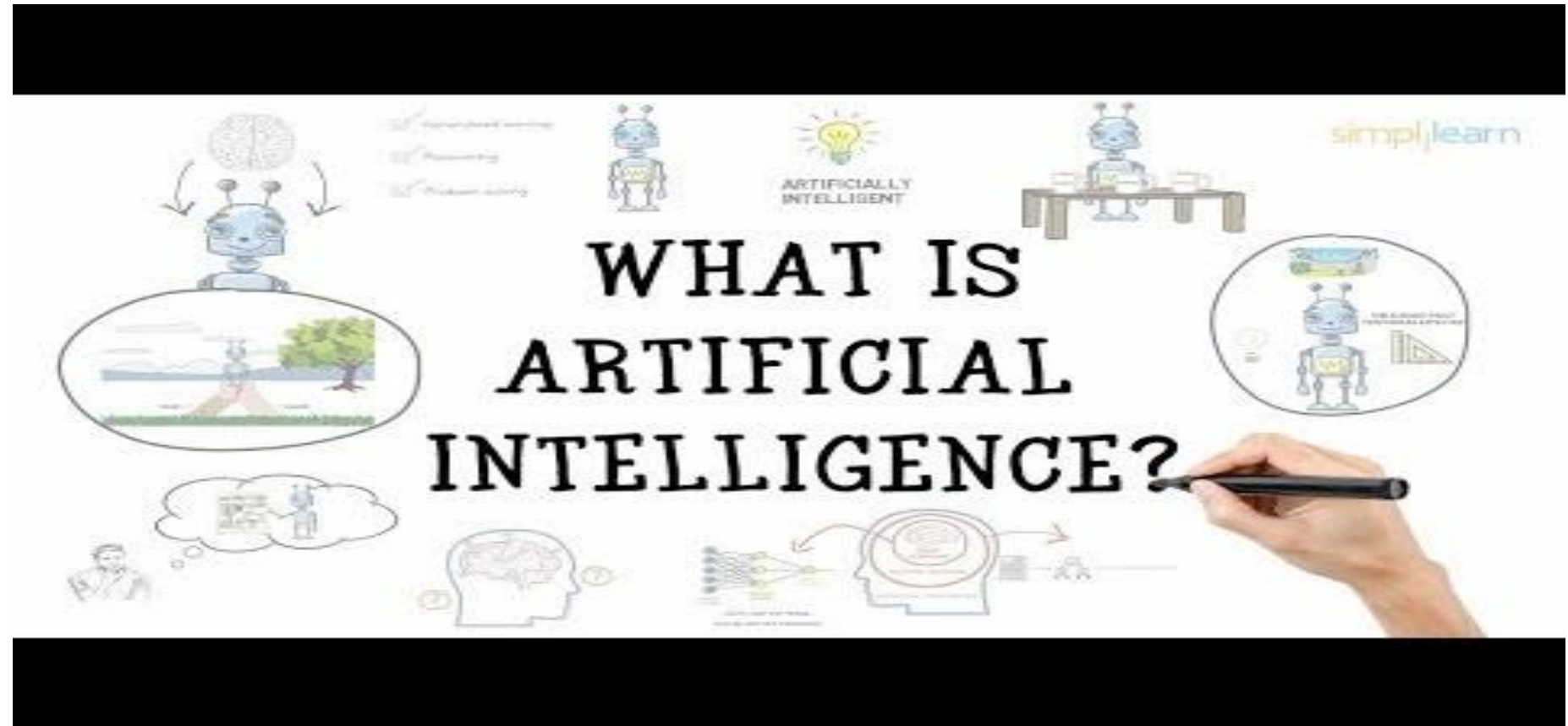
- 1 Understand what is meant by artificial intelligence (AI)
- 2 Describe the main characteristics of AI as the collection of data and the rules for using that data, the ability to reason, and can include the ability to learn and adapt
- 3 Explain the basic operation and components of AI systems to simulate intelligent behaviour

Notes and guidance

- AI is a branch of computer science dealing with the simulation of intelligent behaviours by computers
- Limited to:
 - expert systems
 - machine learning
- Expert systems have a knowledge base, a rule base, an inference engine and an interface
- Machine learning is when a program has the ability to automatically adapt its own processes and/or data

6.3 Artificial intelligence

<https://www.youtube.com/watch?v=ad79nYk2keg>



6.3 Artificial intelligence

How AI Works

https://www.youtube.com/watch?v=Ok-xpKjKp2g&list=PLzdnOPI1iJNeehd1RXhnVMBFi1WhWLx_Y



6.3 Artificial intelligence

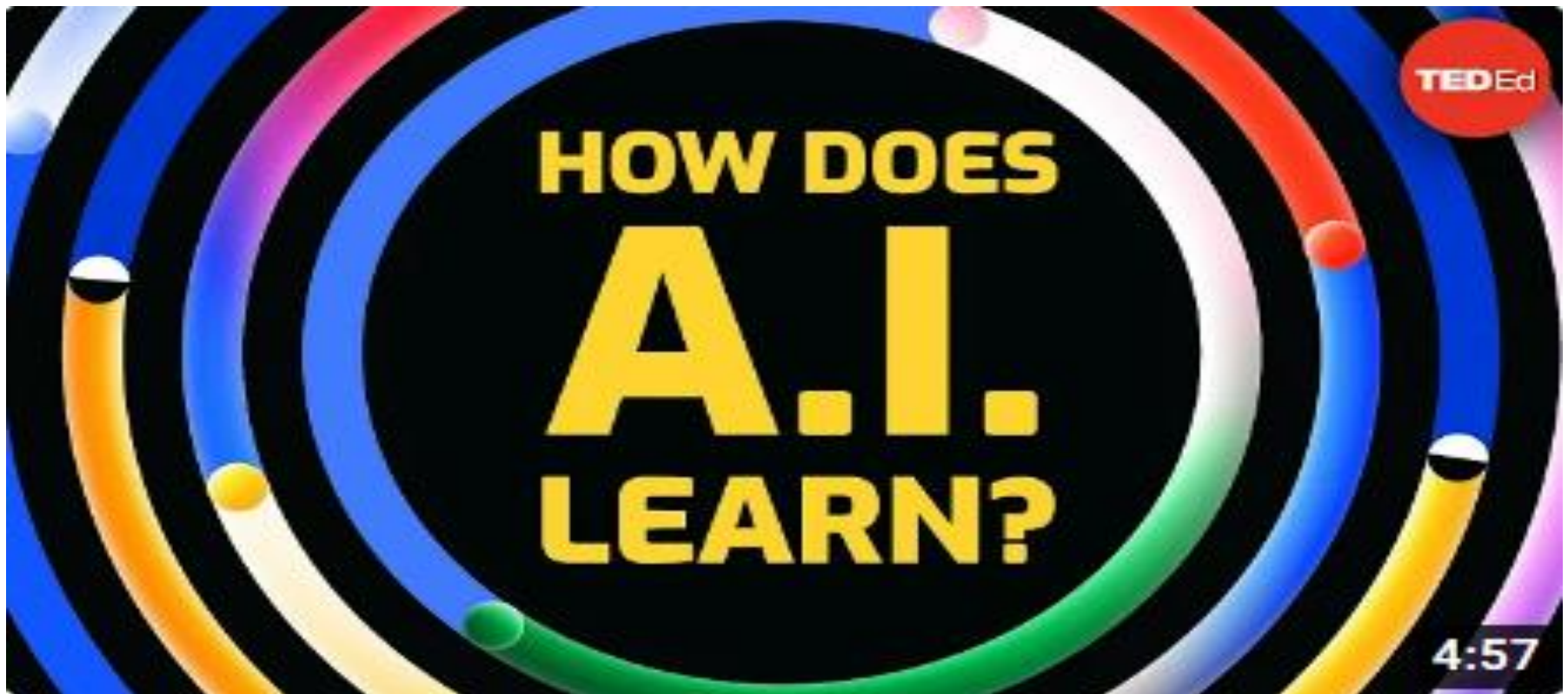
<https://youtu.be/0oRVLf16CMU>



6.3 Artificial intelligence

How does artificial intelligence learn?

<https://www.youtube.com/watch?v=0yCJMt9Mx9c>



6.3 Artificial intelligence

How do all the algorithms, like ChatGPT & Bing, around us learn to do their jobs?

<https://www.youtube.com/watch?v=R9OHn5ZF4Uo>



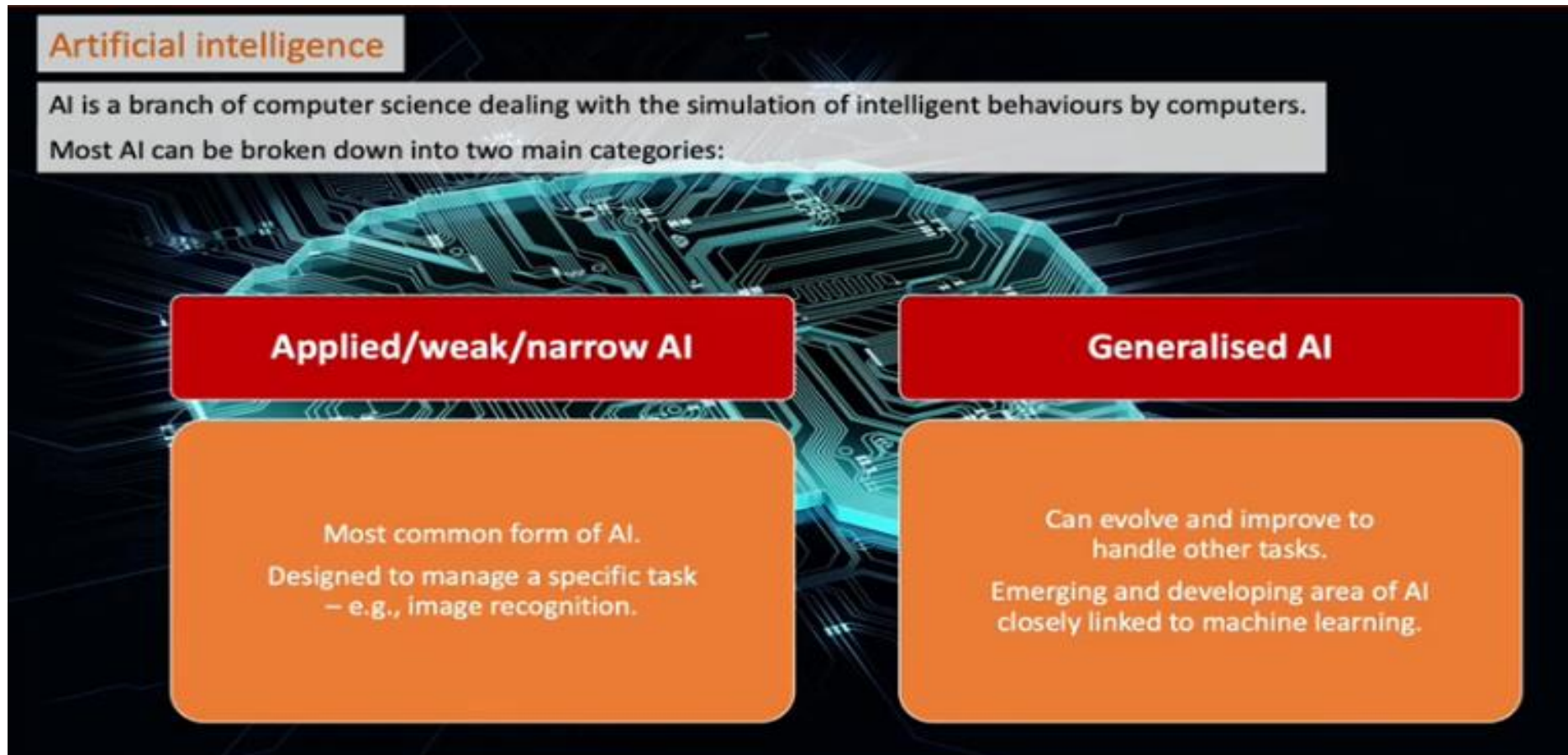
6.3 Artificial intelligence

<https://www.youtube.com/watch?v=rE00609d-f4>



6.3 Artificial intelligence

Understand what is meant by artificial intelligence (AI)



6.3 Artificial intelligence

Understand what is meant by artificial intelligence (AI)

Types of artificial intelligence

Weak AI or Narrow AI

Weak AI also called Narrow AI is AI trained and focused to perform specific tasks. Weak AI drives most of the AI that surrounds us today. 'Narrow' might be a more accurate descriptor for this type of AI as it is anything but weak; it enables some very robust applications, such as Apple's Siri, Amazon's Alexa, IBM Watson, and autonomous vehicles.

General Intelligence (AGI)

Is a theoretical form of AI where a machine would have an intelligence equaled to humans; it would have a self-aware consciousness that has the ability to solve problems, learn, and plan for the future.

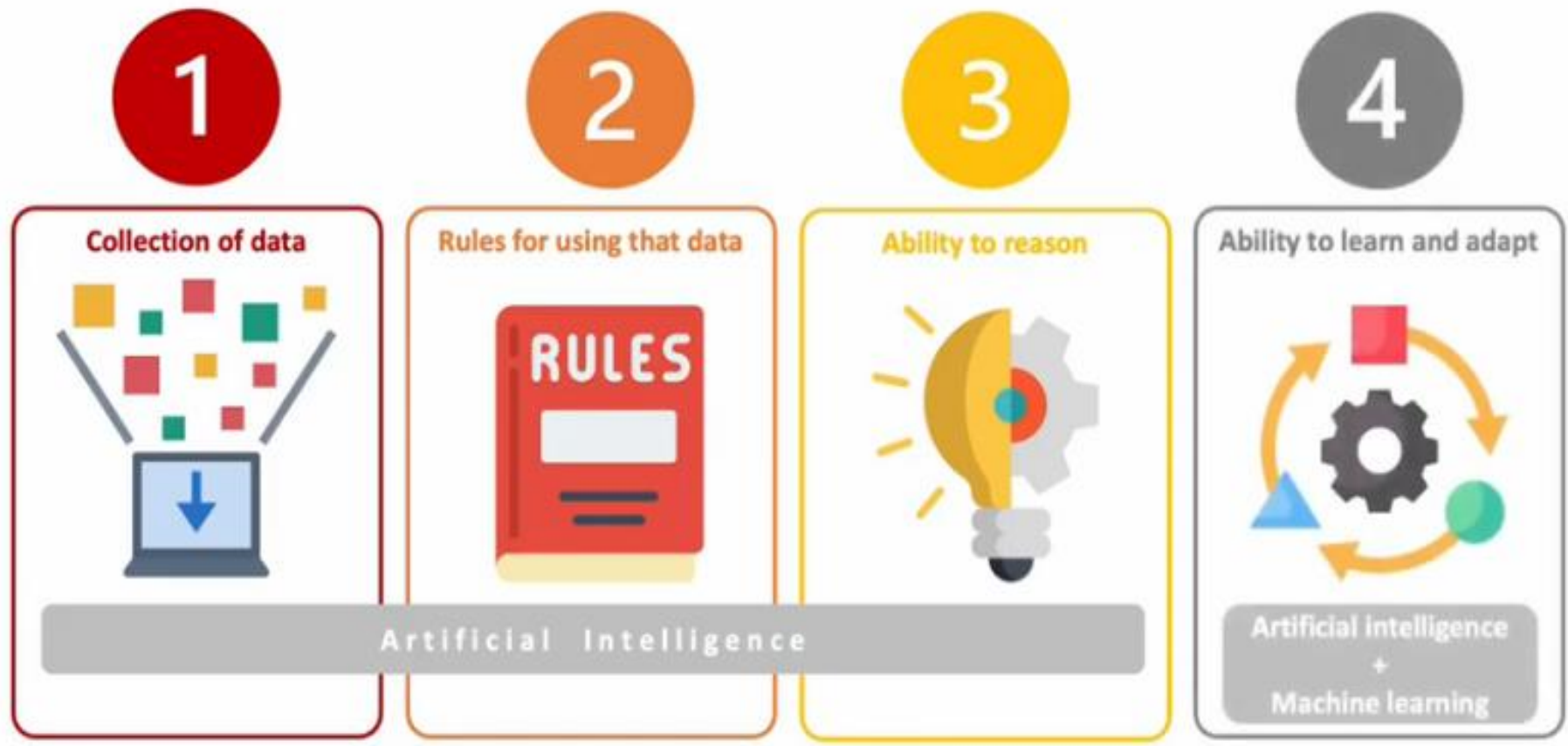
Strong AI

.Artificial Super Intelligence (ASI) also known as superintelligence would surpass the intelligence and ability of the human brain.

6.3 Artificial intelligence

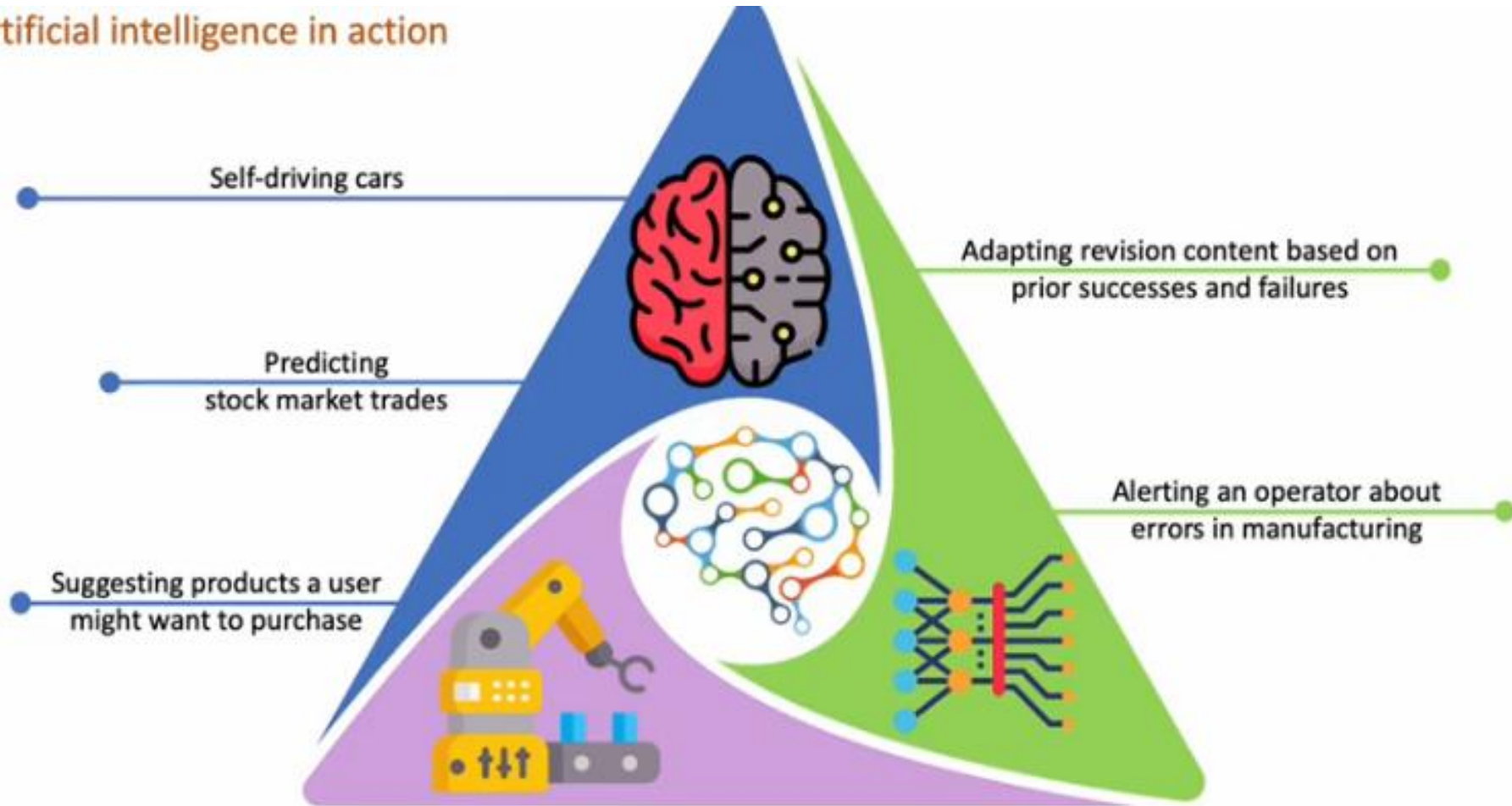
Main characteristics of artificial intelligence

In very simple terms, you can think of the main characteristics of AI as falling into the following categories:



6.3 Artificial intelligence

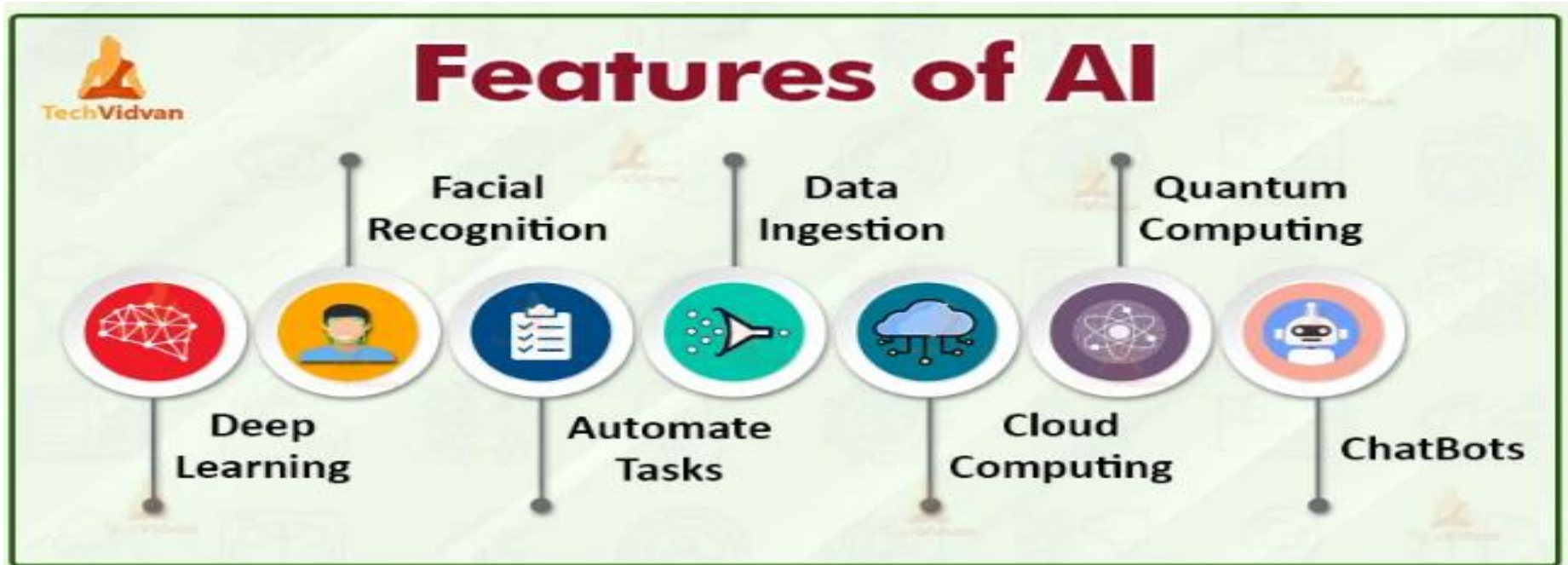
Artificial intelligence in action



6.3 Artificial intelligence

Examples of AI include:

- » news generation based on live news feeds
- » smart home devices (such as Amazon Alexa, Google Now, Apple Siri and Microsoft Cortana):
 - the AI device interacts with a human by recognizing verbal commands
 - it learns from its environment and the data it receives
 - the device becomes increasingly sophisticated in its responses, thus showing the ability to use automated repetitive learning



6.3 Artificial intelligence

Examples of AI include:

Deep learning: is a machine learning technique that teaches computers to do what comes naturally to humans, to learn. Take an example of **Self Driving feature in cars like Tesla(Autopilot)**

Facial Recognition: Artificial Intelligence has made it possible to recognize individual faces using biometric mapping. For example, Clearview AI, an American technology company, offers surveillance technology for law agencies to monitor entire cities with a network of CCTV.

Automate Simple and Repetitive Tasks: AI has the ability to execute the same kind of work over and over again without breaking a sweat.

Data Ingestion: With every passing day, the data that we are all producing is growing exponentially, which is where AI steps in. Instead of manually feeding this data, **AI-enabled** not just gathers this data but also analyzes it with the help of its previous experiences.

Chatbots: are software to provide a window for solving customer problems' through either audio or textual input. Earlier the bots used to respond only to specific commands. If you say the wrong thing, it didn't know what you meant.



Hello! I am the Hodder Education Chatbot. How can I help you?

I ordered the new A Level computer science textbook – when will it be delivered?



Can you give me your order number please and I will check for you?

CSH123456. Thank you!



6.3 Artificial intelligence

Examples of AI include:

Quantum Computing: AI is helping solve complex quantum physics problems with the accuracy of supercomputers with the help of quantum neural networks. This can lead to path-breaking developments in the near future for example, A pioneer in this field is Google AI Quantum

Cloud Computing: Next Artificial Intelligence characteristics is Cloud Computing. With such a huge amount of data being churned out every day, data storage in a physical form would have been a major problem for example Microsoft Azure is one of the prominent players in the cloud computing industry. It offers to deploy your own machine learning models to your data stored in cloud servers without any lock-in.

6.3 Artificial intelligence



Artificial intelligence

Artificial intelligence: A broad term that describes any machine programmed to think, work and react like humans.

- Applied/narrow AI is designed to manage a specific task.
- Generalised AI can evolve and improve to handle other tasks.

Artificial intelligence has the following characteristics:

- The ability to collect data.
- A set of rules for using that data.
- The ability to reason

An artificial intelligence can be described as **machine learning** if it possesses the ability to learn and adapt.

6.3 Artificial intelligence

<https://www.youtube.com/watch?v=T-X2VhflzCw>



6.3 Artificial intelligence

Basic operations and components of AI systems

Artificial intelligence systems

There are many different types of AI systems.

The two you need to know about for the exam are **expert systems** and **machine learning**.

Artificial intelligence

Technique that enables machines to mimic human behaviour.

Expert systems

Subset of AI that mimics the decision-making ability of humans using expert knowledge.

Machine learning

Subset of AI that uses data to enable machines to be trained and learn from past experiences.

6.3 Artificial intelligence

Basic operations and components of AI systems



Expert systems

Expert systems attempt to mimic human knowledge and experience. In essence, they use a combination of knowledge and inference to solve problems or answer questions in a similar way to a human expert.

- Identifying animals, plants, chemical and biological compounds.
- Diagnosing medical problems.
- Financial calculations – e.g., tax.
- Logistics – e.g., efficient routing of parcels for delivery.
- Diagnosing faults in mechanical and electronic equipment.
- Suggesting potential new sites for resource extraction.

6.3 Artificial intelligence

Basic operations and components of AI systems



Expert systems

Expert systems have many advantages.

- High level of accuracy and expertise.
- Traceable solutions and diagnostics.
- Consistent results.
- Faster response times than human experts.
- Can store large amounts of facts and data.
- Unbiased reporting and analysis based on facts.

6.3 Artificial intelligence

Basic operations and components of AI systems



Expert systems

Expert systems have some disadvantages too.

- Only as good as the data entered into the system.
- Correct, optimal use requires training.
- Assuming the system is infallible can lead to poor decisions.
- Responses can be cold and lack the humanity of a real expert.

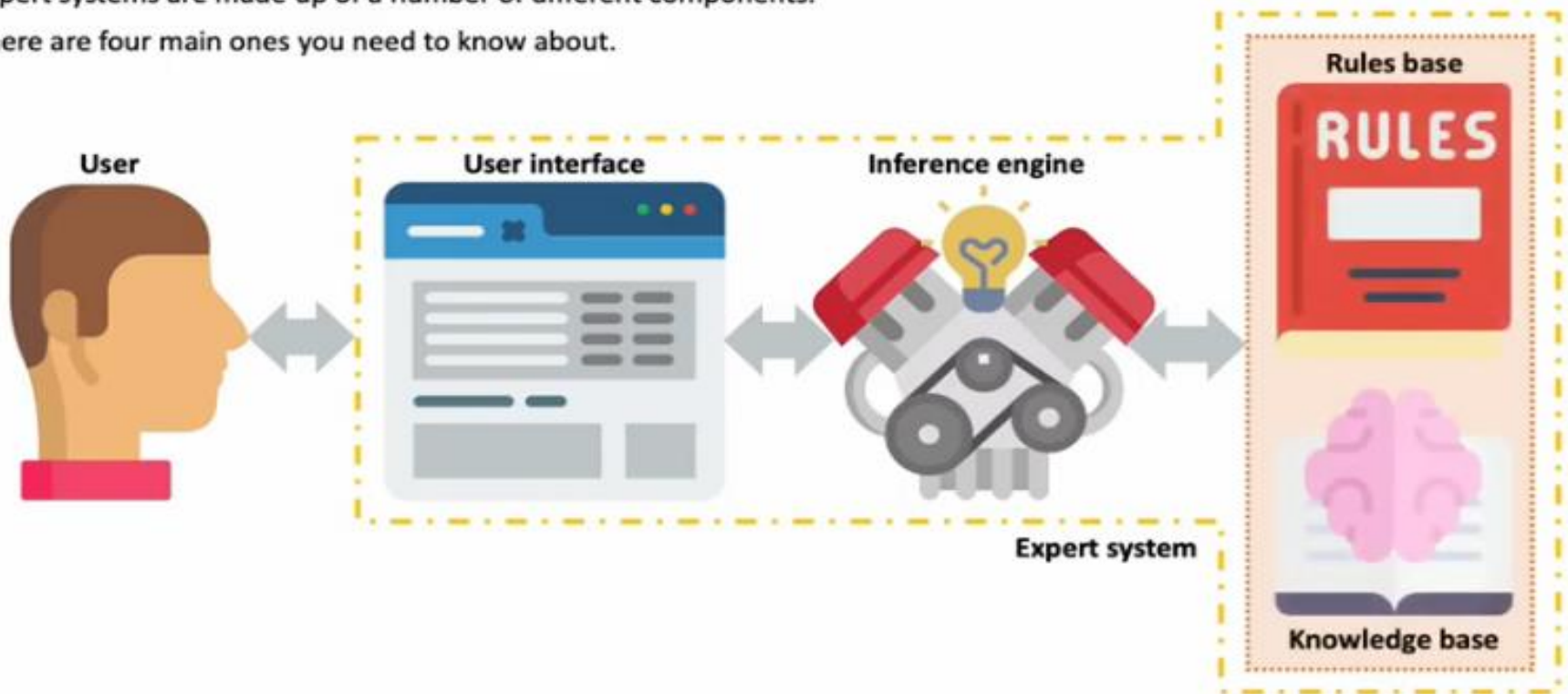
6.3 Artificial intelligence

Basic operations and components of AI systems

What makes up an expert system?

Expert systems are made up of a number of different components.

There are four main ones you need to know about.



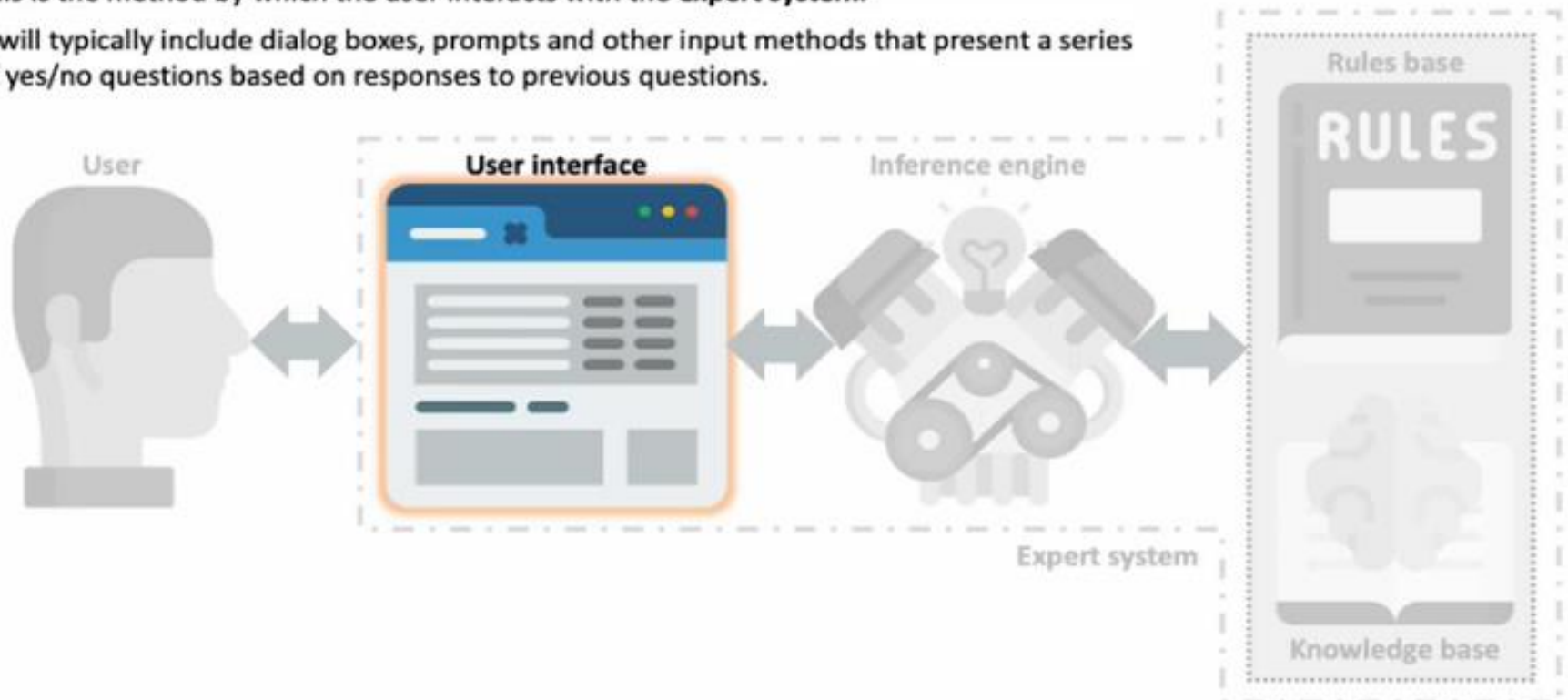
6.3 Artificial intelligence

Basic operations and components of AI systems

User interface

This is the method by which the user interacts with the **expert system**.

It will typically include dialog boxes, prompts and other input methods that present a series of yes/no questions based on responses to previous questions.



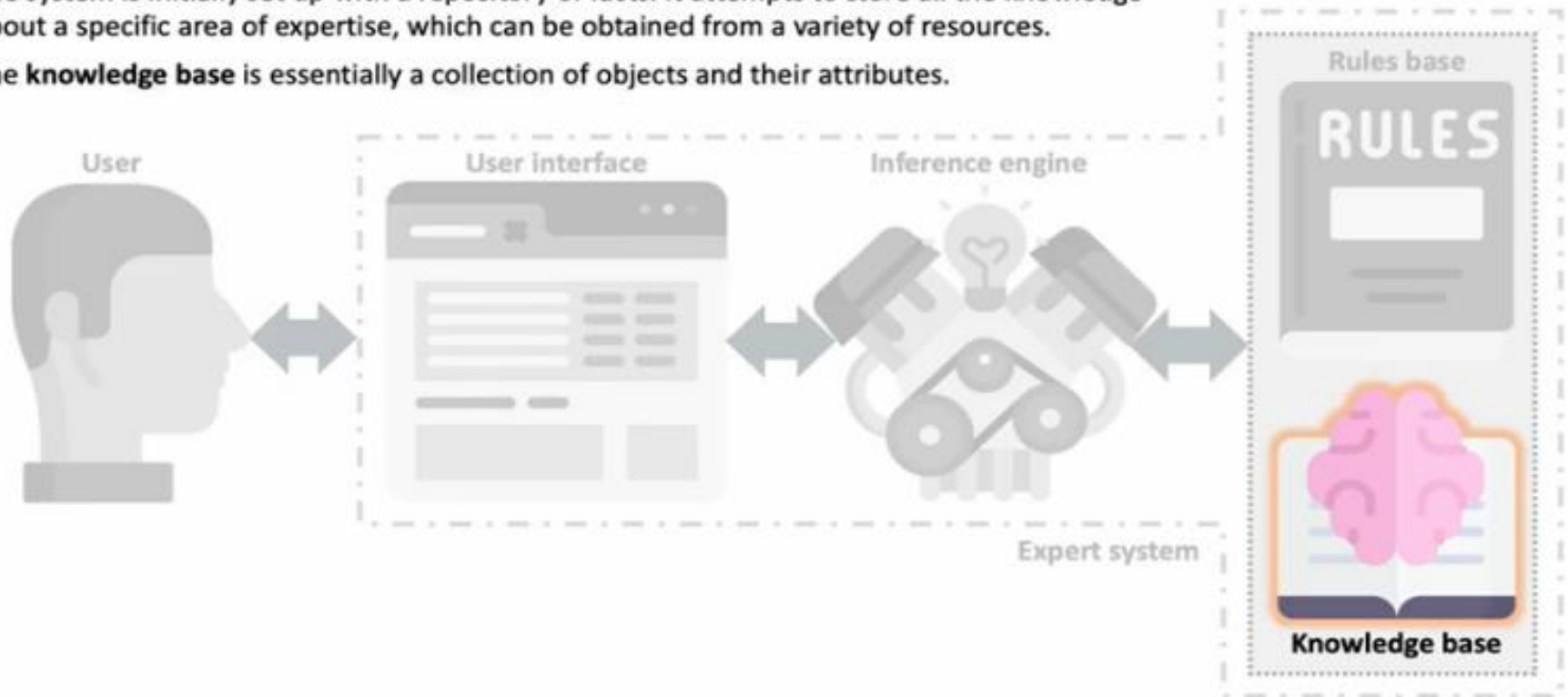
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Basic operations and components of AI systems

Knowledge base

The system is initially set up with a repository of facts. It attempts to store all the knowledge about a specific area of expertise, which can be obtained from a variety of resources.

The **knowledge base** is essentially a collection of objects and their attributes.



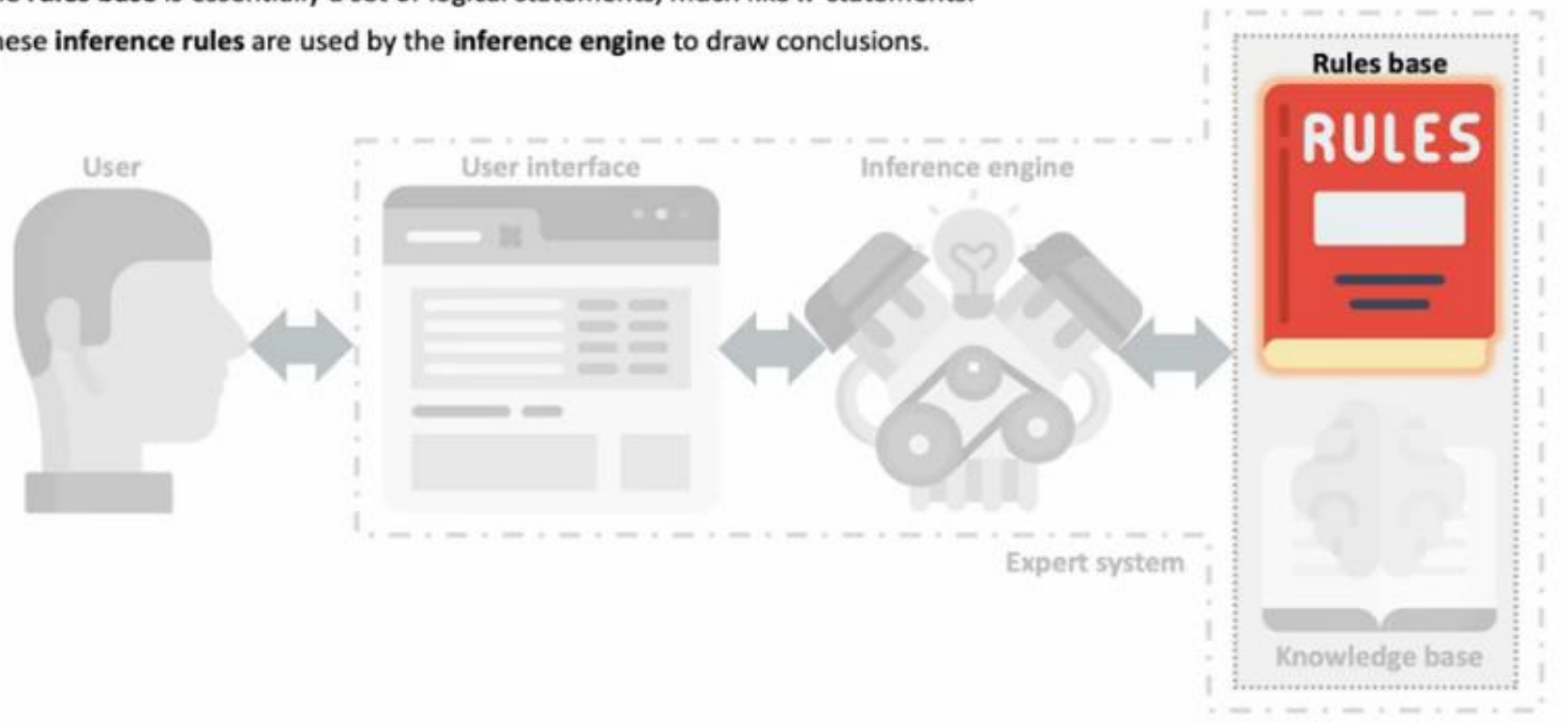
6.3 Artificial intelligence

Basic operations and components of AI systems

Rules base

The **rules base** is essentially a set of logical statements, much like IF statements.

These **inference rules** are used by the **inference engine** to draw conclusions.



6.3 Artificial intelligence

Basic operations and components of AI systems

Inference engine

The **inference engine** is the main processing component of an **expert system**.

In essence, it acts like a search engine. It searches the **knowledge base**, using the **inference rules** from the **rules base** to make decisions.



6.3 Artificial intelligence

Basic operations and components of AI systems

Simple example of an expert system

Object	Attribute 1 (Travels land/air/water)	Attribute 2 (Method of power)	Attribute 3 (Number of wheels)
Train	Land	Engine	Many
Car	Land	Engine	Four
Helicopter	Air	Engine	None
Cruise ship	Water	Engine	None
Bike	Land	Manual	Two
Plane	Air	Engine	Many
Rowing boat	Water	Manual	None
Skateboard	Land	Manual	Four

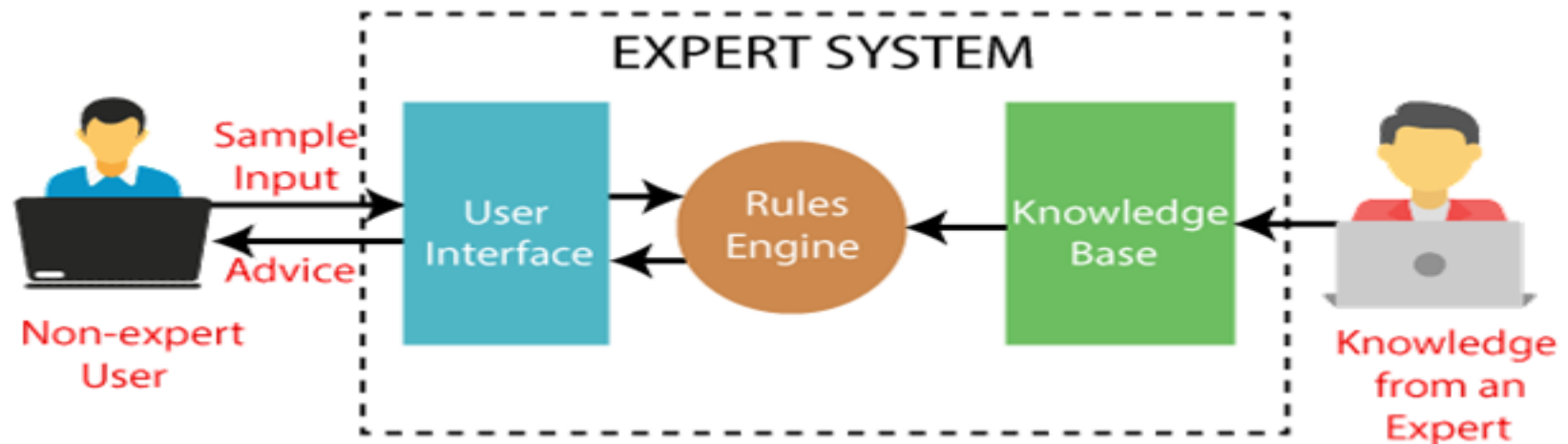
Does it travel on water? **NO**
Does it travel in the air? **NO**
Is it powered manually by a human? **YES**
Does it have two wheels? **YES**
Conclusion: It is a bike.

6.3 Artificial intelligence

Basic operations and components of AI systems

An expert system: is a computer program that uses artificial intelligence (AI) technologies to simulate the judgment and behavior of a human or an organization that has expertise and experience in a particular field.

Expert systems are usually intended to complement, not replace, human experts.



Note: It is important to remember that an expert system is not used to replace the human experts; instead, it is used to assist the human in making a complex decision. These systems do not have human capabilities of thinking and work on the basis of the knowledge base of the particular domain

6.3 Artificial intelligence

Basic operations and components of AI systems

An expert system:

Below are some popular examples of the Expert System:

DENDRAL: It was an artificial intelligence project that was made as a chemical analysis expert system. It was used in organic chemistry to detect unknown organic molecules with the help of their mass spectra and knowledge base of chemistry.

MYCIN: It was one of the earliest backward chaining expert systems that was designed to find the bacteria causing infections like bacteremia and meningitis. It was also used for the recommendation of antibiotics and the diagnosis of blood clotting diseases.

There are many applications that use expert systems:

- » oil and mineral prospecting
- » diagnosis of a patient's illness
- » fault diagnostics in mechanical and electronic equipment
- » tax and financial calculations
- » strategy games, such as chess
- » logistics (efficient routing of parcel deliveries)
- » identification of plants, animals and chemical/biological compounds.

6.3 Artificial intelligence

Basic operations and components of AI systems

An expert system:

Expert systems have many advantages:

- » they offer a high level of expertise
- » they offer high accuracy
- » the results are consistent
- » they have the ability to store vast amounts of ideas and facts
- » they can make traceable logical solutions and diagnostics
- » it is possible for an expert system to have multiple expertise
- » they have very fast response times (much quicker than a human expert)
- » they provide unbiased reporting and analysis of the facts
- » they indicate the probability of any suggested solution being correct.

Expert systems also have disadvantages:

- » users of the expert system need considerable training in its use to ensure the system is being used correctly
- » the set up and maintenance costs are very high
- » they tend to give very 'cold' responses that may not be appropriate in certain medical situations
- » they are only as good as the information/facts entered into the system
- » users sometimes make the very dangerous assumption that they are infallible.

6.3 Artificial intelligence

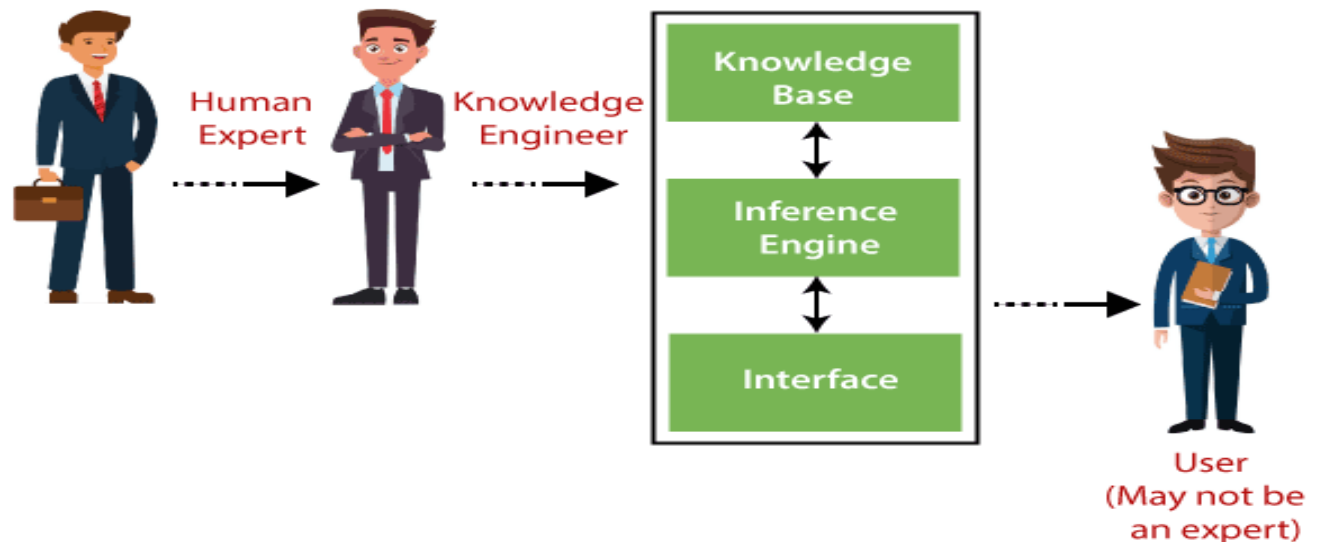
Basic operations and components of AI systems

An expert system:

Components of Expert System

An expert system mainly consists of three components:

- 1- **User Interface**
- 2- **Inference Engine (Rules of Engine)**
- 3- **Knowledge Base**
- 4- **Rules base**

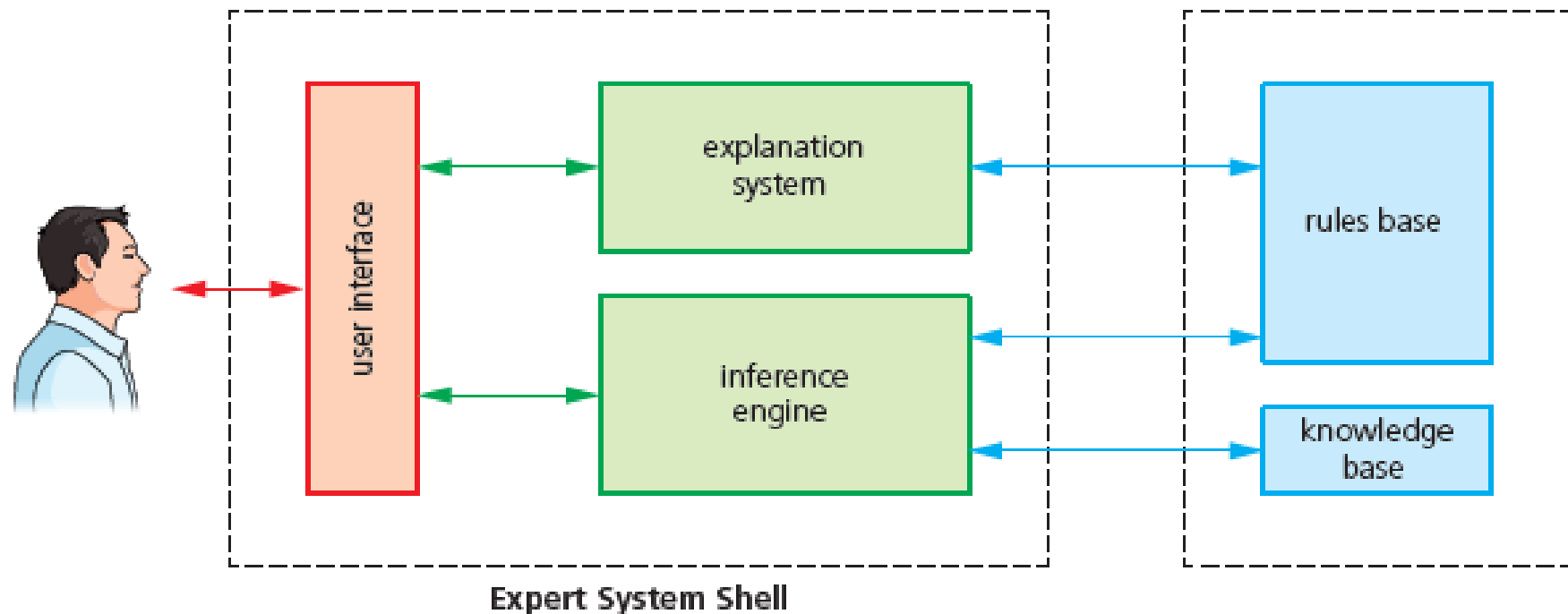


6.3 Artificial intelligence

Basic operations and components of AI systems

An expert system:

Structure of an Expert System:



6.3 Artificial intelligence

Basic operations and components of AI systems

An expert system:

1- User Interface

With the help of a user interface, the expert system interacts with the user, takes queries as an input in a readable format, and passes it to the inference engine. After getting the response from the inference engine, it displays the output to the user.

» interaction can be through dialogue boxes, command prompts or other input methods.

» the questions being asked usually only have Yes/No answers and are based on the responses to previous questions.

6.3 Artificial intelligence

Basic operations and components of AI systems

An expert system:

2. Inference Engine(Rules of Engine)

- The inference engine is known as the brain of the expert system as it is the main processing unit of the system.
 - » this is the main processing element of the expert system
 - » the **inference engine** acts like a **search engine** examining the **knowledge base** for information/data that matches the queries
 - » it is responsible for gathering information from the user by asking a series of questions and applying responses where necessary; each question being asked is based on the previous responses.
 - » the inference engine is the problem-solving part of the expert system that makes use of **inference rules** in the **rules base**.
 - » since the knowledge base is a collection of **objects** and **attributes**, the inference engine attempts to use information gathered from the user to find an object that matches (making use of the rules base to find a match).

6.3 Artificial intelligence

Basic operations and components of AI systems

An expert system:

3. Knowledge base

- » the knowledge base is a repository of facts
- » it stores all the knowledge about an area of expertise obtained from a number of expert resources.
- » it is basically a collection of objects and their attributes; for example:

Object	Attribute 1	Attribute 2	Attribute 3	Attribute 4	Attribute 5	Attribute 6
dog	mammal	can be a pet	lives on land	makes bark sounds	body is covered in fur	walks on 4 legs
whale	mammal	not a pet	lives in water	makes sonic sound	body covered in skin	swims; no legs
duck	bird	not a pet	lives in water	makes quack sounds	body covered in feathers	swims; has two legs

- is it a mammal? YES
- can it be a pet? NO
- does it live in water? YES
- does it make sonic sounds? YES
- is its body covered in skin? YES
- does it have any legs? NO

conclusion: it is a WHALE.

6.3 Artificial intelligence

Basic operations and components of AI systems

An expert system:

4. Rules base

- » the rules base is a set of inference rules
- » inference rules are used by the inference engine to draw conclusions (the methods used closely follow human reasoning)
- » they follow logical thinking like the example above; usually involving a series of 'IF' statements, for example:

IF continent = "South America" AND language = "Portuguese" THEN country = "Brazil"

6.3 Artificial intelligence

Basic operations and components of AI systems

An expert system:

Setting up an expert system

- » information needs to be gathered from human experts or from written sources such as textbooks, research papers or the internet
- » information gathered is used to populate the knowledge base that needs to be first created
- » a rules base needs to be created; this is made up of a series of inference rules so that the inference engine can draw conclusions
- » the inference engine itself needs to be set up; it is a complex system since it is the main processing element making reasoned conclusions from data in the knowledge base
- » the user interface needs to be developed to allow the user and the expert system to communicate
- » once the system is set up, it needs to be fully tested; this is done by running the system with known outcomes so that results can be compared and any changes to the expert system made.

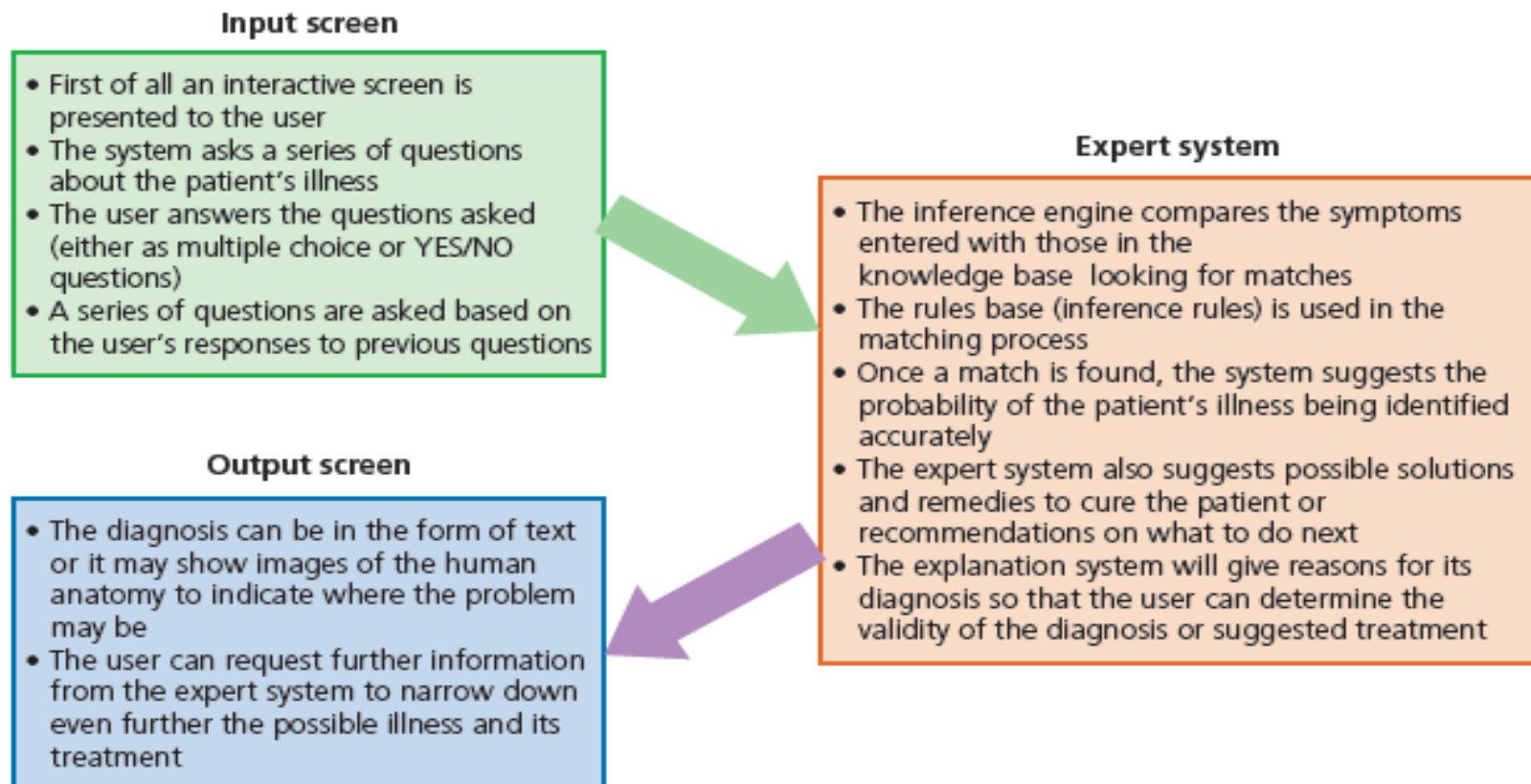
6.3 Artificial intelligence

Basic operations and components of AI systems

An expert system:

Setting up an expert system

Example use of an expert system (medical diagnosis)



6.3 Artificial intelligence

Basic operations and components of AI systems

Machine learning

Machine learning is a subset of AI – in other words, it is one way to achieve artificial intelligence.

Machine learning is the science of training a device or application to perform a task and improve its capabilities – this is achieved by feeding it data, information and scenarios so it can learn over time from its past experiences.

KEY CHARACTERISTICS

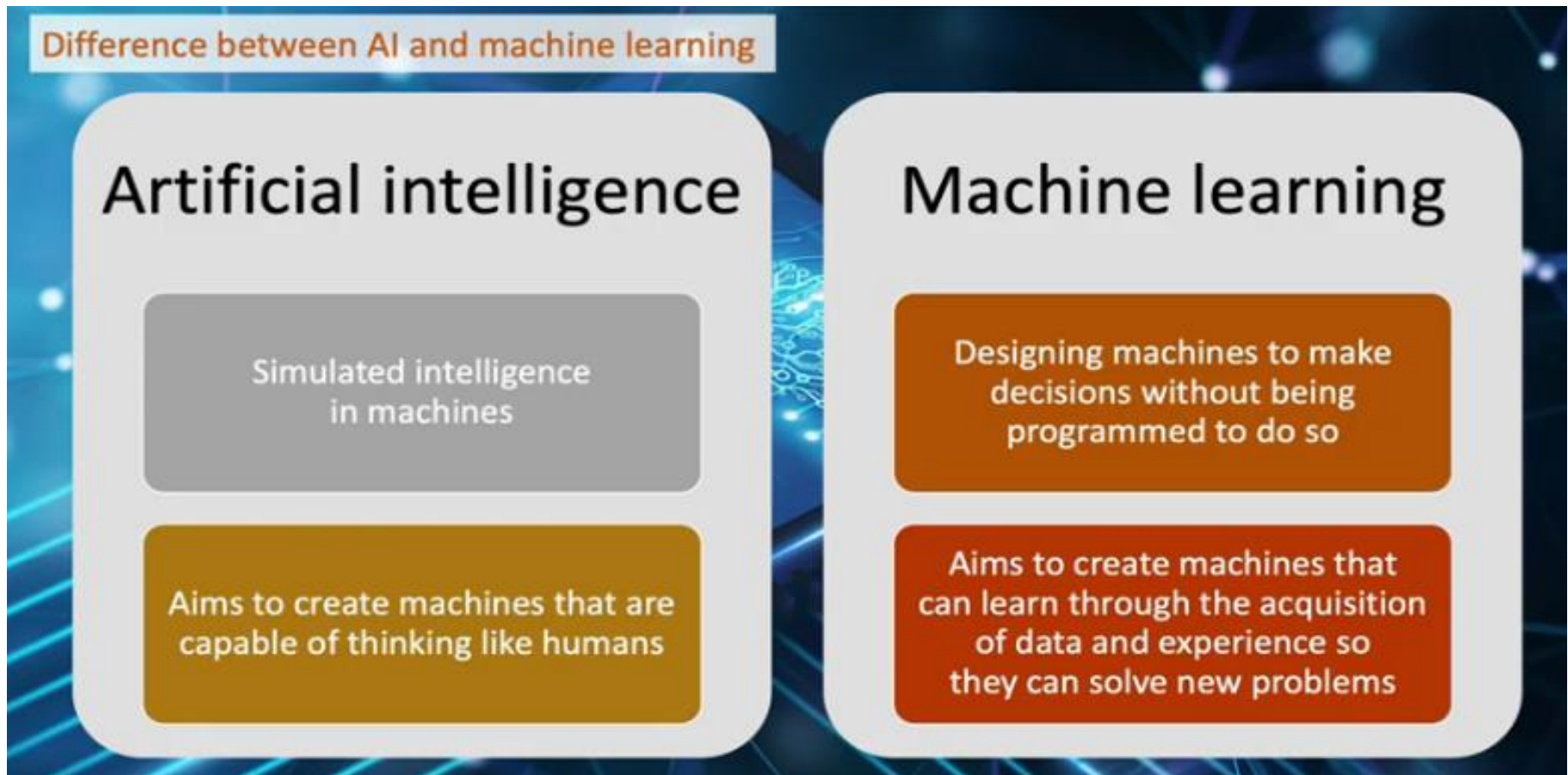
Possible to make predictions and decisions based on previous scenarios

Ability to manage and analyse vast volumes of complex data

Can provide fast, accurate outcomes

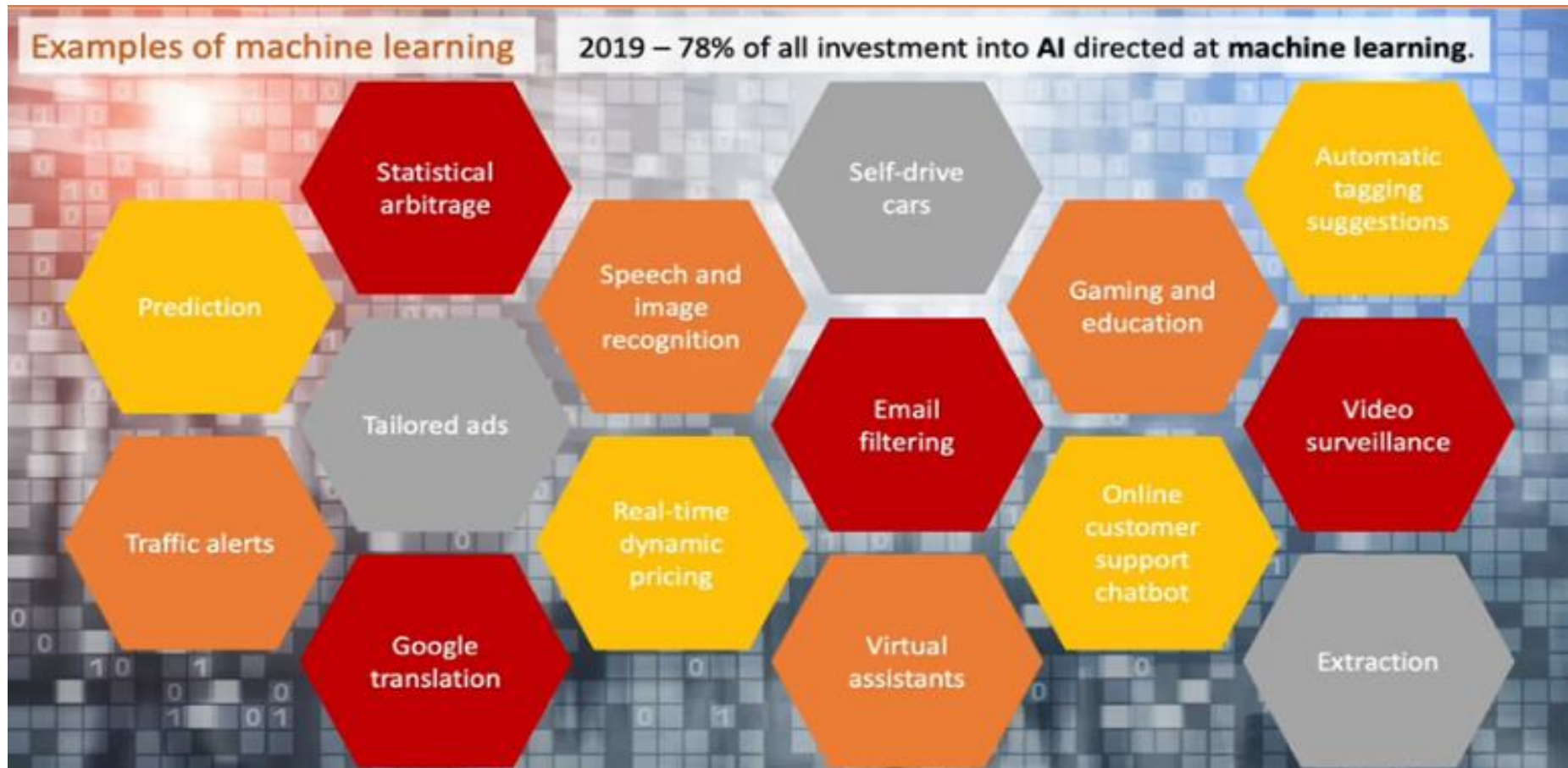
6.3 Artificial intelligence

Basic operations and components of AI systems



6.3 Artificial intelligence

Basic operations and components of AI systems



6.3 Artificial intelligence

Basic operations and components of AI systems

Examples of machine learning

A great example of **machine learning** is the automatic tag suggestion feature found on many social media sites such as Facebook™.

Facebook™ can automatically suggest friends to tag in photos you upload using its proprietary machine learning facial recognition system, DeepFace.

The system uses a nine-layer neural network with over 120 million connection weights. It was initially trained using over four million images uploaded by users and continued to learn as more images were added.

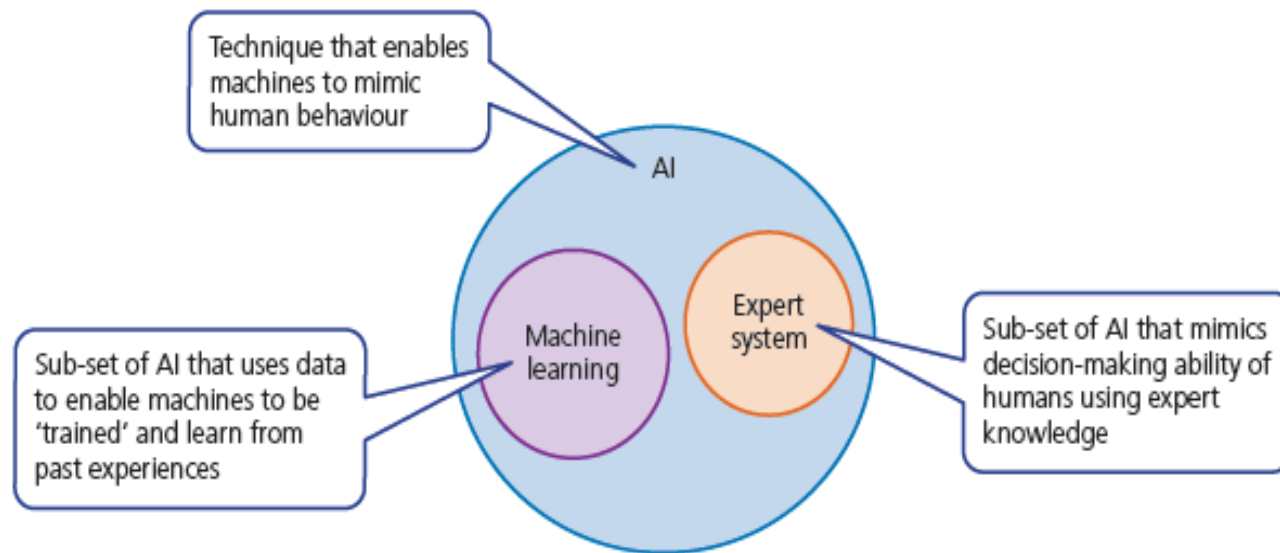
Human accuracy	97%	FBI software	85%
DeepFace (Facebook)	97%	FaceNet (Google)	99%



6.3 Artificial intelligence

Basic operations and components of AI systems

Machine learning



Mimics = to imitate or copy in action, speech, etc., often playfully

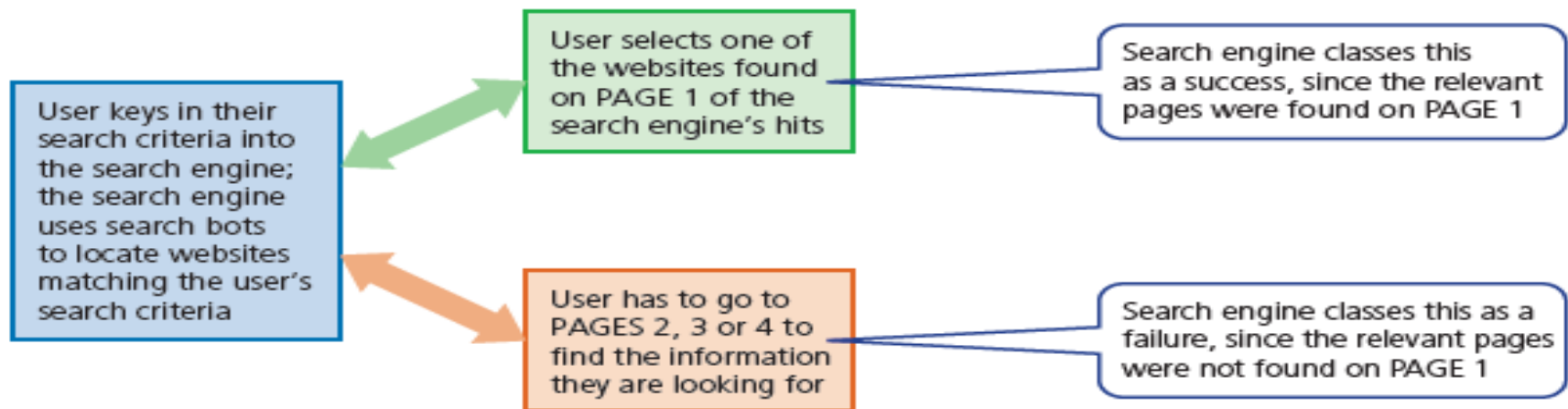
6.3 Artificial intelligence

Basic operations and components of AI systems

Machine learning

Machine learning is a subfield of artificial intelligence, which is broadly defined as the capability of a machine to imitate intelligent human behavior. Artificial intelligence systems are used to perform complex tasks in a way that is similar to how humans solve problems.

Example that uses machine learning are the most sophisticated **search engines** or **Image recognition** (x-ray as cancerous or not)



The search engine will learn from its past performance, meaning its ability to carry out searches becomes more and more sophisticated and accurate.

6.3 Artificial intelligence

Basic operations and components of AI systems

Machine learning

Difference between AI and machine learning

AI	Machine learning
represents simulated intelligence in machines	this is the practice of getting machines to make decisions without being programmed to do so
the aim is to build machines that are capable of thinking like humans	the aim is to make machines that learn through data acquisition, so that they can solve new problems

6.3 Artificial intelligence

Basic operations and components of AI systems

Machine learning

Examples of machine learning



Example 1: Categorising email as spam

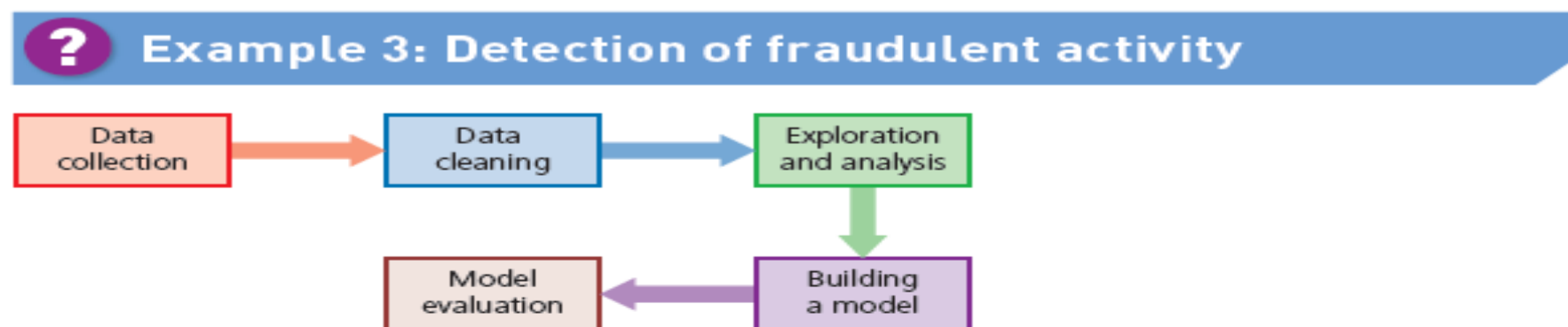
Consider email messages such as '*You have won \$2 million in the National Lottery*'; how can machine learning determine that this email should be put into your spam folder?

- » A machine learning algorithm collects data about emails, such as email content, headers, senders name/email address and so on.
- » It carries out a 'cleaning' process by removing **stop words** (for example, the, and, a) and punctuation, leaving only the relevant data.
- » Certain words/phrases are frequently used in spam (for example, lottery, earn, full-refund) and indicate that the incoming email is very likely to be spam.
- » The machine learning model is built and a 'training data set' is used to train the model and make it learn using past email known to be spam.
- » Once it is evaluated, the model is fine-tuned and tested live.

6.3 Artificial intelligence

Basic operations and components of AI systems

Machine learning



▲ **Figure 6.27** Machine learning model development

- » Data is gathered by a survey or **web scraping**; for example, to detect credit card fraud, information about customers is gathered, such as types of transactions (does the customer buy designer clothes?), shopping habits and certain personal data.
- » Redundant data is then removed; this needs to be carefully done to remove the possibility of wrong predictions.
- » The most important machine learning step: the algorithm is trained through real examples of customer purchasing behaviour.
- » A model is built based on learning from the training data, and the machine learning algorithm can now be used to detect fraud (for example, if a customer spends an unusual amount on a piece of jewellery, there is a high chance a fraudulent activity has taken place).
- » The machine learning model is then fully tested with known data and known outcomes; the system is modified if it hasn't met its criteria to detect fraudulent activity.

6.3 Artificial intelligence

Basic operations and components of AI systems



Operations and components of AI systems

Expert systems

- Simulates a human expert and consists of the following components:
 - Knowledge base
 - Rule base
 - Inference engine
 - Interface
- Used in areas such as medical diagnosis and telecommunications.

Machine learning

- A subset of artificial intelligence – in other words, it is one way to achieve artificial intelligence.
- The science of training a device or application to perform a task and improve its capabilities.
- Achieved by feeding it data, information and scenarios so it can learn over time.

6.3 Artificial intelligence

Basic operations and components of AI systems

Click the link below to practice Machine learning using python programming

https://www.w3schools.com/python/python_ml_getting_started.asp